COMPLEXITY AND STABILITY FOR EPISTEMIC AGENTS: THE FOUNDATIONS AND PHENOMENOLOGY OF CONFIGURATIONAL ENTROPY

A Thesis

Submitted to the Faculty

in partial fulfillment of the requirements for the

degree of

Doctor of Philosophy

in

Physics & Astronomy

by

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DARTMOUTH COLLEGE

Hanover, New Hampshire

October 19, 2017

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Acknowledgments

To my parents, Ryszard and Helena, who instilled in me an awe towards learning and set me on a lifelong path of asking *why?* and *how?*

To my advisor, Marcelo Gleiser, whose experience and kindness guided me during my time in the wilderness.

To all my teachers, from Thea and Anne to Robert and Miles and Lorenza, who have been, and are, my torchlight through the darkness.

To everyone who has changed me, and by extension, the world I partake in; I would not be who I am if not for you.

To Annie, Frank, Sylvia, and Vivian: you shaped my childhood.

To the Pages, the Cables, and the Agrawals: you shaped my adolescence.

To Shirley and Mari: you have been a foundation, peers to look up to, and to laugh and drink and think with throughout my years.

To Sarah, Zajj, and Brian: you reshaped me when I thought I was all grown up, and taught me how to plant gardens when all I knew was how to build skyscrapers.

Lastly to Daniel, around whom our dialogues danced.

Abstract

This thesis lays down the epistemic foundations, and explores some of the phenomenology, of the information theoretic measure known as Configurational Entropy(CE). The second chapter is on the connection between logic, belief, and information, placing a firm emphasis on why, and how, the Bayesian interpretation of probability theory allows for an epistemic interpretation of information and entropy. The third chapter develops the analytical form of CE, and then examines its properties in both discrete form and the continuum. The fourth chapter delves into using the measure to understand phase transitions in the Ising Universality Class through an informational narrative. CE is used to describe how information is stored and transferred in systems close to, and far from, criticality. The fifth chapter explores the relationship between CE and instability.

Preface

This works consists of a majority of the research I worked on at Dartmouth College over the course of three and a half years. It was a difficult start, since my background when I arrived at graduate school was mostly in Cosmology. Marcelo Gleiser introduced me to CE in 2012, and we quickly started to apply it to a broad range of phenomena. Lacking any sort of theoretical basis for the measure, however, troubled me; undoubtedly it was this itch at the back of my mind that eventually turned into this thesis.

I delved deeply into information theory, and realized how important an understanding of probability is to justify any of it. When I began research I was trapped in the frequentist viewpoint, which made the interpretation of informational measures taste... unsavory. Fortunately, none of us work in a vacuum, and as I dug deeper into what others had done before, I was converted to the Bayesian school. I am indebted to the writings of Ariel Caticha, who has done amazing work on the connection between Classical and Quantum Mechanics and Bayesian inference. My journey found bedrock in the works of Cox and Jaynes. There, finally, I saw a way to interpret information theory in a way that fits in with our intuitive understanding of what information is. Once that justification is there, the remarkable efficacy of CE in the phenomenology explored is much clearer.

This thesis is my attempt to bridge ideas that stem from epistemic and ontic considerations about the world. It is *a* path from language and epistemology to information and physics. Not all the trees have been cleared, and some of the bridges I've built can certainly be improved. It is by no means exhaustive, nor is it complete, but it is a path that I am thrilled to walk with you through. I hope you enjoy the scenery.

Damian Sowinski, 2016

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Chapter 1

Going Through the Motions

We live in the *Age of Information*. As a child I witnessed the rapid expansion of the World Wide Web; I grew up alongside the Internet, watched it rapidly penetrate into nearly every aspect of my daily life, and become not just a luxury, but a necessity for modern living. Whenever I want to know something, whether it be a flight of fancy or a thirst for facts, computers connect me to massive online encyclopedias; I have access to nearly unlimited amounts of *information* at the press of a button. This is where the namesake for our Age comes from: simple access to vast repositories in whose depths lay the accumulated wealth of recorded human knowledge. The process can be overwhelming, and, not on a rare occasion, have I found myself staring blankly at words and numbers and figures, clicking mindlessly from one page to another, pushing a boulder made of bits like some modern incarnation of Sisyphus.

Day in and day out, I read papers, listen to podcasts, and have discussions with other human beings, each time leaving me feeling that the activity I just partook in has somehow changed me; I feel *informed*. Is this because the experiences themselves have information in them? It's difficult to define exactly what information is without

some recourse to *data* [29][27]. The experiences I have are undoubtedly filled with data, but there is a vast gulf between information and data. A book written in Persian contains a lot of data, but it is almost completely uninformative to me because I do not know how to read it. If I look through the pages of the book, I may start to recognize repeated characters, notice the overall structure in which the symbols are placed on the pages, and so on. But someone who reads it and can contextualize the symbols as a work of Jalād ad-Dīn Muhammad Rūmī, the 13^{th} century poet, will be far more informed than I. Similarly, I can watch a movie in Polish, and understand the dialog, making the experience of watching both informative and entertaining. I doubt it would be either to someone who does not understand Polish; just a series of images where actors act out their roles to some tale whose premise is an enigma.

Even if I have the proper prior knowledge to contextualize data and gleam information from it, there is still an ambiguity. Reading a novel for the first time leaves me feeling informed as I assimilate all the data in it within the context of my prior knowledge. As the themes in the novel become apparent through storytelling, I process that newly acquired knowledge by comparing it to my own past experiences to relate it to what I believe about the world. When I read that same book again I do not feel informed by the same amount as I did the first time. I have the same prior knowledge as before, so what has changed? If two people, twins who have grown up identically and shared all the same experiences, read the same book, are they equally informed by it? If they have all the same prior knowledge, do they contextualize the data in the book in the exact same way? With regard to both these lines of questioning, I would argue that no two experiences are ever the same. When I reread the book, the book may not have changed, but I certainly have: my prior knowledge now includes

the experience of having read the book. Therefore the process of contextualization will occur differently than when I read the book for the very first time. In the case of the twins, they occupy different places at the same time, making the experience surrounding the experience of reading the book different for both. A slight breeze in the room felt by one of them while reading a rather ominous passage, but not by the other, could make all the difference in how they contextualize the data they experience. It is as the presocratic philosopher Heraclitus put it, *On those stepping into rivers staying the same other and other waters flow.* [31]

Experiences are unique because they do not occur in a vacuum. This string of moments ends up molding each of us into the individuals that we become. Each experience provides context for the next, affecting how the following experience is contextualized in an endless chain of being and becoming. The amount of information that we gain in our encounters with data becomes history dependent [28]. At every moment, who we will become depends on who we are as much as it depends on the experience currently playing out before us. Herein lays the distinction between data and information that we have been seeking. Data is experience, information is experience that induces change within us [13][14].

A change in what though? What is being described here is the subjective experience of thinking things such as ourselves. It is that internal ability to contextualize experience that changes during an informative experience. The notion of an uninformative experience, then, is an idealization. All experience, at some level, changes the way we perceive the following moments and assimilate the data contained in them. The distinction is then between the level of informativeness in experience. Those experiences that change us drastically are the most informative, those that

barely perturb us, the least. Information exists solely through it's ability to make a difference [74].

If information can be found in the change that is induced in a thinking thing's ability to contextualize experiential data, then, at its heart, information is an epistemic, not ontic, quality. Information is not found in the world, it is found in our interactions with the world. This is in stark contrast to the way in which information was introduced in Claude Shannon's *A Mathematical Theory of Communication*; the paper that gave birth to the field of information theory [81]. It is important to recall that Shannon was interested, as his paper bluntly puts it, communication. Shannon considered a string of data being sent from a source to a receiver, over what he dubbed a communication channel. We can think of the source and the receiver as you and I, with the air between us being the channel. Or we can think of two telephones, connected by a long cable. Or two cell phones, connected by a series of communication satellites. The possibilities are nearly limitless, but the idea is generic: data is sent from a source to a receiver over some channel. To Shannon information is the reliable transmission of messages over that channel.

Shannon wanted to know what maximum number of messages per unit time could be supported by the channel, which he called the *channel capacity*. He considered the case of both noiseless and noisy channels. The noiseless case is important for considerations of compression, but it is the noisy case that had a stunning result. Imagine that you and I are at a concert. We've both been looking forward to hearing a new Morrissey cover-band, but given that our thoughts have been focused on the epistemological foundations of information for the past few days, you cannot help yourself from relaying to me a key insight you just had. *Everyday is Like Sunday*

begins playing, and the singer's voice is like butter to the ears. You encode your thoughts in sound waves, not unlike the singer, your vocalizations reaching my ears, *trudging slowly over wet sand*. The band is really good, and I can't help but hear the lyrics as I listen to your words. I mishear some of the words you say. You realize I'm not being very attentive, so you begin repeating yourself. This, of course, makes the rate at which you're transmitting thoughts to me sluggish. Shannon investigated this relationship between the fidelity of a message and the rate messages are sent. There seemed to be a direct relationship between the rate messages could be sent and the errors in them. It appeared that if a sender wanted to send messages with fewer errors, they would have to correspondingly decrease the rate at which they sent messages, in the limit that perfect fidelity corresponded to a rate of no message transmission. You would never be able to tell me anything if you had to repeat each word an infinite number of times.

What Shannon discovered was that a noisy channel has a non vanishing message transmission rate with perfect fidelity, which he called the channel capacity. A qualitative description of Shannon's result is shown in figure 1.1 There exist coding schemes that allows a sender to send messages with perfect fidelity at finite rates. Those rates are bounded by the channel capacity. A channel can relay messages perfectly at a rate no greater than its channel capacity, given the optimal coding scheme. Here we glimpse a thread that connects Shannon's view of information as being a reliable transmission of messages, and the epistemically subjective notion of information which we began to develop: Shannon's result relies on the existence of an internal mechanism, an encoding. Shannon proved the existence of such codes, but several decades passed before Gallager implemented the first parity check code that

approached the Shannon channel capacity [34].

In the noiseless case, Huffman showed that there are encoding schemes that work better than others, which we shall briefly discuss [51][18]. Given some alphabet of symbols, $\mathcal{A} = \{a_1, a_2, \ldots, a_N\}$, that optimal encoding relies on the frequency with which the symbols are used in messages. For example, in English the letter e appears approximately once in every 8 letters. The letter q, on the other hand, appears once in every 1000 letters [83]. If we used the same number of bits for both letters, then we would be very wasteful. Since e appears so often, it is better to encode it using fewer bits. The rarity of q means we can encode it with a lot of bits. Shannon's result then tells us that if we use the natural frequencies of letters in a language, then the maximal rate at which messages can be sent is related to the *Shannon Entropy*:

$$S = -\sum_{a_i \in \mathcal{A}} p(a_i) \log p(a_i). \tag{1.1}$$

This optimal rate relies on knowing the frequencies with which letters appear in a language using the alphabet $\mathcal{A}$. Here, again, we see subjectivity crawling into Shannon's results. Information, as measured by the optimal rate of message transmission, relies on an internal knowledge of the frequencies of letters.

For both the noisy and noiseless coding theorems of Shannon, a knowledge of the distribution of letters in a language is necessary. This knowledge is encoded in a probability distribution over the alphabet. For practical purposes, these distributions can be found by examining a large corpus of literature, and extracting the frequencies of each letter. This is a statement concerning contextuality. Without the context of the language being used, the maximal rate of message transmission is meaningless.

Let us turn then to the context, namely the probability distribution of letters in

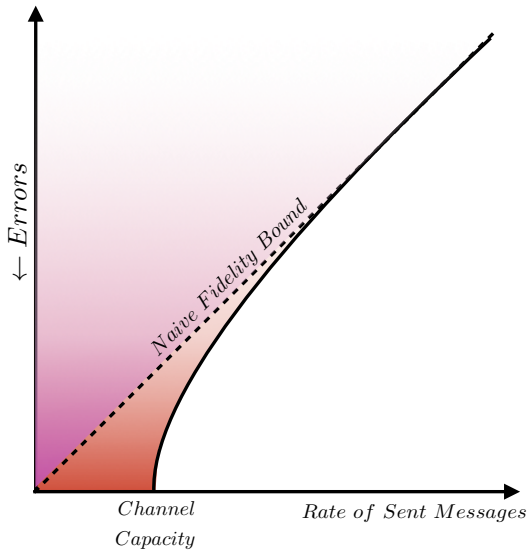

Figure 1.1: The impact of Shannon's coding theorem was to show that an information channel can transmit messages with perfect fidelity at a non zero rate known as the channel capacity. The dashed line represents a naive fidelity bound where errors are corrected for by repetition. The higher the rate of errors, the more repetition must be used to correct for them. The solid line represents the Shannon bound on the rate of message transmission that beats the naive fidelity bound.

the alphabet. Here the term probability relies on a frequentist interpretation: probabilities are the frequencies with which letters occur. The frequentist interpretation of probability is useful in this context, as it is in games of chance and any other repeatable experiences. However, it relies on a sort of unrealistic notion wherein the full corpus of all possible messages is used to define the probability. A sender and receiver are unlikely to have this full corpus. They need to use probability distributions that are inferred from incomplete data.

Probability distributions are inferred from data. There is something important in this statement, for it is a statement about the epistemic state of the sender and the receiver. The connection here goes beyond probabilities of letters, and brings us to the Bayesian interpretation of probability. Unlike the frequentist point of view, the Bayesian perspective concerning probabilities is that they are not absolute quantities forced on thinking things by the external world, but epistemic measures of the strength of belief thinking creatures like ourselves have about the world [55]. This interpretation has great strength, because it opens the possibility of defining the probability of not just repeatable events, but of unique ones. The connection between statements about the world, whether they concern repeatable or unique events, and epistemic states will be explored in Chapter 2. Propositional logic will be developed to understand how an epistemic agent talks about the world [24][22]. Following the work of Cox, probability theory will be derived from an epistemic theory about belief, and information will be shown to emerge from processes in which epistemic agents have experiences that alter their beliefs [21].

In this way, the central thesis of the beginning of this chapter, that *information is contained in experiences that change us* will be put on a solid quantitative foundation.

Going Through the Motions

We will derive Shannon's information measure on purely epistemic grounds, by considering how information is hidden from epistemic agents by the world and revealed through experience. A novel relation between hidden information and entropy will be examined from which the additive property of entropy, typically assumed as one of the Shannon-Khinchin axioms, will follow [58]. To make the connection between the coding theorems a bit more explicit, it will be apt to think of the sender of messages as the ontological substrate that an epistemic agent is embedded in, making the epistemic agent the receiver of the messages. The transmission of messages is then the continuous conversation that a thinking thing has with the world; the messages are the experiences that the agent has. The results of the coding theorems then become statements concerning the optimal rate at which epistemic agents can extract information from the continuous stream of experience. Unlike the receiver in the theorems, however, epistemic agents do not have a fixed set of beliefs, a probability distribution. Their interaction with information will cause their beliefs to change.

Once we have established how epistemic agents acquire information, and update their beliefs, we will consider how they do so while embedded in an ontological substrate that sends messages at different spatial scales. In Chapter 3 the informational measure of configurational entropy will be developed, allowing epistemic agents to recognize and focus on scales in which *interesting events* are taking place. This chapter will focus on the epistemic reasons for constructing the measure given its success in a wide array of applications studied by my advisor, Marcelo Gleiser, as well as Nikitas Stamatopoulos, Nan Jiang, Noah Graham, and Ruy Alberto Corréa [19][20] [39][37] [40] [41] [38] [84] [85] [86]. Once the theoretical foundations of this measure are established, it will be applied to specific phenomenological cases in Chapters 4

and 5.

Chapter 4 will focus on phase transitions to explore the ability of the measure to allow the epistemic agent to recognize which scales are relevant in the system they are contained in. By discerning the informativeness of certain scales over others, a rich informational narrative emerges for the epistemic agent. The results mirror standard approaches of statistical mechanics in describing phase transitions. The novelty comes from interpretation of the way in which information describing the system is perceived by the thinking thing interacting with it. Information, or more precisely, the hidden information in the world that is inaccessible to the epistemic agent, will be seen to minimize as natural scales emerge close to criticality. A closer examination will reveal that the density of hidden information becomes turbulent at scales that are larger than the interesting ones.

The full power of an informational narrative in the context of phase transitions has only just started to be explored, however I would like to take a moment to argue for the relevance of having such a narrative. Biological systems have shown us that the standard mechanistic narrative of physical phenomena is not flawed, but severely lacking in explanatory power. Take for example the process of protein synthesis. One could, for all intents and purposes, describe the process by specifying the positions and velocities of all the molecules inside the cell, and how, through mechanistic collisions and energetics, molecules such as ATP are formed. Though complete, this description is mostly data with very little information. By instead considering the informational narrative, one is lead to a coarse grained, yet much more informative description of protein synthesis. The information contained in DNA is read and transcribed into RNA in the Nucleus. RNA leaves the nucleus and the information it contains is

translated by ribosomes, which use it to build long strings of amino acids. When a chain of amino acids is complete, it folds into a protein. The informational narrative reveals the teleological mechanisms inside a cell, explaining far more than a simple accounting of the phase space evolution of all the atoms in a cell. Examining the storage, transfer, and processing of information by biological matter could be the key to understanding biogenesis [73] [90] [89] [91]. To complement the informational narrative of biological systems, the narrative must also be extended to those systems we do not consider as *alive*.

Configurational entropy provides an informational narrative for phase transitions, describing how information is both stored and transferred in the modes of the underlying order parameter field. Furthermore, the alternate form of configurational entropy known as deviation from expectation is shown to provide an interesting informational narrative for localized objects that approach instability. Chapter 5 explores both non-topological solotonic configurations in classical field theory known as Q-balls, as well as polytropic stellar configurations that are good models for stable and unstable degenerate stars known as white dwarfs. It is argued that the measure captures the inherent instability associated with objects that are close to being unbound energetically.

Chapter 2

Something to Sing About

Our protagonists have just finished a home cooked meal. The other guests have left, and they're taking a moment prior to cleaning up.

Oontolicus: That was truly a most entertaining dinner party, my old friend.

Epistemicus: Why yes, I'm glad you thought so highly of it.

Oontolicus: Well, it wasn't just me. Everyone who was there enjoyed the courses you paraded before our tastebuds. The duck was succulent. The risotto was delicate. The soufflé was superb. Truly masterfully done.

Epistemicus: Your compliments are too much...

Oontolicus: Don't be humble now, putting that much effort into arranging the meal must have had you in the kitchen all day. Such effort deserves praise. Would I have been able to create such an array of culinary adventures? I think not.

Epistemicus: Now who's being humble? You were always better at doing things than I, old friend.

Oontolicus: And you were always able to think things through to a degree that made me wonder. I guess our strengths have rubbed off on one another.

Epistemicus: It's funny how friendships work out that way...

In this chapter I will look at the epistemic foundations of information theory. This will create a foundation for the Bayesian interpretation of probability, which will lead naturally to a definition of information that is in line with common epistemic notions of what information should be. The chapter begins with a discussion of *epistemic agents*: idealized thinking things. Sentential logic will then be reviewed to develop the language that epistemic agents use to discuss the world [24]. Belief functions will then be discussed, and their connection to probability will be discovered through the Cox axioms [21] [13]. These will be used to quantify how information is hidden from epistemic agents interacting with the world, and a derivation of general entropic measure will follow [81]. A novel relation between entropy and hidden information will form the core of this derivation.

2.1 Epistemic Agents

A theory of thought has at its core a idealized thinking thing: an Epistemic Agent (EA). Edwin Jaynes referred to these inferential automatons as *robots*, while Ariel Caticha as *idealized rational agents*. [55] [14] I prefer the term EA since it places emphasis on two aspects that are central to the following: thought and agency. The first of these is encapsulated by an EA having a web of consistent beliefs. [13] The latter is found in an EAs ability to have experiences: to generate experiments that evolve its beliefs. This development will not focus on epistemic paradoxes concerning issues of what is knowable. [47]

Like human beings, EAs are entities that use language to talk about, and concep-

tualize, the world around them. Unlike human beings, EAs are infallible in the use of language. Their understanding of language is completely structural, and when they analyze the truth or falsity of a statement, they do so in the context of all possible statements that exist in the language. When a particular statement's truth value is established by an EA, all statements that are logically connected to that statement instantly *feel* that update. An EA cannot hold within itself contradictions, because all possible statements that could be said in the language are always present within the *mind* of the EA. Humans are not so lucky; we do not have the cognitive resources that an idealized EA has.

The ability for an EA to have access to the complete set of statements has profound consequences on the way that they make inferences during experiences. For us, an experience does not change all our beliefs in an instantaneous fashion, but for an EA this is not the case. Learning that *it is raining* has no effect on our belief that *the Yen is the currency used in Japan.* An EA does not see these statements as being completely independent. By pinning down the truth value of rain occurring at this very moment, the EAs beliefs instantly respond, no matter how tenuously connected they may seem to us: *It is raining because an abnormal wind system has caused a storm* $\rightarrow$ *This abnormal wind system is the result of a volcano exploding* $\rightarrow$ *Fujiyama has errupted* $\rightarrow$ *There is major tectonic activity occuring along the Ring of Fire* $\rightarrow$ *Massive earthquakes have caused most of Honshu to have submerged* $\rightarrow$ *Japan no longer exists as a sovereign state* $\rightarrow$ *The Yen is not the currency used in Japan, because Japan no longer exists.* This chain is so implausible that the average human being wouldn't register differences in belief in the links beyond the first. The effect of an experience to an EA is a change, however seemingly insignificant, to its

belief in everything else. It doesn't matter how far fetched a line of reasoning goes, all that matters to an EA is the structure of language; an EA responds to experience through an instantaneous updating of belief. We are not so lucky as to have this type of inferential acumen.

We require time to process experience and fully understand the implications it has had on our beliefs. Because processing takes time, experiences for us overlap with one another, blurring the borders between moments. Data from multiple experiences might get mixed up; leading to mistakes in our inference schemes. This is not the world that EAs experience. Their ability to instantly process ensures that any experiences separated by finite time intervals, however small, will update the complete set of beliefs the EAs have in the order in which they happen. It is important to keep these things in mind in the coming sections. Sometimes I will fall back into the habit of talking about a *me*, *we*, *he*, or *she* to illustrate a point. Rather than taking these statements literally, it is important to always step back and remind oneself that the subject is an EA, and not an actual human.

2.2 Language and the World

Language is that which is used to convey states of the world, both to ourselves, and to other epistemic agents. A language has the capacity to talk about not just the state of the world, but all possible states of the world, and therefore all possible worlds. A particular state in the world relies on pinning down the truthfulness of certain statements in the language, which is accomplished via a truth assignment. A language must also have the capacity to bring together statements to construct new statements, which is accomplished via logical connectives. A language is rational over

the possible worlds on which its logical connectives are consistent. Possible worlds in which this does not hold are irrational with respect to the language: the language is incapable of consistently describing those worlds. Hence even though the machinery of a language can be used to talk about all possible worlds, only those worlds on which the language can say things clearly can truly be spoken of.

One mustn't confuse the use of the term *language* in what follows as a case of a natural language such as English, which will be used in examples. The language that EAs use is, like them, an *idealization*. It is assumed that this language is sufficient at describing all the experiences that an EA can have. I will limit the discussion to the propositional calculus, otherwise known as zeroth-order logic. [69] The lack of quantifiers is precisely why paradoxes concerning knowability can be ignored.

2.2.1 Definitions

A *language* is a tuple, $\mathcal{L} = \langle \mathcal{S}, \Phi \rangle$. $\mathcal{S}$ is a set of objects, called *statements*, and $\Phi : \mathcal{S}^n \to \mathcal{S}$ is a set of n-ary maps into S, called *logical connectives*. The set of statements can be decomposed into a disjoint union, $\mathcal{S} = \mathcal{A} \cup \mathcal{C}_{\mathcal{A}}$. These are the atomic statements, $\mathcal{A}$, and the compound statements, $\mathcal{C}_{\mathcal{A}}$. The former is constructed by choosing a subset of statements so that no application of logical connectives acting on the set has an image which intersects the set.

An *epistemic state*, $\mathcal{E}_\tau$, is a fixed assignment of a truth function, $\tau : S \to \{0, 1\}$, $\mathcal{E}_\tau \subset \mathcal{S} \times \{0, 1\}$ such that for all $s \in \mathcal{S}$ only either $(s, 0)$ or $(s, 1)$ are in the state. Allowing for all possible truth assignments, we call the bundle of epistemic states the *epistemic realm*, $\mathcal{R}_\mathcal{E} = \cup_\tau \mathcal{E}_\tau$.

An epistemic state is *consistent* with the language $\mathcal{L}$ iff for each n-ary logical

connective, ϕ, there exists a function, ψ, such that for all statements, $s_i \in \mathcal{S}$, we have that $\tau \circ \phi(s_1, s_2, \ldots, s_n) = \psi(\tau(s_1), \tau(s_2), \ldots, \tau(s_n))$. The subset of the epistemic realm that is consistent with the language is the *realm of discourse*, $\mathcal{R}_\mathcal{D} \subset \mathcal{R}_\mathcal{E}$. A language is *rational* on its realm of discourse. Figure 2.1 concisely displays these definitions.

2.2.2 Unary Logical Connectives

First we will consider the possible unary logical connectives. Let s be a statement in a language with consistent truth functions. Then we have that:

$$\tau(\phi(s)) = \psi(\tau(s)). \tag{2.1}$$

The LHS can have two possible values, whereas the argument of ψ can also have two possible values. Hence only 4 possible ψ's can exist. We can label each possible ψ with a corner of the unit square, each vertex being an instance of $(\psi(0), \psi(1))$. Two of these instances, namely $(0, 0)$ and $(1, 1)$, are truth functions that are argument independent and can be trivially admitted into any language over epistemic states which include at least one false or one true, respectively, truth assignment. They are, for all intents and purposes, uninteresting, so we throw them out of consideration.

Of the remaining two possibilities $(0, 1)$ leads us to the logical connective satisfying: $\tau \circ \phi = \tau$. This function can mix up any statements that have the same truth value in the epistemic state. Since the number of statements with a particular truth value is not invariant over every epistemic state, the only such function on the largest subset of these in the epistemic realm is the identity: $\phi(s) = s$. Once again, an uninteresting logical connective which we throw out.

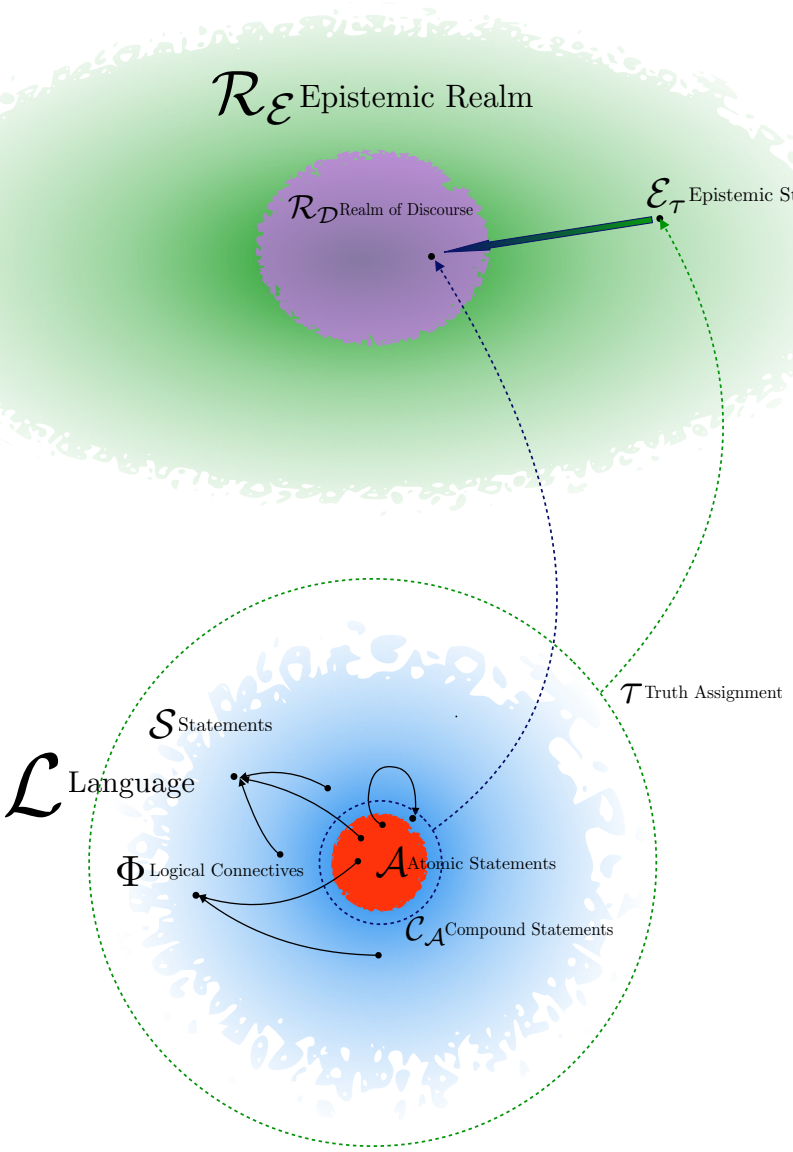

Figure 2.1: A Language consists of statements and logical connectives. A truth assignment on the set of all statements is an epistemic state. The set of all epistemic states is the epistemic realm. For a given set of logical connectives, a choice can be made of a special set of statements from which all others can be generated. The former are atomic, and the latter compound. Truth assignments that obey the consistency of the logical connectives map to rational epistemic states, which belong to the realm of discourse. Language can only say things about possible worlds. Impossible worlds cannot be spoken of.

The final possibility, $(1, 0)$, leads to a far more interesting logical connective, one which must satisfy $\tau \circ \phi \circ \phi = \tau$. Once more, for this property to hold on the largest subset of the epistemic realm, the logical connective must be an involution: $\phi \circ \phi(s) = s$. Furthermore, the epistemic states that admit this logical connective will be ones which for every $s \in \mathcal{S}$ there exists a dual statement, $\neg s = \phi(s) \in S$, with opposite truth value. We call the pair of statements *negations* of one another. We call the unique unary connective $\neg$ *negation*.

2.2.3 Binary Logical Connectives

We now construct all the binary connectives which satisfy

$$\tau(\phi(s_1, s_2)) = \psi(\tau(s_1), \tau(s_2)). \tag{2.2}$$

Once again, the LHS admits two values, and there are four possible pairs of arguments for ψ leading to 16 candidate functions. These correspond to the vertices of a $4-d$ hypercube, $(\psi(0,0), \psi(1,0), \psi(0,1), \psi(1,1))$. Once again the antipodal vertices $(0,0,0,0)$ and $(1,1,1,1)$ are argument independent, so we need not consider them further. This brings us down to 14 possibilities.

Since we already have the unary function at our disposal, we can further reduce the number of unique possibilities by identifying ψ and ψ' if $\psi' = \neg\psi$. Antipodal vertices can be transformed into one another via negation, so of the 14 candidates, 7 remain. Next we note that the application of negation to the arguments generates equivalences. For example, functions ψ_i and ψ_j are equivalent if any of the following

hold:

$$\psi_i(x, y) = \psi_j(1 - x, y)$$
$$\psi_i(x, y) = \psi_j(x, 1 - y)$$
$$\psi_i(x, y) = \psi_j(1 - x, 1 - y).$$

This equivalence relation partitions the remaining into 4 classes. Two of these classes, however, are really unary logical connectives acting on only one of their arguments. We throw out these disguised functions, leaving 2. The final condition to show that these two are actually on and the same is a bit tricky to find on one's own, but the final result is:

$$\psi_2(x, y) = \psi_1(1 - \psi_1(x, 1 - y), 1 - \psi_1(1 - x, y)). \tag{2.3}$$

All of the binary logical connectives can be written in terms of a single binary connective and negation [45]. The outline of the argument is shown in figure 2.2. The choice of the binary connective is not unique, however we take the pragmatic route rather than generality, and pick the $(1, 0, 0, 0)$ operator. We denote this binary connective as $\psi(x, y) = \tau(x \wedge y)$, and read it as statement x *and* statement y. It will also be

referred to as the *conjunction* of statements. It has several nice properties.

$$s_1 \wedge s_2 = s_2 \wedge s_1 \qquad \text{Commutativity} \qquad (2.4)$$

$$s_1 \wedge (s_2 \wedge s_3) = (s_1 \wedge s_2) \wedge s_3 \qquad \text{Associativity} \qquad (2.5)$$

$$s \wedge s = s \qquad \text{Idempotentcy on } \mathcal{L} \qquad (2.6)$$

$$s \wedge \neg s = F \qquad \text{Non-Contradiction} \qquad (2.7)$$

The last of these is very useful to an epistemic agent. It rules out a great many statements in the language on the grounds that they are contradictory, and thus prunes the epistemic realm into the realm of discourse. A language can be thought of as projecting out those worlds in the epistemic realm that it deems contradictory. It is instantiating the Wittgensteinian notion that the limits of language are the limits of the world [95].

It is important to consider how negation and conjunction work with one another. Though it is not an independent logical connective, we can introduce the *disjunction* operator, $\vee$ (read as *or* and living on $(1, 1, 1, 0)$). It is related to conjunction via *de Morgan's Laws*, which state that $\neg(x \wedge y) = \neg x \vee \neg y$. This is an instance of one of the conditions we used to reduce the number of binary connectives down to a unique one, though it has a long history in logic and Boolean algebra, with references going back not just to medieval, but ancient greek logicians [10]. Conjunction and disjunction also distribute over one another, $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ and $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.

One final comment in this section is necessary. We constructed our logical connectives first by considering unary ones, and then binary ones. Had we started with

binary ones first, it is possible to find a single one that implements all the others as well as negation. It is the well known *nor* operator. This remarkable property gives it the title of being functionally complete. An interesting piece of trivia is that the Apollo flight computer was built using only this single operator, in the form of electronic gates [17].

We eschew the use of a single LC by deferring to the masses. The negation and conjunction operators are broadly used by people in commonplace discussions, so it is easier to think of these as being primary. Statements such as *I will not have coffee* supersede *I will have coffee nor will I have coffee* in the everyday, though they are logically equivalent.

2.2.4 Higher Arity Logical Connectives

One might think that the process can continue logical connectives of arbitrarily high arity. The work to find the number of unique ones would have to be tremendous. Just considering the form of equation 2.3, one begins to think the general problem would be insurmountable. As it so happens, the answer turns out to be trivial: there are none [70]. The story behind this result is a bit of Polish history involving Banach (of Banach-Tarski fame) [96].

2.2.5 Generating Compound Statements

We have decomposed the language we are using to talk about the World into atomic statements and compound statements. I want to note once again that this decomposition is by no means unique. This fact was brought to my attention by Kenny Easwaren who has been working on such matters since we first met at Berkeley. Let

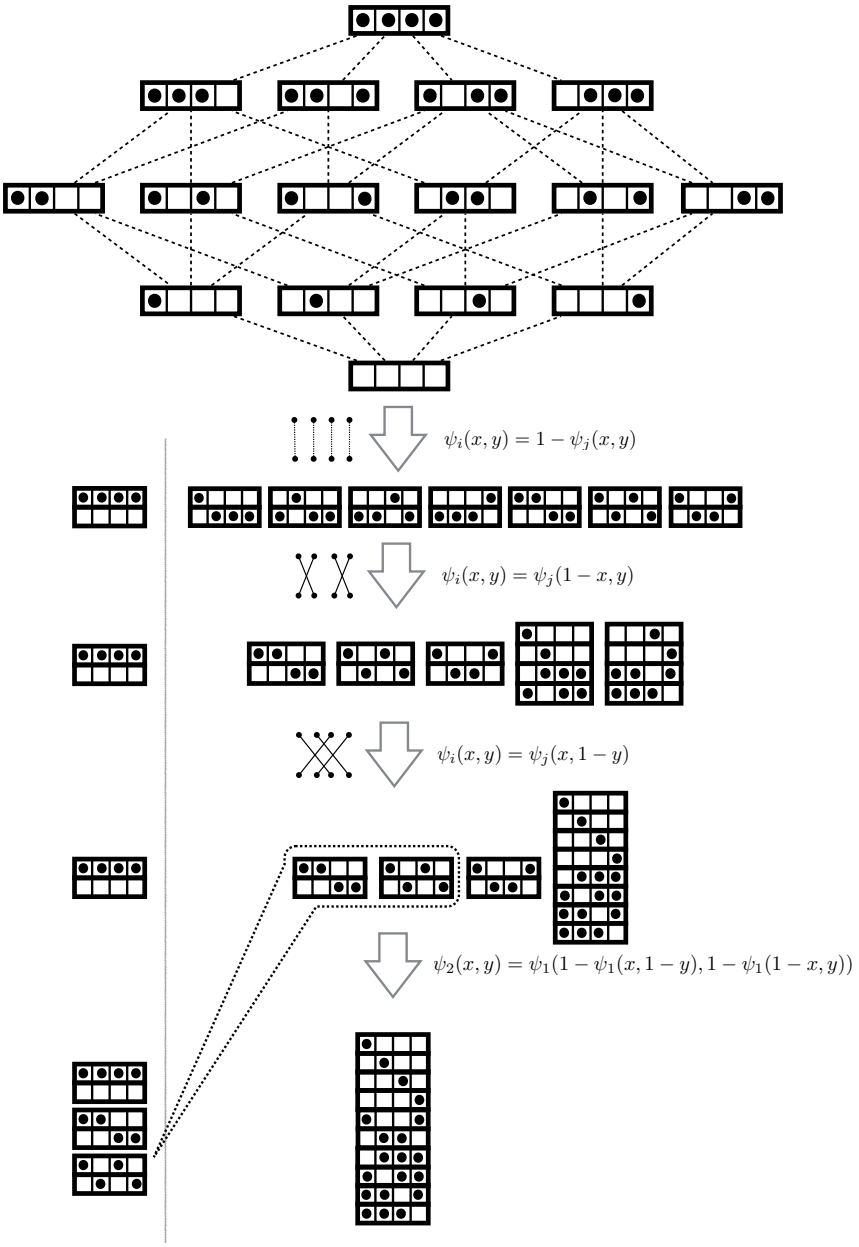

Figure 2.2: The outline of the argument showing the uniqueness of a single binary logical connective using negation and composition.

us see how two EAs can choose a different basis of atomic statements for the same language.

Consider one EA, Alice, who chooses statements $\{a, c\}$ as atomic. She uses logical connectives to form the new statement

$$\begin{aligned} b &= \neg(\neg(a \wedge c) \wedge \neg(\neg a \wedge \neg c)) \\ &= (a \wedge c) \vee (\neg a \wedge \neg c) \end{aligned} \tag{2.8}$$

Thus b is a compound statement for Alice. For completeness sake, the above is just one of the other binary connectives that we chose not to use as primary, and is called *nand*. Its location on the LC hypercube is $(1, 0, 0, 1)$.

Now Bob, another EA, considers statements $\{b, c\}$ as atomic. To him, statement a is compound, and he constructs it via

$$a = \neg(\neg(b \wedge c) \wedge \neg(\neg b \wedge \neg c)) \tag{2.9}$$

$$= (b \wedge c) \vee (\neg b \wedge \neg c). \tag{2.10}$$

For every compound statement constructed by Alice using, namely $s = s(a, c, \dots)$, there exists a compound statement constructed by Bob, $s' = s'(b, c, \dots)$, such that $s'(b, c, \dots) = s'(b(a, c, \dots), c, \dots) = s(a, c, \dots) = s'$. Conversely, $s(a, c) = s(a(b, c), c) = s'(b, c)$. Alice and Bob's choice of atomic statements both span the same space of compound statements, and they therefore share the same Realm of Discourse. In the parlance of abstract algebra, the ring freely generated by Bob's atomic statements is also generated by Alice's [30].

Compound statements form the bulk of the statements that are used in common

discussions between humans. If humans disagree on the truth value of even one statement, then they are not talking about the same world. Of course, arguments and experiences may cause us to change truth values. We will need to see how to incorporate these dynamics in epistemic agents. Furthermore, truth values of statements alone are not enough to capture the possibility that we may be ignorant of the truth or falsehood of a statement: I am not sure whether the statement *It will rain tomorrow* is true. These considerations lead us to another layer of epistemology, where we must leave behind absolutes, and consider possibilities.

2.3 Belief and Language

Upon fixing a language, a realm of discourse is established concerning what an epistemic agent may now think about. There are many possible worlds in the realm of discourse. Many such worlds overlap for the same truth value of a particular statement, or set of statements. An epistemic agent prioritizes the value of the truth of statements via a belief function. In this sense, the epistemic agent prioritizes worlds through belief, and considers the ontological state of the world to coincide with the epistemic world which maximizes their belief. What an epistemic agent believes the world to be is that which it is.

Though one may at first think of belief as being binary, one either does or does not believe in s, this is not the case. Beliefs are graded, and the binary nature of belief only comes about due to a self imposed threshhold. An EA says they believe in a statement when their belief function is beyond this threshhold. Thus an EA's belief in a statement could start off at any value. Experience may nudge this value by tiny amounts until finally the thresh-hold is reached, and even then it is possible for the

EA to continue to strengthen their belief in a statement, all the while declaring that they believe. We therefore postulate that the belief function is a continuous function, the first of the Cox axioms [21].

Since beliefs are meant to prioritize statements, they are an ordering imposed by the EA. Beliefs are transitive, in the sense that if the epistemic agent believes A more than B, and B more than C, then they must believe in A more than C. This structure restricts the possible objects that one can use to describe beliefs, which will be a great benefit. The transitive property means that we can represent beliefs as real numbers. We can then define belief to be a function, $b : \mathcal{S} \to \mathbb{R} : s \mapsto b(s|K)$, from the set of all statements to the reals. Here K represents the background knowledge of the EA. Ab initio this background knowledge pertains to the choice of logical connectives chosen by the EA. As we incorporate dynamics in the form of experience, background knowledge will grow to include these experiences.

A Note on Continuity

It may make you wonder if continuity is truly necessary. We have made an implicit assumption that the Realm of Discourse has an infinitude of statements inside of it, which is reasonable until one realizes that this would only be the case if we allowed for either an infinite number of atomic statements, or compound sentences of non-finite length. The latter of these possibilities could never be the case for thinking human beings. Our Realm of Discourse may have a combinatorial explosion of statements in it, but we can only create a compound sentence that takes as long to say as the longest living human being. In finite domains Halpern constructed a counterexample contradicting Cox's results [44]. For the results to hold, the Realm of Discourse

must be dense. Continuity, therefore, throws out the possibility of a finite number of statements, and avoids such contradictions. This poses no problem philosophically, since even though we cannot create infinitely long compound statements, it is perfectly justifiable to have a subset of atomic statements indexed by a rational (or real) number.

Returning to our main argument, common sense indicates that our belief in the negation of a statement is somehow related to our belief in the statement. Common sense also tells us that our belief in the conjunction of a statement should be related to our beliefs in the original statements, as well as our beliefs in the conditioned statements. Formally, there exist endomorphisms on the space of beliefs, f and g, such that:

$$b(\neg s|K) = f[b(s|K)] \tag{2.11}$$

$$b(s_1 \wedge s_2|K) = g[b(s_1|K), b(s_1|s_2 \wedge K), b(s_2|K), b(s_2|s_1 \wedge K)]. \tag{2.12}$$

Together, equations 2.11 and 2.12 will be called the Axiom of Common Sense. [21][22]

Lastly, we want the reasoning of an EA to be consistent. This means that if there are multiple ways to derive a compound statement in the realm of discourse, then the axioms of common sense must result in the same beliefs. This is known as the Axiom of Consistency, and will unlock many of the properties of belief functions to which we will now turn. To summarize this section, for an epistemic agent with knowledge K

using $\mathcal{L}$:

There exists a function, $b : \mathcal{S} \to \mathbb{R} : s \mapsto b(s|K)$, called a **belief** such that:

$C1:$ b is continuous

$C2:$ $\exists f : \mathbb{R} \to \mathbb{R},\ g : \mathbb{R}^4 \to \mathbb{R}$ such that $\forall s, s' \in \mathcal{S}$

$$b(\neg s|K) = f[b(s|K)]$$

$$b(s \wedge s'|K) = g[b(s|K), b(s|s' \wedge K), b(s'|K), b(s'|s \wedge K)]$$

$C3:$ f and g must be consistent with Φ.

2.3.1 Consistency of the Common Sense of Conjunction

Let us turn first to equation 2.12. First note that there are four arguments in g. But by the symmetry of conjunction, $g(a, b, c, d) = g(c, d, a, b)$, so we may be overcounting the number of necessary arguments. To see if we are we can consider the seven functions with up to four of these arguments. A full accounting of all these forms is done by Ariel Caticha. [14] The first and last argument are shown to lead to a nontrivial result, with the remaining clusters of arguments giving either the same result as this pair or trivial relationships. Using this simplification we write:

$$b(s_1 \wedge s_2|K) = g[b(s_1|K), b(s_2|s_1 \wedge K)]. \tag{2.13}$$

Consistency is a very powerful tool, so next we use the fact that conjunction is associative, $s_1 \wedge (s_2 \wedge s_3) = (s_1 \wedge s_2) \wedge s_3$. Denoting $x = b(s_1|K)$, $y = b(s_2|s_1 \wedge K)$,

and $z = b(s_3|s_1 \wedge s_2 \wedge K)$, consistency demands that

$$g[x, g[y, z]] = g[g[x, y], z] \tag{2.14}$$

This is known as the associativity functional equation, and it is very difficult to solve. Cox originally showed that if g is twice-differentiable, then there exists a strictly monotonic function $\rho : \mathrm{R} \to \mathbb{R}$, such that

$$g[x, y] = \rho^{-1}(\rho(x)\rho(y)), \tag{2.15}$$

which one can easily confirm by plugging into the associativity equation 2.14 [21]. Since then proofs have been found that do not even require the differentiability condition [2]. The solutions, however, are more easily interpretable using disjunction as the unique binary logical connective [13]. For a thorough treatment of this, and the many other functional equations we will encounter, I would refer the reader to Janos Aczel's work [1]. The function ρ is the first of two *regraduations* that will *normalize* belief functions across EAs, allowing comparisons.

2.3.2 The First Regraduation: Pinning Down Truth

Up until now we have not mentioned the range of the belief function. In fact, every EA can have a different range of belief. All that matters to an EA is the ordering of beliefs. Furthermore we have not stated whether beliefs that map to larger numbers are more believable, or those that map to lesser numbers. This freedom in choosing a range needs to be remedied, since in the end we will want EAs to be able to compare their beliefs. Changing the scale of belief is accomplished via regraduation. This

is very similar to choosing a different temperature scale in thermodynamics. One chooses the scale in order to make things as simple as possible; scales are chosen on pragmatic grounds.

We use the freedom to regraduate here to make conjunction appear as simple as possible. We regraduate the belief function via $b \to \tilde{b} = g \circ \rho$. The conjunction rule 2.13 now takes both a very simple, and familiar, form:

$$\tilde{b}(s_1 \wedge s_2|K) = \tilde{b}(s_1|K)\tilde{b}(s_2|s_1 \wedge K) \tag{2.16}$$

We will drop the tilde henceforth, and call our regraduated belief simply belief. What has this regraduation done to our range of belief? Let us call the belief in falsehood $b_F = b(F|K)$, and the belief in truth $b_T = b(T|K)$. By the indempotency of statements with respect to conjunction, we have that:

$$\begin{aligned} b(s|K) &= b(s \wedge s|K) \\ &= b(s|K)b(s|s \wedge K) \\ &= b(s|K)b_T \end{aligned} \tag{2.17}$$

This equation is satisfied if $b_T = 1$. The first regraduation has pinned down the numerical value for the belief in truth. We do not know yet, however, whether less strong beliefs fall above or below 1. Notice, however, if the range is below 1, then it must have a minimum at 0, otherwise multiplying negative probabilities could potentially mix the two domains. The range of belief, D must be either $[0, 1]$ or $[1, \infty) \cup \{\infty\}$.

2.3.3 Consistency and Negation

Moving along to the first axiom of common sense, we denote $x = b(s|K)$. Since double negation is identity, consistency demands that the relationship f must satisfy

$$x = f[f[x]] = f^2[x]. \tag{2.18}$$

This is the special case of a functional equation known as Babbage's equation, $x = f^n[x]$, for $n = 2$ [6].

It can be shown that 2.18 implies the existence of a family of functions, parametrized by $\alpha \in \mathbb{R}^*$, such that

$$f[x] = (1 - x^\alpha)^{\frac{1}{\alpha}} \tag{2.19}$$

The parameter is restricted depending on the domain of belief. For $D = [0, 1]$, $\alpha > 0$, and for the other case, $\alpha < 0$. The choice of this parameter will pin down the range of belief, and leads us to the second regraduation of belief [13].

2.3.4 The Second Regraduation: Pinning Down Falsehood

Once again we return to the idea that at some point it would be nice for EAs to compare beliefs. This freedom that each EA has in constructing their belief function will now be removed through a second regraduation. We scale our belief function to a new belief function via $b(s|K) \to \tilde{b}(s|K) = b(s|K)^\alpha$. Under this regraduation, $[0, 1] \to [0, 1]$, while $[1, \infty) \cup \{\infty\} \to [0, 1]$. The degeneracy in the range has been removed, and the beliefs of different EAs now all fall on the same scale.

We drop using the tildes, and look at the belief in falsehood

$$b_F = b(F|K) = b(\neg T|K) = 1 - b(T|K) = 1 - 1 = 0$$

As was probably expected, belief in a falsehood falls on the bottom of the range of belief. Furthermore, we have a normalization condition on belief in statements and their negations:

$$b(s|K) + b(\neg s|K) = 1 \tag{2.20}$$

All of this is starting to look incredibly familiar, is it not? The rules for manipulating beliefs are starting to look very similar to the rules for manipulating probabilities. Let's see if other familiar results follow.

An Excursion into Disjunction

What is the rule for manipulating disjunctions of statements, $b(s \vee s'|K)$? As always, we apply consistency and the normalization and conjunction rules

$$\begin{aligned} b(s \vee s'|K) &= b(\neg(\neg s \wedge \neg s')|K) \\ &= 1 - b(\neg s \wedge \neg s'|K) \\ &= 1 - b(\neg s'|K)b(\neg s|\neg s' \wedge K) \\ &= 1 - b(\neg s'|K)[1 - b(s|\neg s' \wedge K)] \\ &= 1 - b(\neg s'|K)[1 - b(s|K)\frac{b(\neg s'|s \wedge K)}{b(\neg s'|K)}] \\ &= b(s'|K) + b(s|K)(1 - b(s'|s \wedge K)) \\ &= b(s|K) + b(s'|K) - b(s \wedge s'|K) \end{aligned} \tag{2.21}$$

The disjunction rule of probability theory is recovered. The conjunction rule and normalization are enough to show the equivalence with the rules of probability. In probability theory the conjunction rule is referred to as Bayes' Rule, for Thomas Bayes, an English statistician living in the eighteenth century [56]. The remarkable equivalence between beliefs and probabilities was first shown by Cox in 1946. [21] Henceforth we will use p instead of b, and use the terms *belief* and *probability* interchangeably.

2.4 Information and Belief

Information is that which changes belief. [14] For an experience to be informative, the Epistemic Agent having the experience must be changed upon its conclusion. Let

us take a look again at the process of having an experience. An EA starts with a prior set of beliefs based on her background knowledge, $p(s|K)$. After the experience, she has a posterior set of beliefs where the experience has been incorporated into her background knowledge, $p(s|e \wedge K)$. These two are related by the multiplication rule of belief:

$$\begin{aligned} p(s|e \wedge K) &= \frac{p(s,e|K)}{p(e|K)} = \frac{p(e|s \wedge K)}{p(e|K)} p(s|K) \\ &= \mathcal{L}(e; s, K) p(s|K) \end{aligned} \tag{2.22}$$

The probability ratio on the right is called the *likelihood*. It is not a probability distribution, since clearly it is not normalized. It is, however, a very important quantity epistemically. Note that for a given statement, if it is greater than one then it lifts the EAs belief in the statement. If it is less than one then it depresses it. It acts on the old belief distribution multiplicatively to create the new belief distribution.

This update procedure can be generalized to a time series of experiences. Consider an EA living her life which consists of a sequence of experiences: $E = \{e_t\}_{t=0}^{t=T}$. Her beliefs evolve over time as well:

$$p(s) \to p(s|e_0) \to p(s|e_0 \wedge e_1) \to \cdots \to p(s|e_0 \wedge e_1 \wedge \cdots e_T) \tag{2.23}$$

There are two ways to think about this chain of experiences. First is to treat each one with Bayes' rule. The second is to treat all the experiences as a single experience.

The first way leads to:

$$\frac{p(s|E \wedge K)}{p(s|K)} = \mathcal{L}(e_0; s, K)\mathcal{L}(e_1; s, e_0 \wedge K)\mathcal{L}(e_2; s, e_0 \wedge e_1 \wedge K) \cdots , \tag{2.24}$$

while the latter leads to

$$\frac{p(s|E \wedge K)}{p(s|K)} = \mathcal{L}(E; s, K). \tag{2.25}$$

The equivalence of these two is a statement of the *holistic nature of experience*: an epistemic agent may partition her experiences in any way she chooses, but this does not affect her final belief. Clearly this is pointing us at what is missing in the theory, since if we are to assume that the beliefs of an EA require a physical substrate, then there must be some ontological cost to changing belief. Moving through a series of in between beliefs will then not necessarily have the same cost as moving from the initial to final belief. For example, an EA which throws a die will experience a change in belief due to the outcome. They may then draw a card from a deck, which, too, will change their belief function. The resulting belief after both experiences must be the same in both sequential and simultaneous updating. In the case where both are performed sequentially, we would expect that the ontological substrate would have undergone more changes to get its final state than if they were performed at once. The excess change in sequential versus simultaneous updating would be the source for this cost difference, which may not be negligible. We will not delve into this further, but spending a moment to think about the implications does point to uncharted territory.

Assuming the holistic nature of experience, we can also consider how independent experiences affect the likelihood. Independent events are ones whose outcomes do not

affect each other. For example, throwing a die and drawing a card are independent. The order of the events is irrelevant. Picking a dessert to eat and eating a dinner can be dependent experiences, since their order influences one another. For independent events, the likelihood factors quite nicely:

$$\mathcal{L}(\{e_1, e_2\}; s, K) = \mathcal{L}(e_1; s, K)\mathcal{L}(e_2; s, K) \tag{2.26}$$

The fact that the two functions on the right hand side commute is another way of restating that the order of the events makes no difference to the epistemic agent.

Returning to the question of information, this is precisely what we need. If information is that which changes belief, then information and likelihood must be related to one another! The information contained in the experience is some function of the likelihood! We posit that the information contained in an experience, e, about a statement, s, to an EA with background knowledge, K, is

$$i_s(e|K) = f(\mathcal{L}(e; s, K)). \tag{2.27}$$

We can quickly say quite a few things from this general statement. First off, since an uninformative experience doesn't change belief, it must contain no information. Secondly, as the likelihood changes by an infinitesimal amount, we do not expect the amount of information contained in the experience to change discontinuously, so f must be continuous. Lastly, the information gathered from independent events must reflect the commutativity of their temporal ordering. Let e_1 and e_2 be independent events, denote $x = \mathcal{L}(e_1; s, K)$ and $y = \mathcal{L}(e_2; s, K)$. The above statements imply that

information must satisfy the functional constraints:

$$f(xy) = f(x) + f(y) \quad \text{and} \quad f(1) = 0 \quad \text{and} \quad f \text{ is continuous,} \tag{2.28}$$

which we will now investigate.

2.4.1 The Functional Form of Information

Information is a function of the subjective likelihood of experience, satisfying the functional constraints 2.28 for experiences that are independent. First note that for any positive integer, $n \in \mathbb{Z}^+$:

$$\begin{aligned} f(x^n) &= f(x^{n-1}x) \\ &= f(x^{n-1}) + f(x) \\ &\vdots \\ &= nf(x). \end{aligned} \tag{2.29}$$

From the boundary condition, we can extend this to negative integers:

$$0 = f(1) = f(x^n x^{-n}) = f(x^n) + f(x^{-n}) \Rightarrow f(x^{-n}) = -nf(x)$$

Consider now $q \in \mathbb{Q}$, st $\exists n, m \in \mathbb{Z}, q = \frac{n}{m}$, and define y such that $y^m = x^n$:

$$nf(x) = f(x^n) = f(y^m) = mf(y) = mf(x^{\frac{n}{m}}) \Rightarrow f(x^q) = qf(x)$$

Finally we apply continuity of f and the fact that rationals are dense in the reals to extend this property to all reals, $r \in \mathbb{R}$. Using $y = x^r$:

$$\frac{d}{dr}\left[r f(x) = f(x^r)\right] \Rightarrow f(y) = f'(y) y \ln y$$

The resulting differential equation is almost elementary and has a general solution

$$f(x) = A \ln x. \tag{2.30}$$

With this we can now write the analytical formula for the epistemic notion of information as:

$$i_s(e|K) = A \ln \mathcal{L}(e; s, K). \tag{2.31}$$

With A an arbitrary integration constant.

We can write the likelihood in terms of prior and posterior:

$$i_s(e|K) = A \ln p(s|e \wedge K) - A \ln p(s|K), \tag{2.32}$$

expressing the information in terms of the EAs beliefs.

2.4.2 Information Hidden in the World

This motivates the definition of a new quantity, the *hidden information*:

$$h(s|K) = -A \ln p(s|K) \tag{2.33}$$

The information in the event is revealed to the EA during the experience. Since we would like the total information hidden by the world to be positive, this means $A > 0$. Furthermore, the arbitrariness of the constant can be absorbed into the arbitrariness of the base of the logarithm being used. We therefore write

$$h(s|K) = -\log p(s|K) \Rightarrow i_s(e|K) = -\Delta h = h(s|K) - h(s|e \wedge K) \tag{2.34}$$

The information in an event about a statement is the decrease in the hidden information about that statement due to the experience for the EA.

An experience changes not only the belief in a statement, but also belief in the statement's negation. The total hidden information should depend on both $h(s|K)$ and $h(\neg s|K)$. We need to think about how both of these contribute to the total. If the EA has a strong belief in s then there is very little hidden information in the world concerning s. On the other hand, if the EA has a very low belief in $\neg s$, there is a lot of information hidden in the world concerning $\neg s$. These statements seem contradictory. Is there a little or a lot of hidden information in the world concerning the pair $\{s, \neg s\}$?

This point is made more apparent if we consider the sum of hidden informations. Letting $p = p(s|K)$;

$$h(s|K) + h(\neg s|K) = -\log p - \log(1-p) = -\log p(1-p). \tag{2.35}$$

This function is symmetric around $p = 1/2$ and gets larger as we move away from this value, diverging as $p \to 1$ or $p \to 0$. So as the EA becomes more certain, the amount of hidden information increases. This is absurd, and we must remedy it to

properly take into account the ignorance of our EA.

2.4.3 The Functional Form of Hidden Information

We want a measure that describes the total amount of information hidden in the world from the EA, and present here a novel proof of that measure. We want to be able to quantify the *ignorance* of our EA. Up until this point we have been considering the pair $\{s, \neg s\}$, and want a measure of total hidden information $H[s, \neg s|K]$ conditioned on knowledge, which has the property that as the belief pair becomes more polarized, the measure should get smaller. When the gap between the two beliefs approaches zero, it is clear that the amount of hidden information should maximize. The state with maximal hidden information corresponds to the maximally ignorant epistemic state where the EA has no preference whatsoever in belief or non-belief.

In more generality, we can consider an arbitrary set of N exhaustive mutually exclusive statements, $\{s_i|s_i \wedge s_j = F \ \forall i \neq j, \vee_i s_i = T\}$. The exhaustive condition ensures that beliefs are normalized, while the exclusive statement restricts conditional beliefs:

$$\sum_i^N p(s_i|K) = 1 \quad \text{and} \quad p(s_i|s_j \wedge K) = \delta_{ij} p(s_i|K) \tag{2.36}$$

We denote $p_i = p(s_i|K)$, and want a measure of hidden information $H[s_i|K]$. What properties should H have? Prior knowledge means that it should be a function of beliefs in the statements, $H = H[s_i|K]$. We now assume that the total hidden information can be factored into a piece that depends explicitly on the prior knowledge, and another piece, *the entropy* that depends implicitly on prior knowledge: The only

possible candidate for the explicitly dependent function if belief itself. We have:

$$H[s_i|K] = p(K)S[p_i] \tag{2.37}$$

Typically entropy and hidden information are synonymous [53] [54]. This novel factorization is far more useful, however, and we will see how the additive property of entropy follows from it. We will now review the other properties of total hidden information, properties typically referred to as the Shannon-Khinchin axioms [81] [58]. H should vanish when belief polarizes, $S[1, 0, 0, \ldots, 0] = 0$. It should be continuous in belief: the amount of hidden information should not change much if the EAs beliefs are just a little bit different. Secondly, it should maximize when there is no prior knowledge to distinguish beliefs, $H_0 = \max H[s_i] = S[\frac{1}{N}, \frac{1}{N}, \ldots, \frac{1}{N}] = f(N)$. As we increase the number of statements that the EA should consider, the amount of hidden information should increase. Hence f should be monotonically increasing. Lastly, hidden information should be recursive under a coarse graining.

What does that last statement mean? It involves how an EA can take statements and group them together, consider the hidden information in the coarser beliefs, and ignore the hidden information contained in the finer beliefs. For example, take the statement $c =$*Sarah likes chocolate*. There is some hidden information concerning the pair,$\{c, \neg c\}$ in the world about Sarah that you can discover by asking her whether she likes chocolate. But now consider the statement $a =$*Sarah is allergic to chocolate*, and the triple, $\{c, \neg c \wedge a, \neg c \wedge \neg a\}$. More information is hidden by the triple than by the pair. The triple is finer, the pair is coarser. Nonetheless we see that the information could be found concerning the triple by first finding out whether Sarah likes chocolate. If she does then we have discovered all the hidden information needed

to polarize belief in the triple. If Sarah did not like chocolate, then we would still have information hidden in the question of whether or not she is allergic to chocolate. Hence the hidden information in the triple is equal to the hidden information in the double plus the hidden information in the refinement conditioned on whether Sarah likes chocolate.

$$H[c, \neg c \wedge a, \neg c \wedge \neg a|K] = H[c, \neg c|K] + H[a, \neg a|\neg c \wedge K] \tag{2.38}$$

Denoting $p = p(c|K)$ and $q = p(a|\neg c \wedge K)$, we also have

$$\begin{aligned} H[c, \neg c \wedge a, \neg c \wedge \neg a|K] &= p(K)S[p, 1-p] + p(\neg c \wedge K)S[q, 1-q] \\ &= p(K)S[p, 1-p] + p(\neg c|K)p(K)S[q, 1-q] \\ &= H[c, \neg c|K] + p(\neg c|K)p(K)S[\frac{(1-p)q}{1-p}, \frac{(1-p)(1-q)}{1-p}] \\ &= H[c, \neg c|K] + p(\neg c|K)H[a/\neg c, \neg a/\neg c|K] \end{aligned} \tag{2.39}$$

The conditioned statement $a/c = a \wedge (\tau(c) = T)$ has been introduced as a way to remove the statement c from the prior knowledge. Comparing equation 2.38 with 2.39, we see:

$$H[a, \neg a|\neg c \wedge K] = p(\neg c|K)H[a/\neg c, \neg a/\neg c|K] \tag{2.40}$$

Conditioning hidden information scales it by the probability of the conditioning statement. This is a novel interpretation of hidden information, relating belief and knowledge, a direct result of the factoring introduced in 2.37.

What if we cared about Sarah's allergy independently of whether she likes choco-

late? Then the statements we'd be interested in thinking about are $\{a \wedge c, a \wedge \neg c, \neg a \wedge c, \neg a \wedge \neg c\}$. Coarse graining over a we find

$$\begin{aligned} & H[a \wedge c, a \wedge \neg c, \neg a \wedge c, \neg a \wedge \neg c|K] \\ & \quad = H[c, \neg c|K] + p(c|K)H[a/c, \neg a/c|K] + p(\neg c|K)H[a/\neg c, \neg a/\neg c|K] \end{aligned} \tag{2.41}$$

and similarly for a coarse graining over c. Getting rid of the normalization factor $p(K)$, and denoting $p_{11} = p(c \wedge a|K)$, $p_{12} = p(c \wedge \neg a|K)$, $p_{21} = p(\neg c \wedge a|K)$, $p_{22} = p(\neg c \wedge \neg a|K)$ the entropy satisfies:

$$\begin{aligned} S[p_{11}, p_{12}, p_{21}, p_{22}] &= S[p, 1-p] + pS[\frac{p_{11}}{p}, \frac{p_{12}}{p}] + (1-p)S[\frac{p_{21}}{1-p}, \frac{p_{22}}{1-p}] \\ &\geq S[p, 1-p] \end{aligned} \tag{2.42}$$

where $p = p_{11} + p_{12}$ as before. This is a particular case of the *additive property of entropy*. The inequality is there to emphasize that coarse graining cannot increase the entropy. Coarse graining reduces hidden information.

This property generalizes for arbitrary coarse graining. Let

$$\{s_{11}, s_{12}, \ldots, s_{1N_1}, s_{21}, s_{22}, \ldots, s_{2N_2}, \ldots s_{MN_M}\} \tag{2.43}$$

be an exhaustive set of mutually exclusive statements. There are M groupings, with N_i statements in each grouping, with corresponding beliefs p_{ij}. If we coarse grains over the second index, the corresponding beliefs are $p_i = \sum_j^M p_{ij}$. Then the additive

property of entropy is

$$S[\{p_{ij}\}] = S[\{p_i\}] + \sum_{i=1}^{M} p_i S[\{\frac{p_{ij}}{p_i}\}] \tag{2.44}$$

This functional equation will unlock the functional form of entropy.

Let each of the $p_{ij} = 1/NM$ be equiprobable; all M groupings have the same N number of elements. Then $p_i = 1/M$, the coarse grained statements are also all equiprobable. Furthermore, in each subgrouping, $p_{ij}/p_i = 1/N$ are all equiprobable. Thus the additive property tells us:

$$f(NM) = f(M) + \sum_{i}^{M} \frac{1}{M} f(N) = f(N) + f(M). \tag{2.45}$$

We have already encountered this equation and know its solution, but let's keep f general. Now we consider arbitrary groupings so that $p_{ij} = 1/N_{tot}$ and $p_i = N_i/N_{tot}$. Then 2.44 gives us:

$$\begin{aligned} f(N_{tot}) &= S[\{p_i\}] + \sum_{i=1}^{M} p_i S[\{\frac{1}{N_i}\}] \\ &= S[\{p_i\}] + \sum_{i=1}^{M} p_i f(N_i) \end{aligned} \tag{2.46}$$

Isolating the entropy we find:

$$\begin{aligned} S[\{p_i\}] &= \sum_{i=1}^{M} p_i (f(N_{tot}) - f(N_i)) \\ &= -\sum_{i=1}^{M} p_i \log p_i. \end{aligned}$$

Putting it all together, the total hidden information in the world to an EA with background knowledge K concerning a set of statements $\{s_i\}$ is

$$H[\{s_i\}|K] = p(K) \sum_i p(s_i|K) h(s_i|K), \tag{2.47}$$

or, dropping the normalization, the entropy of a belief distribution is

$$S[\{p_i\}] = \sum_i p(s_i|K) h(s_i|K). \tag{2.48}$$

The entropy is the expected hidden information.

Concavity of Entropy

Here we simply list some of the properties of the entropy function S. S is a concave function of the probabilities, which can easily be checked by finding the determinant of the Hessian matrix:

$$\frac{\partial S}{\partial p_i \partial p_j} = \frac{\delta_{ij}}{p_j} > 0 \Rightarrow |\nabla_i \nabla_j S| = \prod_i p_i^{-1} > 0 \tag{2.49}$$

It attains a global maximum when $p_i = p_j \ \forall i, j = 1, \ldots, N$, and its value there is

$$S_0 = \log N. \tag{2.50}$$

2.4.4 The Success of the Bayesian Interpretation

The importance of the correspondence between belief and probability cannot be understated. It cuts to the core of two schools of thought concerning the interpretation

of probabilities. Probability theory was initially developed in the correspondences between Blaise Pascal and Pierre de Fermat in the 1650s [23]. The *frequentist* interpretation of probability came from thinking about probabilities as the limiting ratios of the outcomes of experiments such as throwing die or drawing cards.

This interpretation works perfectly well for cases involving repeatable experiments, and Bayes' Rule is a consequence of the frequentist viewpoint. The problem comes when one begins to ask questions about experiments that are not repeatable. The *probability that it will rain on July* 6^{th}*, 2016* is not an experiment that can be performed a sufficient number of times for a frequency to be established that approximates that probability. That experiment is performed once, and only once. Under the frequentist interpretation, the probability is either 0 or 1.

Similarly, the statement *what is the probability of life on Mars?*, is ludicrous to analyze in the frequentist viewpoint. Unless one is capable of creating a myriad of Solar Systems, each with its own red planet, and capable of waiting for an infinite amount of time to see on which of these instances of Mars life develops, the frequentist has no answer to the question.

Herein lays the strength of the Bayesian' interpretation of probability. Rather than being a frequency achieved through repeated experiments, to a Bayesian a probability is a strength of belief. It is a subjective quantity, which may seem strange when one is talking about such objective things as coins, but captures the essence of what we mean when we discuss the probability of events that are not repeatable. The prior knowledge of an EA enters into the evaluation of such a probability, so a planetary scientist may have a completely different answer to the question of whether there is life on Mars compared to a creationist. Both use their prior knowledge to evaluate

the question, and both have different types of prior knowledge.

With regards to questions concerning repeatable experiments, the Bayesian is not in contradiction with the frequentist. To see this let's consider a simple experiment where a coin is tossed N times. The EA has a set of models parametrized by f, wherein the coin has a probability f of coming up heads, and a probability $1-f$ of coming up tails, and that these probabilities do not change with each tossing of the coin. If, during the course of the experiment, there are n heads, and $N-n$ tails, which models should the Bayesian believe in more strongly? The likelihood of a particular sequence of tosses, the data $\mathcal{D} = \{HTT\cdots H\}$, given a model is

$$\mathcal{L}(\mathcal{D}|f) \propto f^n(1-f)^{N-n}. \tag{2.51}$$

The likelihood has a maximum at

$$\frac{d\mathcal{L}}{df}|_{\hat{f}} = 0 \quad \Rightarrow \quad \hat{f} = \frac{n}{N}. \tag{2.52}$$

Recall that a belief function is updated by Bayes' Rule

$$p(f|\mathcal{D}) = \mathcal{L}(\mathcal{D}|f)p(f) \tag{2.53}$$

Assuming an ignorant prior, $p(f) = constant$, after the experiment the EA's belief in a model where $f = \frac{n}{N}$ will be strengthened the most. The Bayesian formalism for belief updating agrees with the frequentist interpretation of probability.

Furthermore, the EA can also compare their belief in different models by looking

at a belief ratio:

$$\begin{aligned}\frac{p(f+\delta f|\mathcal{D})}{p(f|\mathcal{D})} &= 1+\frac{1}{2}\frac{1}{\mathcal{L}}\frac{d^2\mathcal{L}}{df^2}\delta f^2+\mathcal{O}[\delta f^3] \\ &= 1+\frac{1}{2}\frac{d^2\log\mathcal{L}}{df^2}\delta f^2+\mathcal{O}[\delta f^3] \end{aligned} \tag{2.54}$$

The EA finds that models deviating by more than $|\delta f| \approx \sqrt{-2(\frac{d^2\log\mathcal{L}}{df^2})^{-1}}$ aren't at all credible based on the data. If we assume a Gaussian profile for the likelihood, then this is simply the standard deviation of that Gaussian, in agreement with a frequentist setting bounds on the probability from a finite sampling.

Beyond the realm of repeatable experiments, the EA can also attack such esoteric questions as *How probable is life in the (uni/multi)verse?* This famous question can be attacked by for the Universe by the Drake equation, and for the Multiverse by the Generalized Drake Equation [36]. The equation is a belief function for a conjunction of statements of the form *A star is located in some volume of space* $\wedge$*The star has planets* $\wedge$*A planet around a star has liquid water and sufficient concentrations of specific atoms* $\wedge$*Complex compounds form in the water and create sustained, self-replicating chemical networks that are robust under environmental perturbations* $\wedge$ etc. An EAs belief in each member of this conjunction is a direct application of Bayes' rule, and each term can be evaluated with sufficient background knowledge. To a frequentist, this is impossible without omnipotence. The Bayesian formalism is limitless in its ability to attack such questions precisely because it is a framework for inference; it doesn't tell an EA *what* to believe, it tells an EA *how* to believe.

The Problem of Fundamentalism

There is a slight problem with Bayes' rule referred to as the Problem of Fundamentalism [13]. Belief in tautologies or contradictions is a statement concerning the internal logic of the language being used. It is perfectly acceptable for beliefs in these statements to take on the values of 0 or 1. What happens when the EA believes in a non-tautological(non-contradictory) statement with a strength of 0, $p(s|K) = 0$?

$$p(s|e \wedge K) = \mathcal{L}(e|s \wedge K)p(s|K) = 0 \tag{2.55}$$

The EA's belief in s doesn't change with experience. Just imagine anyone who doesn't believe in climate change. No amount of data can sway them; they are experiencing the problem of fundamentalism in belief updating. To ensure that our EAs are not fundamentalist, beliefs are always in the range $(0, 1)$. Extreme beliefs are not allowed.

2.4.5 Setting Priors

Once we have a language, we can consider the world as a particular epistemic state in the realm of discourse. Certain things are true in a world, while others are false. The world is not static; it changes through time. Truth values are then not necessarily static: it may be the case that my refrigerator is bereft of milk this morning, but that was not true last night. As truth values change over time, the epistemic state that represents the world jumps from one state to the next.

Now it is also the case that an epistemic agent has a belief function on the realm of discourse. They may not know which epistemic state corresponds to the real world, but they believe in certain worlds more than other worlds. We have already seen that

Bayes' rule tells us how the EA's beliefs evolve as she experiences the world: how experience changes prior into posterior. How though, does a newly placed EA in the changing world choose a prior?

The EA has some prior knowledge, K, from which they must decide the strength of their beliefs. If the EA does not want to be biased, then they must assume they are as ignorant as possible. This means that out of all the priors they can choose that satisfy the constraints of their prior knowledge, the one that maximizes the amount of hidden information should be prioritized over those that do not. This method is referred to as the Maximum Entropy principle, or simply MaxEnt. [53][54]

To establish this principle we need a bit of machinery concerning how to set up prior knowledge.

Expectation

Let's say we have a set of statements concerning the outcomes of measurements some physical quantity, Q. The N possible outcomes are denoted Q_i, so that based on prior knowledge K, the EA has a set of beliefs about the statements $s_i =$*The measurement of* Q *will result in* Q_i, namely $p_i = p(s_i|K)$. They measure Q somehow, and find Q_j. They do it again and find another value (perhaps the same, perhaps not). After M measurements, they have a sequence of measured values. Assuming the measurements are independent, what is the estimated hidden information revealed in

each measurement? Let's call the number of occurrences of Q_i, m_i.

$$\begin{aligned} p(Q_{i_1} \wedge Q_{i_2} \wedge \cdots \wedge Q_{i_M}|K) &= \prod_i^N p_i^{m_i} \quad \text{where} \quad \sum_i^N m_i = M \\ &= \prod_i^N e^{-m_i h_i} \\ &= e^{-\sum_i^N m_i h_i} \\ &= e^{-M \sum_i^N \frac{m_i}{M} h_i} \\ &= \left(e^{-\hat{S}}\right)^M \\ &\Downarrow \\ \hat{S} &= \sum_i^N \frac{m_i}{M} h_i \end{aligned} \tag{2.56}$$

This is an estimator for the hidden information based on a finite number of measurements. The hidden information for a particular outcome is being weighed by the frequency ratio of that outcome. Furthermore, we know that in the limit of a large number of measurements, this estimator should approach the entropy 2.48. We can extend this idea to define an estimator for the quantity Q such that it approaches the expectation value:

$$\hat{Q} = \sum_i \frac{m_i}{M} Q_i \rightarrow_{M\to\infty} \langle Q \rangle = \sum_i p_i Q_i \tag{2.57}$$

This is an empirical constraint that an EA can use to choose a prior.

The Principle of Maximum Entropy can then be instantiated by introducing a

constrained hidden information functional:

$$S[\{p_i\};\lambda] = -\sum_i p_i \log p_i + \lambda \left(\hat{Q} - \sum_i p_i Q_i \right) \tag{2.58}$$

Here λ is a Lagrange multiplier, which enforces the constraint. Maximizing this functional with respect to the belief distribution then ensures that the EA is maximally ignorant of everything but the empirical constraints. We will use this formalism to derive the belief distributions of the Landau model in the chapter four.

Chapter 3

I've Got A Theory

Our protagonists have been hiking in the White Mountains, and take a moment to enjoy the fantastic view.

Oontolicus: Why look over there, you won't believe it, it's quite marvelous that its there!

Epistemicus: What, where? What is it?! What did you see?

Oontolicus: Right along the side of that mountain. Look there, just stare at it.

Epistemicus: Sometimes you really are quite diff... Oh! Well, would you look at that.

Oontolicus: It's not every day that you see a rock outcropping that big in the shape of a human face. Just look at him, all granite-y and grey. Surely he's an old, old man?

Epistemicus: The Old Man of the Mountain... seems a fitting name for something so grand.

Oontolicus: It's a good thing he's so big, otherwise I might not have spotted him, let alone been able to point him out to you.

Epistemicus: Why yes, I do think that large things are always the easiest to see. Not like pesky small things... sometimes I look away for a moment, and upon returning my gaze the small things that were there are no longer at my attention.

Oontolicus: Oh, well you won't have to worry about that here. If there's one thing the Old Man is not, it is small. He will always get your attention whenever you look back.

Epistemicus: Now that's a comforting hope... but to all things there is a season...

Oontolicus: Hush now! He'll be there till the end of time.

In this chapter we will introduce the main informational measure to be used throughout the rest of this work, Configurational Entropy (CE). First we will motivate the definition of CE by considering how EAs interact with phenomena at different scales in the world. We will develop both the continuum and discrete formulations of CE, discussing the nuances that come with both. Finally we will discuss a possible biological origin of CE to complement the derivation in the preceding section.

3.1 Relevance of Scale

Confronted with sensory data, an epistemic agent is continuously updating their beliefs about the world. The inferences made are a result of applying the Bayesian formalism on experience to generate posteriors, which are then the input for the next round of experience. The flux of belief is a signature of there being information in the experiences; the experiences are relevant to the agent. As experiences cause certain beliefs to polarize, the amount of hidden information decreases, corresponding to the rising certainty that the agent has about statements.

3.1 Relevance of Scale

An EA in the world will come to believe that certain things happen more regularly than others and that certain objects will appear more frequently in their day to day than others. I myself, no matter how much I would like it to be the case, am pretty sure that on my daily walk to the office I will not have the pleasure of seeing a Triceratops. I am however pretty certain that I will come across some cars, maybe a bus, and quite a few people. Those events that occur on the same scale as myself tend to leave me with very polarized beliefs concerning them. Those things that happen on scales much smaller or larger leave me with a sense of uncertainty as to their realizability.

We can go further by considering our daily interactions with objects which we perceive on length scales close to our own. Imagine entering a room for the very first time. The walls are painted in a slightly off red color, and there is a desk with a computer. There are shelves along the walls filled with books, and posters on the walls. Next to the corner closest to us we notice a small spider, dangling from a silken thread, its legs moving frantically in ascent. We turn and look more closely at the shelves, noticing books about physics and philosophy, and when we focus back to the desk we become interested in papers on it which seem to be covered with scribbles. Closer inspection reveals equations, and as we touch the papers, a pencil rolls from underneath them to the edge of the desk, and falls off the side. We pick it up, put it back on the desk, and then stand back, panning our gaze over the room as a whole. We imagine the academic that must call this room their home, and all the questions we'd like to ask them about the books and the papers. We're standing now close to the corner next to the entrance, a memory tickles us, and we turn back to see the spider. It isn't there. We panic just a bit, brushing ourselves in the event that the

spider has had the means to get onto our body in the brief moments that we were taking stock of the office. Several brushes and a quick inspection leaves us confident that the spider is most likely not on us, but as we look around frantically we cannot find it. Realizing we're being foolish, we smile and look back at the shelves to see if any of the books might jump out at our interests, we note a few and remember them. If we ever see the owner of the office we will have to ask them about those books. We turn and go from whence we came.

What can we learn from this story? There are a few things that seem obvious, but should be stated explicitly. When we first entered the office we were attentive to the existence of the shelves and the desk, to the colors of the walls. The largest things in the room caught our attention, and created the skeletal structure of the model in our minds of the room we were in. Once our beliefs were polarized about the general shape of the room and its coarsest features, we began to pay more attention to smaller things. The types of books and papers on the desk. The motion of the spider alerts us to its presence even though it is much smaller than the scales we are currently investigating. After taking note of it we turned to the equations which, too, were small relative to the other objects in the room. Why do we do things in this order? Why do we investigate starting at the coarsest scales first, and then move to finer scales? How can instability, as exemplified by the unexpected presence of the spider, derail us from this general mode of investigation?

What is the informational narrative of the above process? We began with quite a bit of hidden information when we entered the room. Experience quickly remedied that, but why? Coarse-graining is a theme which we will return to often, and it is where we turn to for an answer. A coarse-grained distribution has less hidden

information, in general, in it than a fine distribution. Pinpointing a value in such a distribution seems to not decrease the amount of hidden information by much. However, the act of conditioning a fine-grained distribution based on the information from a coarse-grained experience will significantly decrease the hidden information in the fine distribution. Thus the act of minimizing hidden information by forcing experiences through actions (looking around the room, gazing at what is on the shelves and on the desk), can be accomplished more efficiently by pinning down coarse-grained features first, and using them to throw away irrelevant fine features.

Of course the spider was not a coarse grained object in the room, so noticing it seems to be at odds with the interpretation that gaining information in the most efficient manner should progress from coarser to finer scales. What made something at a finer scale more relevant to us? Well, first off it was in motion. The rest of the office was in a static state, and so the best way to proceed in reducing hidden information would be the process of conditioning on coarser experiences. Things that move capture our attention because of the instability that change introduces to the static world. We will return to the notion of instability later, but now back to scale.

Ignoring the particulars of different objects that are the same size, what is a good measure for the amount of hidden information to an EA which has just been introduced into a room that contains objects, or distributions of objects, at many different scales? *Correlations* between the positions of objects will need to be considered, since orientations contribute to the overall information that the EA wants to discover about the general shape of the space that they're in. Scales at which correlations are the largest will cause the EA's belief about objects at those scales to polarize the fastest. Smaller and larger scales relative to this correlation scale will contain much more

hidden information. We are forced to consider a measure that can capture spatial correlations, such that less correlated scales imply more hidden information from the EA. This will be essential as we move to applications later on.

3.2 Configurational Entropy and the Continuum

3.2.1 The Correlation Function

To capture the correlation between the positions of objects, and hence define a scale for an EA, we are led to the idea of a correlation function. The usefulness of this object cannot be overstated, and it is used to analyze spatial correlation in fields as diverse as cosmology to population dynamics [77][49].

Consider a giant volume filled with some scalar quantity, ρ with vanishing mean. The vanishing mean indicates that we are talking about fluctuations in the scalar quantity. One could imagine the quantity being the density or the temperature of the air in a room. Both have a mean value, but there exist fluctuations around that mean value. Some parts of the room may be slightly warmer or cooler, or denser or rarer, than others. How similar is the value of the quantity at one point, $\mathbf{r}'$ to that at another some fixed distance, $\mathbf{r}$, away? A simple solution is to start by taking a product of the two: $\rho(\mathbf{r}')^*\rho(\mathbf{r}'+\mathbf{r})$. If the resulting sign is positive, then we know that the quantity at both points had the same sign. If the resulting sign is negative, then they must have had the opposite sign. Of course doing this to two points tells us nothing about the entire space, so we take this product for every single point $\mathbf{r}'$, and normalize using the volume integrated over. What we have outlined here is just taking the volume averaged convolution of the quantity with itself, $\frac{1}{V}\rho \star \rho$. We have

the definition of the 2−Point Correlation Function:

$$C_2(\mathbf{r}) = \frac{1}{V}(\rho \star \rho)(\mathbf{r}) = \frac{1}{V}\int \mathrm{d}^d\mathbf{r}'\ \rho(\mathbf{r}')^*\rho(\mathbf{r}'+\mathbf{r}) \tag{3.1}$$

Notice that if we had chosen the separation, $\mathbf{r}$, to be nil, then we would just be taking the sum of the square of the quantity at every point: an estimator for the variance of the quantity (because we assumed vanishing mean), $C_2(0) = \sigma_\rho^2$.

The two point correlation function is used to find the total correlation between two points via the well known Ornstein-Zernike relation [76]. It is assumed that the total correlation function is written as a *direct* piece, which is the 2-point function, and an *indirect* piece:

$$C(\mathbf{r}) = C_2(\mathbf{r}) + \alpha \int \mathrm{d}^d\mathbf{r}'\ C_2(\mathbf{r}')C(\mathbf{r}-\mathbf{r}') \tag{3.2}$$

with α an expansion parameter. The indirect piece is just the direct correlation between one point and another arbitrary point, times the total correlation between that arbitrary point and the final point, summed up over all possible arbitrary points. This expression is a recursion relation which allows one to write the total correlation between two points as a series in the two point correlation function, and has analytical solutions for certain systems such as hard spheres [94], [78]. We will therefore restrict our attention to the 2-point correlator.

One might consider building a belief distribution from this quantity, but it does not have the correct properties. Note that in the limit that the quantity becomes completely random and uncorrelated, this function collapses to zero everywhere but at the origin. If we built a belief distribution out of it, it would have no hidden

information in the random state. This is at odds with the intuition that a very random assortment of objects in space of all sizes would require quite a bit of investigation at all scales to fully take in. Even more importantly, the function is not positive definite, so we can't even normalize it. We have a measure of correlation, but it has the exact opposite properties that we would expect if we used it as a belief function for a measure of hidden information. What to do?

3.2.2 The Power Spectrum

Very closely related to the 2 point correlation function is its Fourier Transform, the Power Spectrum. For a more in depth introduction see [93]. We write a Fourier pair of functions $(f, \mathcal{F}f)$ as:

$$(\mathcal{F}f)(\mathbf{k}) = \int \mathrm{d}^d\mathbf{r}\ f(\mathbf{r})e^{2\pi i\mathbf{k}\cdot\mathbf{r}} \tag{3.3}$$

$$f(\mathbf{r}) = \int \mathrm{d}^d\mathbf{k}\ (\mathcal{F}f)(\mathbf{k})e^{-2\pi i\mathbf{k}\cdot\mathbf{r}}. \tag{3.4}$$

Denoting the Fourier transform at a wave mode $\mathcal{F}\rho(\mathbf{k})$, one can easily show using a variant of the convolution theorem [57], that

$$\begin{aligned}\mathcal{P} &= \mathcal{F}C_2 \\ &= \mathcal{F}[\frac{1}{V}\rho \star \rho] \\ &= \frac{1}{V}(\mathcal{F}\rho)^*(\mathcal{F}\rho) \\ &= \frac{1}{V}|\mathcal{F}\rho|^2.\end{aligned} \tag{3.5}$$

The power spectrum is positive definite, and therefore normalizable, and a candidate for a belief function. Furthermore, for a random field the power spectrum is a uniform distribution. Using it as a belief function, this would imply maximal hidden information when correlations are not present. Furthermore, when the quantity is almost homogenous throughout space, the correlation length is the size of the volume. The power spectrum is then tightly peaked at the lowest modes which correspond to the largest length scales. This fits in well with intuition because if an EA sees that there is one giant space filling object, then its belief in objects at smaller scales should be very small. There is no room for them. Hidden information should be minimized.

3.2.3 Configurational Information

Since the power spectrum is normalizable and fits well with our intuition for how an information measure about scales should behave, we're just about done. Consider the class of statements that an EA could have, $s(\mathbf{k}) =$ *The most interesting configurations are at a scale* k *with window* $\mathrm{d}k$. We propose that the belief an EA has in these, $p(k, \mathrm{d}k)$ should be some function of the power spectrum, and construct it as follows.

Normalization turns out to be very simple. By the Plancheral theorem [11], we have that

$$\int \mathrm{d}^d\mathbf{k}\ \mathcal{P}(\mathbf{k}) = \frac{1}{V}\int \mathrm{d}^d\mathbf{k}\ |\mathcal{F}\rho(\mathbf{k})|^2 = \frac{1}{V}\int \mathrm{d}^d\mathbf{r}\ |\rho(\mathbf{r})|^2 = \sigma^2. \tag{3.6}$$

The normalization constant is just the estimator of the variance of the distribution of the quantity.

We define the *anisotropic modal fraction* as the resulting probability distribution:

$$p(\mathbf{k}, \mathrm{d}^d\mathbf{k}) = \frac{|\mathcal{F}\rho(k,\Omega)|^2}{V\sigma^2}\mathrm{d}^d\mathbf{k} = f(\mathbf{k})\mathrm{d}^d\mathbf{k}. \tag{3.7}$$

If we want to look at a length scale, we marginalize over the angles. This leads to:

$$p(k, dk) = dk\frac{k^{d-1}}{V\sigma^2}\int \mathrm{d}\Omega\ |\mathcal{F}\rho(k,\Omega)|^2 = f(k)dk. \tag{3.8}$$

For a particular mode, the *Configurational Information* is just the hidden information (2.33) in that mode. The function $f(k)$ is the *isotropic modal fraction*, and a slightly different form was initially introduced in [41] to construct an entropic measure that could quantify the fitness of approximate solutions to a specific class of field theoretic problems involving vacuum interpolating bubbles and kinks in field theory.

Since we are dealing with a continuum quantity here, it is important to specify a background distribution to compare this distribution to. This is one of the nuances in moving from a discrete set of statements to a continuous one, which we shall now dive into.

3.2.4 Continuous Beliefs and Regularization

Consider a continuous set of fine grained statements, x. Finely grained statements that are very close to one another are indiscernible to a resolution Δx, so we write the belief in all the statements between x and $x + \Delta x$ as $p(x, \Delta x) = f(x)\Delta x$. f is the probability density function (pdf) for p. Now we will look into our coarse statements and make discernments to access the finer statements. We will do this over and over again, moving to finer and finer distributions. What happens when we try to write the

limiting entropy of this distribution when our resolution becomes arbitrarily small, $\Delta x \to 0$?

$$\begin{aligned} S &= -\sum_x p(x, \Delta x) \log p(x, \Delta x) \\ &= -\sum_x \Delta x f(x) \log f(x) \Delta x \\ &= -\log \Delta x - \sum_x \Delta x f(x) \log f(x) \\ &\to \lim_{\Delta x \to 0} \log \frac{1}{\Delta x} - \int \mathrm{d}x \; f(x) \log f(x). \end{aligned} \tag{3.9}$$

Formally the entropy of a continuous distribution diverges because of the first term. Finer distributions come from having many choices to choose from; the more choices to choose from the potentially more hidden information. With an infinitum of choices, there must be an infinite amount of total hidden information for an EA. To get around this we should recall that for any finite system there is a maximum amount of entropy (otherwise the MaxEnt method wouldn't work, and our EAs could never establish priors). That entropy, under no information constraints, for a system of N mutually exclusive choices is $\log N$, which comes from the uniform distribution. Clearly this diverges as well if we allow $N \to \infty$.

We can use the divergence in the uniform distribution to regularize the divergence in the entropy. We simply note that the domain of x is unchanging as we fine grain our distributions, so that at each refinement $L = N\Delta x$ is a constant. We regulate the refinements by comparing the entropy of a distribution to the maximal possible

entropy that it can have. The regulated continuum entropy reads

$$\begin{aligned} S &= \lim_{N\to\infty} S_{uniform} - S[f] \\ &= \log L + \int \mathrm{d}x\ f(x) \log f(x) \\ &= \int \mathrm{d}x\ f(x) \log \frac{f(x)}{L^{-1}}. \end{aligned} \tag{3.10}$$

This is no longer the amount of total hidden information in the system, but a measure of the difference between maximal and actual hidden information. We didn't have to compare to the uniform distribution (whose pdf is L^{-1}). Any other distribution could have done just as well, and the resulting measure would have been:

$$D_{KL}(f||g) = \int \mathrm{d}x\ f(x) \log \frac{f(x)}{g(x)}. \tag{3.11}$$

It was first introduced by Solomon Kullback and Richard Leibler in 1951 as a *distance or divergence* between statistical populations. [61] Given that it is not symmetric in its arguments, $D_{KL}(f||g) \neq D_{KL}(g||f)$, it is typically referred to a the Kullback-Leibler Divergence. One way to think about this measure is as the difference in hidden information between using a *true* distribution f, and a *false* distribution g. Take the two to be discrete, the entropy of the former is $-\sum_i f_i \log f_i$. We then measure the hidden information in the false distribution, assuming the true distribution, $-\sum_i f_i \log g_i$. The difference between the two is a measure of error in assuming the false distribution. The convenient thing about this definition is that it instantly generalizes to continuous distributions without a need to regularize.

Returning to our discussion of scale, we can now define a deviation from the maximum total hidden information, 2.38, contained in all the scales present to an EA

by making use of the KL-divergence, 3.11:

$$S_C[\rho] = -\int \mathrm{d}k \; f(k) \log f(k). \tag{3.12}$$

For continuous systems this measure of information is called *Configurational Entropy* using an ignorant prior regularization.

Regularizing via comparison with an ignorant prior is not the only way to regularize continuous CE. Another way is by using *deviation from expectation*. In this case the EA may have already noticed a particular scale of interest and is now focusing on an object at that scale. Since she has changed the domain of examination, there is now an infrared cutoff in the calculation of the CE, to take into account that objects outside that scale are no longer relevant to the EA. Secondly, because that scale has already peaked the interest of the EA, the EA expects it to be the most interesting scale even as she surveys the object. The deviation from expectation is the difference between the hidden information at subscales and the expected hidden information at the largest scale which defines the size of the new region of focus:

$$\begin{aligned}
\mathcal{D}(k||k_*) &= \frac{S[f] - h(k_*)}{f(k_*)} \\
&= -\frac{1}{f(k_*)} \sum_{\Delta k} f(k)\Delta k \log f(k)\Delta k + f(k) \log f(k_*)\Delta k \\
&= -\int \mathrm{d}k \; \frac{f(k)}{f(k_*)} \log \frac{f(k)}{f(k_*)}.
\end{aligned} \tag{3.13}$$

This measure will be useful in analyzing localized objects that have already been picked out from the background. It's form was originally introduced on phenomenological grounds, though now, in the context just stated, there is a proper formal

reason as to why it should be used to study localized objects [84].

3.3 Configurational Entropy and the Lattice

Before investigating more of the properties of the CE, 3.12, we will rework the formalism of the last section for discrete systems. This will be crucial in implementing CE in simulations, which are by necessity finite and discrete.

3.3.1 A Short Detour into Quantization and Periodicity

In going from the continuum to a lattice, two effects need to be taken into account. The first is the effect that periodic boundary conditions have. The second is the effect that quantization of the continuum into a lattice has. We will see that the two are intimately related to one another. Consider first the case of a periodic space of dimensions $\mathbf{L}$, so that in particular $f(\mathbf{r}) = f(\mathbf{r} + \mathbf{L})$. Using this relation in 3.4, we get the condition that $\mathbf{k} \cdot \mathbf{L}$ must be an integer. Hence for each dimension $k_i = n_i/L_i$, where n_i is an integer. The possible wave modes are constrained to be integer multiples of the inverse size of the world. Incorporating periodicity in space imposes quantization of the possible modes in Fourier space.

This means that the inverse Fourier transform, 3.4 can incorporate periodicity via:

$$\begin{aligned}\int \mathrm{d}k\ \mathcal{F}f(\mathbf{k})(\mathcal{N}\sum_{\mathbf{n}\in\mathbb{Z}^d}\prod_i^d \delta(k_i - \frac{n_i}{L_i})e^{-2\pi i\mathbf{k}\cdot\mathbf{r}} &= \mathcal{N}\sum_{\mathbf{n}\in\mathbb{Z}^d}\mathcal{F}f(\frac{n_1}{L_1},\frac{n_2}{L_2},\dots)e^{-2\pi i\sum_j \frac{n_i x_i}{L_i}} \\ &= \mathcal{N}\sum_{\mathbf{n}\in\mathbb{Z}^d}\mathcal{F}f_{\mathbf{n}}e^{-2\pi i\sum_j \frac{n_i x_i}{L_i}} \qquad (3.14)\end{aligned}$$

The Fourier transform has become a Fourier series. Since space and Fourier space

transform are complementary, we can make use of this correspondence between periodicity and quantization to see the effect of what happens when the space continuum is replaced by a lattice. Let's say that the lattice spacing has size in each dimension Δx_i. Then $x_i = m_i \Delta x_i$ for some integer m_i. The number of lattice points in each dimension will then be related by $N_i = L_i/\Delta x_i$. The role of Δx_i for spatial quantization is analogous to the role of $1/L_i$ in mode quantization. Hence $1/\Delta x_i$ must play the same role for mode periodicity as L_i does for coordinate periodicity. Since there is an upper limit on the size of a mode, we have that:

$$\frac{1}{\Delta x_i} > k_i = \frac{n_i}{L_i} \Rightarrow n_i < N_i \tag{3.15}$$

Mode space is quantized into just as many lattice points as configuration space. For a periodic continuum, mode space is an infinite lattice. For an infinite lattice, mode space is a periodic continuum. For a periodic lattice, mode space is also a periodic lattice of the same number of lattice points. This feature allows the normalization constant to be determined as $1/\sqrt{N}$ where N is the total number of lattice points.

3.3.2 The Nyquist Wavemode

Periodicity of mode space has some interesting features. First off we need to choose which modes to use. Since the fields we are dealing with are real, $\mathcal{F}f(\mathbf{k}) = \mathcal{F}f(-\mathbf{k})^*$, we might as well choose a symmetric interval centered at $\mathbf{k} = 0$. This mode corresponds to a constant in the field value. If all we are interested in are fluctuations from equilibrium, then this $\mathcal{F}f(0)$ will always vanish. As we move to larger modes we are talking about smaller spatial scales. Having gone out to the largest wavemode, $k_i = 1/2\Delta x_i$, we see that the spatial scales associated with this mode are on the or-

der of two grid spacings. This is the Nyquist scale, k_{NQ}, and represents the smallest spatial scale resolvable by the lattice.

The glaring problem here is that we are trying to describe objects at spatial scales where we only use two grid points. Resolving a coherent object requires sampling more than just two grid points. This is a typical problem when dealing with hydrodynamic simulations. Sampling too few points could result in capturing numerical artifacts instead of physical structures. Requiring too many points severely limits the volume being sampled in momentum space. Results coming from the Phurbas magnetohydrodynamics code used at the American Museum of Natural History indicate that one needs at least 8 grid points to properly resolve structure. For calculating quantities, we will follow suit and always cutoff wavemodes $k > k_{NQ}/4$ [72][71].

3.3.3 Discrete Fourier Transforms

For periodic lattices we have then the analogues of the Fourier transforms, 3.3 and 3.4, are the discrete Fourier transforms (DFT):

$$\mathcal{F}f_n = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} f_m e^{\frac{2\pi i}{N} mn} \tag{3.16}$$

$$f_n = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} \mathcal{F}f_m e^{-\frac{2\pi i}{N} mn} \tag{3.17}$$

Sometimes we will refer to $\mathcal{F}f_n$ as the Fourier coefficients of f. Though the DFT is a linear operator, it is actually not efficient to implement it as pure matrix multiplication using:

$$\mathcal{F}_{nm} = \frac{1}{\sqrt{N}} e^{-\frac{2\pi i}{N} mn} \tag{3.18}$$

Standard matrix multiplication has an $\mathcal{O}[N^2]$ computational complexity, but this can be reduced if we take advantage of the properties of the roots of unity appearing in 3.16 and 3.17. Fast algorithms can be implemented by breaking up the modes into odd and even ones, so that the complexity is bounded by the largest prime factor of N. In our simulations we always take the number of lattice points to be some power of 2, so that the complexity of the DFT is $\mathcal{O}[N \log N]$. [18]

3.3.4 Discrete Power Spectrum and Marginalization

The correlation function is much simpler in the discrete case:

$$C_k = \frac{1}{\sqrt{N}} \sum_n f_n^* f_{n+k} \tag{3.19}$$

and due to the periodicity satisfies $C_k = C_{N-k}$. Expanding the fields in Fourier series gives us:

$$C_k = \frac{1}{\sqrt{N}} \sum_n |\mathcal{F} f_n|^2 e^{-\frac{2\pi i}{N} nk} = \frac{1}{\sqrt{N}} \sum_n \mathcal{F}\mathcal{C}_n e^{-\frac{2\pi i}{N} nk} \tag{3.20}$$

we see now that things are going just as before, and the discreet power spectrum reads $\mathcal{P}_k = \mathcal{F}C_k = |\mathcal{F} f_k|^2$.

We can now define the anisotropic modal fraction

$$f_k^a = \frac{|\mathcal{F} f_k|^2}{\sum_n |\mathcal{F} f_n|^2} \tag{3.21}$$

This is precisely the same as the original definition of the modal fraction in [41]. We

can now construct the configurational entropy as before:

$$S_C^a = -\sum_{|k|<k_{NQ}} f_k^a \log f_k^a \tag{3.22}$$

where we have used the Nyquist frequency to create an ultraviolet cutoff. Of course this is an anisotropic form of 3.12. If we want an isotropic form then we should marginalize over the angles:

$$f_k = \sum_{k'} f_k'^a \mathcal{W}(|k-k'|, \Delta k) \tag{3.23}$$

We have introduced here a window function which is equal to 1 whenever the k''s being summed over have magnitudes within $\Delta k/2$ of k.

With these in hand we can tackle numerical calculations on a periodic lattice, which we will do in the next chapter within the context of a phase transition in the Landau model.

3.4 A Biological Origin?

Though we have argued for CE on the basis of using a function related to the two point correlation function and having the properties expected of a probability measure concerning scales, there is biological motivation for it as well coming from the well known empirical relation known as the Weber-Fechner Law [92] [25] [50]. The law states that just noticeable differences to perception are proportional to the intensity of the perception. Empirically the law has been shown to hold for vision, odor, taste, and sound perception. A brief historical introduction can be founf in Buntain's notes

[12]. It is known to fail at the extremes of perception, but will nonetheless provide a rather interesting derivation of the CE if assumed to hold everywhere [87].

Consider an EA looking at a bunch of objects of different sizes. The objects linear size in one dimension changes by some fixed amount. For the smallest of objects, this change will be quickly perceived since the length difference is comparable with the original size of the object. For the largest of objects, the change is so small relative to the size of the object that it will not register to the EA. Formally, denote the intensity of a scale as $\mathcal{P}$, and the difference in perception of a change in that scale to an EA as de. The Weber-Fechner law then says that

$$de = \alpha \frac{d\mathcal{P}}{\mathcal{P}} \tag{3.24}$$

where α is some constant of proportionality. This relation is almost trivial to solve:

$$e(\mathcal{P}) = \log \left(\frac{\mathcal{P}}{\mathcal{P}_0} \right)^{\alpha} \tag{3.25}$$

The average experience of scale to an EA is then

$$\langle e \rangle = \sum_k p(k|K) e(\mathcal{P}(k)) \tag{3.26}$$

What belief function should the EA have based on their average experience? We use the formalism of Maximum Entropy to answer this, finding the belief function with the highest entropy that satisfies the constraint set by experience, 3.26. The total

hidden information functional reads:

$$H[p|K] \propto -\sum_k p(k|K) \log p(k|K) + \beta(\langle e \rangle - \sum_k p(k|K) \log \left(\frac{P(k)}{\mathcal{P}_0} \right)^{\alpha}) \tag{3.27}$$

Varying with respect to p and setting to zero we find

$$p(k|K) \propto \mathcal{P}(k)^{-\alpha\beta} \tag{3.28}$$

For the case $\alpha\beta = -1$ we recover the form of CE.

Though this derivation gets us to CE, there are still many questions that have yet to be explored concerning the process. In a more realistic setting a window function should be taken into account which describes the limits of perception for the EA. Furthermore, the constraint on $\alpha\beta$ is not yet fully understood. Nonetheless, the connection between epistemic belief functions and psychophysical laws is more than tenuous. Exploring this relationship in the future will undoubtedly prove fruitful.

Chapter 4

Under Your Spell

Our protagonists are sitting along the Connecticut River, in the shade of the maple trees.

Oontolicus: As much as I love summer, it does get incredibly hot here.

Epistemicus: I believe there's some ice tea in my bag... let me check...

Oontolicus: Splendid!

Epistemicus: Oh, here it is. I think the ice has melted a bit, but it's still pretty cool.

A small canteen is handed over, the ice clinking on the inside against the metal casing.

Oontolicus: Thank you. It sounds like there's still enough ice in there for a sip to do the trick.

Epistemicus: While you enjoy that, I'm going for a dip in the water.

Epistemicus jumps into the river while Ontolicus enjoys the cold tea. In the bushes nearby there's a rustling sound and two catsnakes emerge, one of them black, the other white.

Oontolicus: There you are, Kodo and Podo. Done frolicking in the woods?

The two little beasts hop playfully towards Oontolicus, and then plop down next to him. The three of them lay there as a small breeze causes the leaves above to rustle ever so slightly. Within a few minutes the three of them are napping on the beach, while Epistemcus swims his way across the river and back again.

The efficacy of Configurational entropy will come to light as we begin to apply it to different phenomena, the first of these being the phase transition in the Landau model. CE will expose a rich informational narrative that will provide an alternate way of describing phenomena from the traditional mechanistic one. We will start by reviewing a general Ising model, which will be coarse grained into the Landau model [52] [75] [46]. To do this we will briefly discuss stochastic difference equations and the fluctuation-dissipation theorem. We will examine the numerical implementation of the Landau model, and analyze its configurational entropy; these results first appeared in my work with Marcelo Gleiser, *Information-Entropic Signature of the Critical Point* [85]. Finally, an analysis of the modes in the model will shed light on the information storage properties of the Ising universality class as a phase transition is approached, as was first shown in *Information Dynamics at a Phase Transition* [86].

4.1 The Ising Model

4.1.1 The Model

Consider a graph, possibly infinite, with vertex set, V, and edge set E, $G = \langle V, E \rangle$. We define a state function on the vertices $\sigma : V \to \{-1, 1\} : v \mapsto \sigma_v$, which naturally extends to the edge set so that $\forall e \in E$ with endpoints i, j, $\sigma(e) = \sigma_i \sigma_j$. This breaks

up the edges into those that have endpoints of the same state, and those which do not. We denote the state of the graph as σ_G.

There are many states that a graph can have, corresponding to all the ways in which we can label the vertices. We wish to have a way to order all the states of a graph, prioritizing some over others. This is accomplished via a cost function, which we will refer to as the Hamiltonian. The simplest such Hamiltonian simply adds up the state values of all the vertices and edges, with appropriate weights on which vertices and which edges are being added:

$$H[G] = c_0 + \sum_{v \in V} c_1(v)\sigma(v) + \sum_{e \in E} c_2(e)\sigma(e) \tag{4.1}$$

If we use $\sigma(e) = \sigma_i \sigma_j$ on the edges this can be rewritten as

$$H[G] = c_0 + \sum_{i=1}^{|V|} c_1^i \sigma_i + \sum_{i=1}^{|V|} \sum_{j=1}^{|V|} c_2^{ij} \sigma_i \sigma_j \tag{4.2}$$

where $c_2^{ij} = 0$ if i and j are not connected by an edge.

To go any further we need to consider what it is that we're trying to model to impose more constraints on this expression. Since we are comparing states with one another, only the differences in their Hamiltonians will matter, so we can set $c_0 = 0$ with impunity. Secondly, we want to treat all the vertices on equal footing which means that the coupling functions should not depend on indices. They should be constants. Taking into account that each vertex is the neighbor of its neighbor, this

reduces the Hamiltonian further to:

$$H[G] = \sum_{i=1}^{|V|} \sigma_i (c_1 + \frac{c_2}{2} \sum_{j \sim i} \sigma_j) \tag{4.3}$$

We can write the vertex contribution to the Hamiltonian as the energy of the vertex:

$$E_i = \sigma_i (c_1 + \frac{c_2}{2} \sum_{j \sim i} \sigma_j) \tag{4.4}$$

The first constant is typically referred to as the *magnetic field* and denoted h. It gives priority to states which have a larger number of vertices that share the sign of this constant. The second constant, simply referred to as the *coupling* and written with an extra negative sign as $-J$, determines whether one should give priority to homogenous or inhomogenous states. If $J < 0$, then edges which have endpoints of the same vertex state will have more energy. This will lead to homogenous graph states being prioritized. If $J > 0$, homogenous states will have less energy.

Henceforth, we write the Hamiltonian of the Ising model as:

$$H[\{\sigma\}] = -\frac{J}{2} \sum_{i,j \sim i} \sigma_i \sigma_j + h \sum_i \sigma_i. \tag{4.5}$$

4.1.2 MaxEnt

Consider an epistemic agent interacting with an Ising model. The EA knows the underlying graph, G, and has a way of measuring the Hamiltonian of the graph. The EA cannot however measure the states of the vertices, they are forever hidden from her prying eyes. The question that the EA must answer is this: Given the information of the energy of the model, what belief should she have concerning the underlying

state of the system?

Posed this way, the EA's beliefs should be maximally ignorant of everything other than the Hamiltonian, or the one determined by MaxEnt. Consider the resulting entropy functional:

$$S[\{p(\sigma_G)\}] = -\sum_{\sigma_G} p(\sigma_G)\log p(\sigma_G) + \alpha\left(1 - \sum_{\sigma_G} p(\sigma_G)\right) + \beta\left(\langle E\rangle - \sum_{\sigma_G} p(\sigma_G)H[\sigma_G]\right) \tag{4.6}$$

The first Lagrange multiplier ensures normalization of probabilities, and the second ensures that the energy constraint is satisfied. The second constraint ensures the connection to phenomenological data. The measurements the EA has made constrain the possible beliefs she may have.

Let's maximize this with respect to the distribution:

$$\frac{\delta S}{\delta p} = 0 \Rightarrow \log p(\sigma_G) + 1 + \alpha + \beta p(\sigma_G)H[\sigma_G] = 0$$

Rearranging this expression to solve for p gives the maximally ignorant distribution:

$$p(\sigma_G) = \frac{1}{\mathcal{Z}} e^{-\beta H[\sigma_G])} \tag{4.7}$$

Here we have absorbed α into $\mathcal{Z}$, the partition function. The partition function IS the normalization constraint:

$$\mathcal{Z} = \mathcal{Z}(\beta) = \sum_{\sigma_G} e^{-\beta H[\sigma_G]} \tag{4.8}$$

The symbol comes from Max Planck, who called it *Zustandesumme* (sum over states)

[7]. The energy constraint fixes β so that

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log \mathcal{Z} \tag{4.9}$$

Furthermore, the entropy functional for this distribution relates the partition function to the energy in another way:

$$\begin{aligned} S &= \beta \langle E \rangle + \log \mathcal{Z} \\ &= \beta(\langle E \rangle - F) \end{aligned} \tag{4.10}$$

where we have introduced the *free energy*, $F = -\frac{1}{\beta} \log \mathcal{Z}$. The entropy is a scaled difference between energy and free energy. Since entropy is nonnegative, this means that the free energy is bounded from above by the average energy. This can be seen more clearly by using the definition of free energy in equation 4.9:

$$\langle E \rangle = F + \beta \frac{\partial F}{\partial \beta} \tag{4.11}$$

showing us that the free energy must be monotonicaly non decreasing with β.

What if the EA has more information at her disposal than just the average energy, for instance, the average magnetization of the system:

$$\mu[\sigma_G] = \frac{1}{|V|} \sum_{i=1}^{|V|} \sigma_i \tag{4.12}$$

This constraint would be imposed by a new Lagrange multiplier, adding a term to

the entropy functional of the form:

$$S \to S + \gamma \left(\langle \mu \rangle - \sum_{\sigma_G} p(\sigma_G) \mu[\sigma_G] \right) \tag{4.13}$$

Playing the same game as before, we find the probabilities, partition function, and constraints to be:

$$\begin{aligned} p(\sigma_G) &= \frac{1}{\mathcal{Z}} e^{-\beta H[\sigma_G] - \gamma \mu[\sigma_G])} \\ \mathcal{Z}(\beta, \gamma) &= \sum_{\sigma_G} e^{-\beta H[\sigma_G] - \gamma \mu[\sigma_G])} \\ \langle E \rangle = F + \beta \frac{\partial F}{\partial \beta} \qquad & \langle \mu \rangle = \beta \frac{\partial F}{\partial \gamma} \end{aligned} \tag{4.14}$$

The max entropy now reads

$$S = \beta \langle E \rangle + \gamma \langle \mu \rangle - \beta F \tag{4.15}$$

Comparing this case to the last is analytically difficult because the two free energies are very different. This formula does give us a prescription for finding the free energy. Notice that maximal entropy is equivalent to minimizing the free energy. This will be important when we tackle the Landau model later on. We also see a general scheme here for incorporating information into the belief distribution that an EA has, which we will briefly discuss.

4.1.3 The Partition Function

Let's generalize the argument from the last section. Our epistemic agent has at her disposal several measurements to probe properties, $\mathcal{P}_i$ of the system. She then uses an estimator from her measurements to pin down the $\langle \mathcal{P}_i \rangle$'s, each of which will contribute a Lagrange multiplier, λ_i to the partition function. The maximal entropy reads

$$S = \sum_n \lambda_n \langle \mathcal{P}_n \rangle + \log \mathcal{Z}(\lambda) \tag{4.16}$$

$$\langle \mathcal{P}_n \rangle = -\frac{\partial}{\partial \lambda_n} \log \mathcal{Z} \tag{4.17}$$

The variance in a property reads

$$\begin{aligned} \mathrm{Var}[\mathcal{P}_n] &= \sigma_n^2 \\ &= \left\langle \mathcal{P}_n^2 \right\rangle - \langle \mathcal{P}_n \rangle^2 \\ &= \frac{1}{\mathcal{Z}} \frac{\partial^2}{\partial \lambda_n^2} \mathcal{Z} - \left(-\frac{\partial}{\partial \lambda_n} \log \mathcal{Z} \right)^2 \\ &= \frac{\partial^2}{\partial \lambda_n^2} \log \mathcal{Z} \\ &\geq 0 \end{aligned} \tag{4.18}$$

This result is telling us that the logarithm of the partition function is convex in each of the parameters, holding the others fixed. One can extend this calculation to extract the full covariance matrix:

$$C_{ij} = \mathrm{Cov}[\mathcal{P}_i, \mathcal{P}_j] = \frac{\partial}{\partial \lambda_i} \frac{\partial}{\partial \lambda_j} \log \mathcal{Z} = [\nabla^2 \log \mathcal{Z}]_{ij} \tag{4.19}$$

which is the Hessian matrix of $\log Z$. The covariance matrix is symmetric, and positive semi-definite, so it has real non-negative eigenvalues. This translates into the Hessian determinant being non-negative so at every point in parameter space the partition function is either convex, flat, or singular. This is our first glimpse of a *phase transition*, signified by the possible existence of singularities in the free energy.

Higher order moment products of properties can all be calculated similarly,

$$\langle \mathcal{P}_{n_1} \mathcal{P}_{n_2} \cdots \mathcal{P}_{n_k} \rangle = \frac{(-1)^k}{\mathcal{Z}} \frac{\partial}{\partial \lambda_{n_1}} \frac{\partial}{\partial \lambda_{n_2}} \cdots \frac{\partial}{\partial \lambda_{n_k}} \mathcal{Z}, \tag{4.20}$$

which means that the partition function contains all the moments inside of it, and can be thought of as the moment generating function:

$$\frac{\mathcal{Z}(\lambda + \delta\lambda)}{\mathcal{Z}(\lambda)} = 1 - \langle \mathcal{P}_n \rangle \, \delta\lambda_n + \frac{1}{2} \langle \mathcal{P}_n \mathcal{P}_m \rangle \, \delta\lambda_n \delta\lambda_n + \cdots . \tag{4.21}$$

Taking the logarithm of both sides and writing both as a series in $\delta\lambda$'s, we discover another property of the partition function. The LHS is a series in the cumulants of the properties,

$$\kappa_{i_1 i_2 \cdots i_k} = \langle (\mathcal{P}_{i_1} - \langle \mathcal{P}_{i_1} \rangle)(\mathcal{P}_{i_2} - \langle \mathcal{P}_{i_2} \rangle) \cdots (\mathcal{P}_{i_k} - \langle \mathcal{P}_{i_k} \rangle) \rangle \, . \tag{4.22}$$

We identify like terms, and see that the logarithm of the partition function is the cumulant generating function:

$$\begin{aligned} \log \frac{\mathcal{Z}(\lambda + \delta\lambda)}{\mathcal{Z}(\lambda)} &= \nabla_n \log \mathcal{Z}(\lambda) \delta\lambda_n + \frac{1}{2} \nabla_n \nabla_m \log \mathcal{Z}(\lambda) \delta\lambda_n \delta\lambda_m + \cdots \\ &= -\kappa_n \delta\lambda_n + \frac{1}{2} \kappa_{nm} \delta\lambda_n \delta\lambda_m + \cdots . \end{aligned} \tag{4.23}$$

Equations 4.17, 4.18, and 4.19 are particular cases of the more general expression:

$$\kappa_{i_1 i_2 \cdots i_k} = (-1)^k \nabla_{i_1} \nabla_{i_2} \cdots \nabla_{i_k} \log \mathcal{Z} \tag{4.24}$$

All of the statistical correlations between possible higher order measurements are contained in the partition function, and this is the strength of the MaxEnt method. It provides predictions for higher order estimators. Our EA is only privy to measurements of the mean of a particular property. MaxEnt then provides her with the maximally ignorant (most hidden information) belief about the system. It also predicts the cumulants of her measurements, which she can then check using estimators. If there are discrepancies, then she can update her entropy functional with more constraints, and introduce more Lagrange multipliers.

4.1.4 Coarse Graining

We have already seen that coarse graining is a way in which a finer distribution can be made coarser, with a loss in entropy due to non-interest in the particulars of the coarse grained bins. In the context of the Ising model, we can clump together states that share some property, and coarse grain over that property: $\{\sigma_1, \sigma_2, \sigma_3, \sigma_4, \dots\} \to \{\{\sigma_1, \sigma_2\}, \{\sigma_3, \sigma_4\}, \dots\} = \{\sigma'_1, \sigma'_2, \dots\}$. We could coarse grain over energy or over position. Some types of coarse graining will turn out to be very useful.

As an example, consider coarse graining over energy. This means every state with

the same energy will be clumped together into the same coarse state, with probability:

$$\begin{aligned} P(E) &= \sum_{\sigma_G} \delta(E - H[\sigma_G]) p(\sigma_G) \\ &= \frac{n(E)}{\mathcal{Z}} e^{-\beta E}, \end{aligned} \tag{4.25}$$

where $n(E)$ is the number of states that have energy E. By the additive property of entropy, 2.44, we have that

$$\begin{aligned} S[P] &= S[p] - \sum_E P(E) S[\sigma_G | E] \\ &= \beta(\langle E \rangle - F) - \sum_E P(E) \log n(E) \\ &= \beta \langle E \rangle - \beta \left(F + \frac{1}{\beta} \langle \log n(E) \rangle \right), \end{aligned} \tag{4.26}$$

where we used the fact that the microstates in the microcanonical ensemble are all equiprobable. The act of coarse graining has effectively lifted the free energy. The uninteresting information incorporates itself into the free energy, closing the gap between free energy and average energy.

What if instead we had coarse grained over magnetization? This situation is much more difficult to analyze because the microstates in each macrostate are no longer equiprobable, but we introduce $n(\mu)$ the number of states with mean field μ, and

$\mathcal{H}[\mu]$, a coarse grained Hamiltonian, and write things similarly to 4.26:

$$S[P] = \beta \left\langle \mathcal{H} - \frac{1}{\beta} \log n \right\rangle - \beta F \geq 0$$

$$\Downarrow$$

$$F \leq \left\langle \mathcal{H} - \frac{1}{\beta} \log n \right\rangle \tag{4.27}$$

This is an upper bound for the free energy. We assume that the distribution is sharply peaked around $\langle \mu \rangle$ and minimize $\mathcal{H} - \frac{1}{\beta} \log n$ to find it. The coarse grained Hamiltonian is approximated by simply replacing lattice values with the mean field, μ. On a d-dimensional lattice with N lattice sites and $h = 0$, the coarse grained Ising Hamiltonian 4.5 is:

$$\mathcal{H} = -2Jd\mu^2 \tag{4.28}$$

Meanwhile that states that have μ as a mean field all have $\frac{1+\mu}{2}N$ sites with $+1$, and $\frac{1-\mu}{2}N$ sites with -1. Using the Stirling approximation for factorials, their degeneracy is:

$$\log \frac{N!}{\left(\frac{1+\mu}{2}N\right)! \left(\frac{1+\mu}{2}N\right)!} \approx N(\log 2 - \frac{1}{2}\mu^2 - \frac{1}{12}\mu^4 + \dots) \tag{4.29}$$

The free energy per lattice site then reads

$$\frac{F}{N} \approx \frac{1}{2}(\frac{1}{\beta} - 2Jd) \langle \mu \rangle^2 + \frac{1}{12\beta} \langle \mu \rangle^4 + \cdots \tag{4.30}$$

and the EAs belief function over coarse grained states is:

$$P(\mu) \propto e^{-\frac{N}{2}(1-2Jd\beta)(\mu-\langle\mu\rangle)^2-\frac{N}{12}(\mu-\langle\mu\rangle)^4} \tag{4.31}$$

Here we see an ambiguity. β is set by the mean energy constraint, however $\langle\mu\rangle$ is set by minimizing the free energy. When $\beta < \beta_c = (2Jd)^{-1}$ this does not pose a problem because there is a unique value for the mean field, $\langle\mu\rangle = 0$. However, when $\beta > \beta_c$, there are two possible values for $\langle\mu\rangle$ and hence two possible belief distributions for the EA. The Maximum Entropy method is not capable of uniquely determining the belief distribution from mean energy data alone. Below the critical β_c, the maximum entropy method provides a unique belief distribution, whereas above β_c, the EA needs an additional bit of information to determine which distribution to use. This discontinuity in hidden information is the telltale sign of a phase transition.

Energy coarse graining and mean field coarse graining have revealed to us some interesting qualities of the Ising model. The next line of attack for us will be focussing on spatial coarse graining. With the first two examples we coarse grained the state space. Now we will venture to coarse grain the lattice that the Ising model is defined on, so that the individual lattice points slowly blur away into a continuum. The result will be the famous Landau model, whose informational properties we will study.

4.2 The Landau Hamiltonian

We will now construct the Hamiltonian for the Landau model. Our goal is a continuous model that is a spatially coarse grained Ising model. The Hamiltonian is constructed by considering the symmetry of the order parameter, to mimic the be-

havior of the Ising model at low β.

Without an external field the sign of the order parameter is an arbitrary degree of freedom. Thus we expect the energy to be invariant under sign changes, $\phi \to -\phi$. We say the Hamiltonian has a $\mathbb{Z}_2$ symmetry. This means that only even powers of the order parameter will enter into the Hamiltonian. Secondly we want local spatial interactions of the field with itself, meaning the Hamiltonian should depend on spatial derivatives as well. The angle from which we gaze at the system should have no effect on its energy. Rotational symmetry imposes then that the only powers of $\nabla\phi \cdot \nabla\phi$ should appear. For simplicity we do not include powers higher than one, nor do we mix derivative terms with nonderivative ones. Our ansatz for the Hamiltonian is then:

$$\mathcal{H}[\phi] = \int \mathrm{d}^d\mathbf{r}\ \frac{\epsilon}{2}\nabla\phi \cdot \nabla\phi + \sum_{n=1}^{\infty} a_n \phi(\mathbf{r})^{2n} \tag{4.32}$$

The first term is referred to as the *gradient energy* of the field. Minimizing this energy involves smoothing out the order parameter. The second term is referred to as the *potential energy* of the field. It is minimized when the order parameter settles at the minimum of this potential. We will be interested in cases where the potential has minima at non zero values. This implies that when a system reaches equilibrium the $\mathbb{Z}_2$ symmetry of the Hamiltonian will be broken.

Of course our system will not be isolated, but in contact with a thermal bath at some temperature. This bath will exchange energy with the system causing the minimum of the potential to no longer be the equilibrium value of the order parameter. This effect will be captured by the Landau Free Energy functional, which will act as an effective Hamiltonian with temperature dependence. The temperature dependent potential energy will display a behavior wherein at a critical temperature the broken

Z_2 symmetry will be restored. The point at which the symmetry is restored displays a plethora of interesting physics, which we shall explore in due time.

4.2.1 Connection to the Ising Model

It is easy to see that ansatz from the last section is identical to Ising by taking the discrete form of the integral. For simplicity, we look at the one dimensional case:

$$\begin{aligned}\int \mathrm{d}x\ \frac{\epsilon}{2}\left(\frac{d\phi}{dx}\right)^2 + a_1\phi^2 &\approx \sum_x \frac{\epsilon}{2}\Delta x\left(\frac{\phi(x+\Delta x)-\phi(x)}{\Delta x}\right)^2 + a_1\Delta x\phi(x)^2 \\ &= \sum_x \frac{\epsilon}{2\Delta x}\left(\phi(x+\Delta x)^2+\phi(x)^2-2\phi(x+\Delta x)\phi(x)\right) + a_1\Delta x\phi(x)^2 \\ &= -\frac{\epsilon}{\Delta x}\sum_x \phi(x+\Delta x)\phi(x) + \left(\frac{\epsilon}{\Delta x}+a_1\Delta x\right)\sum_x \phi(x)^2\end{aligned} \tag{4.33}$$

We see that under the identification $J \leftrightarrow \epsilon/\Delta x$ the first term just becomes the nearest neighbors coupling. The second term is just a constant in Ising since lattice sites are only allowed to take on the values ± 1.

A good way to think about the Landau model is in terms of a coarse graining scale ξ. Imagine taking a very large Ising model and breaking it up into pieces of size ξ. The scale is large enough so that each piece contains in it a vast number of lattice points. Hence the mean field for a piece can take on what appears to be a continuum number of values between -1 and 1. In this way a spatial coarse graining of the Ising model results in a continuous field.

4.2.2 Connection to the Scalar Field

The order parameter can be thought of as a scalar field, so in this section we will examine the connection between classical field theory and statistical mechanics. The action for a scalar field in d-dimensional spacetime with signature $(-,+,+,\cdots)$ reads:

$$S = \int d^d x \ \sqrt{|g|} \left(\frac{R - 2\Lambda}{16\pi G} + \frac{1}{2} g_{\mu\nu} \partial^\mu \phi \partial^\nu \phi - V(\phi) \right) \tag{4.34}$$

Invariance under small changes in the metric results in the Einstein field equations

$$\frac{\delta S}{\delta g_{\mu\nu}} = 0 \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{4.35}$$

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial_\alpha \phi \partial^\alpha \phi - g_{\mu\nu} V(\phi) \tag{4.36}$$

where the right hand side is proportional to the stress energy tensor of the scalar field. Eq. 4.35 is telling us how the stress energy of the scalar field is causing the metric of spacetime to evolve. We do not consider this backreaction of the field on the spacetime, and set our metric to flat Minkowski, $g_{\mu\nu} = \eta_{\mu\nu}$.

The energy density of the scalar field is the projection onto the 4-velocity of an observer in spacetime, $\rho = T_{\mu\nu} u^\mu u^\nu$. Assuming we ARE the observer puts us in the rest frame, and the energy density of the scalar field reads:

$$\begin{aligned} \rho &= T_{00} \\ &= \partial_0 \phi \partial_0 \phi - \frac{1}{2} g_{00} \partial_\alpha \phi \partial^\alpha \phi - g_{00} V(\phi) \\ &= \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\nabla \phi)^2 + V(\phi) \end{aligned} \tag{4.37}$$

and thus our conserved Hamiltonian functional of the scalar field is

$$\mathcal{H}[\phi] = \int \mathrm{d}^{d-1}\vec{r}\, \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi) \tag{4.38}$$

This is precisely the same Hamiltonian as our ansatz for the Landau Model, plus a term proportional to the time variation of the field. This term represents the kinetic energy of the scalar field. The connection here is that the Landau model is describing a system in equilibrium which is no longer changing. To make the full analogy, we will have to figure out a way to dissipate the energy of the scalar field for it to achieve equilibrium. Going back to the full action, invariance under variations in the field leads to the equation of motion

$$\frac{\delta S}{\delta \phi} = 0 \Rightarrow \ddot{\phi} + \nabla^2\phi + \frac{dV}{d\phi} = 0 \tag{4.39}$$

To get a sense for the evolution of the field, it is good to consider what happens close to the minimum of the potential, $\phi_0 = \arg\min V(\phi) \rightarrow V(\phi_0) = 0$. Expanding the field around this value as $\phi = \phi_0 + \delta\phi$, the linearized equation of motion reads

$$\ddot{\delta\phi} + \nabla^2\delta\phi + V''(\phi_0)\delta\phi = 0 \tag{4.40}$$

This is just the Klein-Gordon equation with the curvature of the potential playing the role of a mass, $m^2 = V''(\phi_0)$. Fluctuations evolve like they're in a simple harmonic oscilltor due to this mass, while the presence of the laplacian causes perturbations in one region to propagate outward as waves. Far from the minimum the dynamics are far more interesting. Disturbances still propagate due to the presence of the Laplacian

generating local interactions, but the shape of the potential strongly determines the way the field is pushed in different regions of space. If the potential has multiple minima, then the field could be pushed into these different vacua in different regions of space. This will generate gradients in space resulting a type of topological defect known as a domain wall separating the regions. Overall the dynamics of the scalar field can get quite interesting for non trivial potentials. [88]

4.2.3 Coupling to a Heat Bath

We are not interested in a scalar field in isolation, but one which is interacting with a large heat bath. Why? The scalar field is not in equilibrium due to the extra kinetic energy term present int he Hamiltonian. By coupling it to a heat bath we will allow energy to be exchanged between the scalar field and the bath, thereby allowing the scalar field to approach an equilibrium state. So, how does this coupling change the equation of motion 4.39?

One needs to be careful to incorporate the heat bath correctly. Ideally, contact with the bath must introduce energy exchange between them, allowing the field to both gain energy from the bath, as well as dissipate it back into the bath. The latter is simple enough by incorporating a friction term proportional to the velocity of the field, $-\gamma\dot{\phi}$. We have introduced a positive constant, γ, which is the *viscosity* of the heat bath. ? It represents how strongly the bath is capable of draining energy away from the field. In the limit of vanishing viscosity, the field decouples from the energy sink.

What about energy input into the field? Here is where stochastic considerations prevail. At every point in space, and at every instance of time, the bath's effect is to

kick the field ever so slightly. If the field is stationary, this will instantly introduce field gradients that will increase the energy, which justifies our use of these kicks as a source of energy. We can incorporate these kicks by adding a random variable field at every point in space, $\xi(\mathbf{r}, t, \dots)$, where the elipsis indicates other parameters that the source kicks should depend on. What are they?

First and foremost the is the viscosity. Fluctuations should go hand in hand with the dissipation, so that as the field decouples from the sink it also decouples from the source. Next is the temperature of the heat bath. We expect the kicks to become larger and larger as the temperature increases. These are reasonable properties, and will be formalized in the fluctuation-dissipation theorem soon enough.

Since we do not want the kicks to push the field collectively in any particular direction (in field space), they should have zero mean, $\langle \xi \rangle = 0$. Lastly, for the fluctuations to be truly random across space and time, we need to ensure that they are uncorrelated across spacetime, $\langle \xi(\mathbf{r}, t)\xi(\mathbf{r}', t') \rangle = \sigma^2 \delta(t-t')\delta^d(\mathbf{r}-\mathbf{r}') \approx \sigma^2 \delta_{t,t'}\delta^d_{\mathbf{r},\mathbf{r}'}/\delta t \delta V$, where in the approximation I anticipate the discretization of the field equations and replace delta functions with Kronecker deltas and take into account the spatio-temporal discritization volume. This means that at each point, the fluctuations at each instant of time are drawn from a distribution that has zero mean, and variance σ^2. Since this is all the information we have about the fluctuations we can use MaxEnt to infer the most ignorant probability for a fluctuation between ξ and $\xi + d\xi$:

$$p(\xi, \xi + d\xi) = \frac{1}{\sqrt{2\pi}\bar{\sigma}} e^{-\frac{1}{2\bar{\sigma}^2}\xi^2} d\xi \tag{4.41}$$

This is a Gaussian random process with an effective variance $\bar{\sigma}^2 = \sigma^2/\delta t \delta V$. Note the dependence of the discritization scale. It gets larger on finer and finer scales, so

if we look very closely at the field it will be dominated by these fluctuations.

Since this Gaussian random variable is equivalent to a process in which we draw from the standard unit variance distribution and then multiply by $\bar{\sigma}$, we can write the fluctuation term as $\xi(\mathbf{r}, t) = \bar{\sigma}\eta(\mathbf{r}, t)$ where $\eta(\mathbf{r}, t) \sim \mathcal{N}(0, 1)$. Now what about pinning down the dependence on temperature and viscosity?

Fluctuation-Dissipation

That the fluctuations must be related to both the temperature and viscosity of the heat bath is intuitively clear, but to make the relationship quantitatively explicit requires a story. The story is both simple and beautiful, and starts off, like most stories in physics, with a room full of tiny noninteracting balls bouncing around. Now walls are pesky things at times, so to make things simple, we let the room be the Universe, and then they're no longer much of a problem. Denoting the position of the i^{th} ball at time t as $x_i(t)$, Newton's laws give the equation of motion $m\ddot{x}_i = 0$, the balls fly in straight lines at constant velocity... forever.

Things are pretty boring in this Universe, so we decide to spice things up by filling the Universe with another set of balls... microballs, each one infinitesimal in size compared to the balls of the first set, allowing these to interact with those and one another. The self interaction allows energy to be exchanged so that equipartition holds: thus the microballs have an associated temperature, T. There are so many of them that the first set of balls feels as if it's immersed in a fluid. A ball from the first set tries to plow forward, but ends up hitting countless multitudes of the microballs, slowly losing speed as it's kinetic energy is stolen by the microtheives. Thus the set of microballs has an associated viscosity, γ. I think you see where I'm going with

this: the set of microballs IS a heat bath.

Our original balls now follow a different equation of motion known as a Langevin equation:

$$m\frac{dv_i}{dt} = -\gamma v_i + \xi_i \quad \text{where } v_i = \frac{dx_i}{dt} \tag{4.42}$$

The last term represents the collisions with heat bath microballs.

A reasonable assumption on this stochastic component of the equation of motion is that the collisions be isotropic and homogenous, $\langle \xi \rangle = 0$, where the bracket represents an ensemble average over all the balls. With this assumption, we get from equation 4.42 that any bulk motion of the original balls will decay to $\langle v \rangle = 0$ in a characteristic time m/γ. In fact, if we had ignored the stochastic term, then every ball would settle down to rest in that time. However, this is not the case since constant collisions with the heat bath microballs will ensure that every ball retains some small velocity. By equipartition, we expect the ensemble average of the velocities to read $\langle v^2 \rangle = \frac{d}{m}kT$, where T is the temperature of the heat bath.

Let us now solve 4.42 by using an integration factor of the form $I = e^{\frac{\gamma}{m}t}$. Assuming that all transients have died out the solution is:

$$v_i(t) = \frac{1}{m}\int_{-\infty}^{t} \mathrm{d}t' \; e^{-\frac{\gamma}{m}(t-t')}\xi_i(t') \tag{4.43}$$

Taking the ensemble average of the velocities

$$\begin{aligned}\left\langle v^2(t)\right\rangle &= \frac{1}{Nm^2}\sum_{i=1}^{N}\int_{-\infty}^{t}\mathrm{d}t'\int_{-\infty}^{t}\mathrm{d}t'' e^{-\frac{\gamma}{m}(t-t')}\xi_i(t')e^{-\frac{\gamma}{m}(t-t'')}\xi_i(t'') \\ &= \frac{1}{m^2}\int_{-\infty}^{t}\mathrm{d}t'\int_{-\infty}^{t}\mathrm{d}t'' e^{-\frac{\gamma}{m}(2t-t'-t'')}\left\langle\xi_i(t')\xi_i(t'')\right\rangle \\ &= \frac{\sigma^2}{2\gamma}\end{aligned} \tag{4.44}$$

By equipartition we must have then that $\sigma^2 = d\gamma kT$, a famous result known as the fluctuation-dissipation theorem. This rather hand wavy argument can be made more precise using the methods of the Stochastic calculus, in which it can be derived from the Fokker-Planck equation for the full probability distribution of states of the balls. For this more in depth approach see [65] and for a good review of stochastic methods see [35].

4.2.4 Towards Equilibrium

To incorporate the heat bath, we need to look not at the Hamiltonian, but the free energy

$$\mathcal{F}[\phi] = \int \mathrm{d}^d\vec{r}\,\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{a_1}{2}\phi^2 + \frac{a_2}{4}\phi^4 - TS[\phi] \tag{4.45}$$

where the last term represents the entropy, and represents the coupling to the heat bath. The field begins out of equilibrium and evolve according to the Langevin equation, exchanging energy with the heat bath. As equilibrium is approached the kinetic energy term will approach 0. This does not mean the field will have no velocity, only that it will have 0 mean velocity. At this point the total energy will be bounded

from below by the Landau free energy, where the entropic contribution acts to create a temperature dependence on the coefficients in the potential:

$$\min \mathcal{H} \leq \mathcal{F} = \int \mathrm{d}^d\vec{r}\, \frac{1}{2}(\nabla\phi)^2 + \frac{a_1(T)}{2}\phi^2 + \frac{a_2(T)}{4}\phi^4 + \mathcal{O}[\phi^6] \tag{4.46}$$

The temperature dependence of the coefficients allows them to change sign, just as we saw with the Ising model.

4.3 The Simulation

4.3.1 Dimensionalization and Discretization

The field evolves according to a Langevin equation,

$$\ddot{\phi} = -\gamma\dot{\phi} + \nabla^2\phi - a_1\phi - a_2\phi^3 + \xi \tag{4.47}$$

To simplify numerics we first introduce a change of variables in order to make the equations dimensionless. Letting

$$x^\mu \to \frac{1}{\sqrt{a_1}}x^\mu \tag{4.48}$$

$$\phi \to \sqrt{\frac{a_1}{a_2}}\phi \tag{4.49}$$

where the transformed coordinates and field are now dimensionless. This simplifies the Hamiltonian into a dimensionless form:

$$\mathcal{H}[\phi] = \frac{a_1^{\frac{4-d}{2}}}{a_2}\int \mathrm{d}^d\vec{r}\, \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}\phi^2 + \frac{1}{4}\phi^4 \tag{4.50}$$

The transformations 4.48 and 4.49 also affect the heat bath, transforming the viscosity and temperature of the heat bath as

$$\gamma \to \frac{1}{\sqrt{a_1}}\gamma \tag{4.51}$$

$$T \to \frac{a_1^{\frac{4-d}{2}}}{\sqrt{a_2}}T \tag{4.52}$$

The appearance of the spatial dimensionality here comes from the fluctuation-dissipation relation depending on the discretization volume. With these dimensionless variables, the Free energy now reads:

$$\mathcal{F}[\phi] = \frac{a_1^{\frac{4-d}{2}}}{a_2}\int \mathrm{d}^d\vec{r}\,\frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}(\frac{T}{T_c}-1)\phi^2 + \frac{a_2}{4}\phi^4 + \mathcal{O}[\phi^6] \tag{4.53}$$

where the transformation parameters have consorted with one another to introduce a critical temperature, T_c. Lastly the equations of motion are now:

$$\ddot{\phi} = -\gamma\dot{\phi} + \nabla^2\phi - \phi + \phi^3 + \xi \tag{4.54}$$

The field will obey these equations of motion, which are simply the equations of motion of the 0-temperature Hamiltonian, coupled to a heat bath. This coupling will cause the system to evolve according to an effective Hamiltonian which seeks to minimize the Free Energy.

Lastly, rather than using a single second order equation of motion, we rewrite as

two first order equations:

$$\dot{\psi} = -\gamma\psi + \nabla^2\phi - \phi + \phi^3 + \xi \tag{4.55}$$

$$\dot{\phi} = \psi \tag{4.56}$$

We call ψ the field velocity.

We discretize the spacetime volume into tiny cells of size $\Delta t \Delta x^d$. Each leaf of the temporal foliation is labeled with a discrete t, and the leaves themselves are discretized as a rectangular lattice with grid spacing Δx. For a given leaf, the the field at a particular lattice point is denoted by $\phi(\vec{r}_i, t) = \phi^t_{i_1,\ldots,i_d}$. Now what?

4.3.2 A Short Course on Numerics

Consider a one dimensional function. How do we write approximate the derivative of the function on a discretized domain? Denote $f_n = f(n\Delta x)$, and the k^{th} derivative $f_n^{(k)}$, then we have:

$$f_{n+m} = f(n\Delta x + m\Delta x) = \sum_{k=0}^{\infty} \frac{1}{k!} f_n^{(k)} m^k \Delta x^k \tag{4.57}$$

We have just used the Taylor expansion of f at $n\Delta x$. Now choose a set of coefficents a_m^k, to be determined, and write

$$f_n^{(k)} = \sum_m a_m^k f_{n+m} = \sum_{k'=0}^{\infty} f_n^{(k')} \sum_m \frac{1}{k'!} m^{k'} \Delta x^{k'} a_m^k$$

$$\Downarrow$$

$$\sum_m \frac{1}{k'!} m^{k'} \Delta x^{k'} a_m^k = \delta_{k,k'} \tag{4.58}$$

This is just a set of linear equations that, given an infinite number of grid points could be solved for the set of coefficients. Thus we could represent any derivative as a linear combination (perhaps infinite) of function values at different points. Unfortunately the resources needed to encode a lattice are finite, and a simulation can have up to N grid points. We can therefore we cannot get the entire set of coefficients. Even using all N points, however, is not required. The error in truncating the series will be proportional to some power of the grid spacing, which for fine enough grids, will be small indeed. We refer to the set of points being used to approximate the derivative as a *stencil*.

4.3.3 Difference Methods

If we use a forward or backward stencil, for the first derivative, the resulting approximations are:

$$f_n^{(1)} = \frac{1}{\Delta x} f_{n+1} - \frac{1}{\Delta x} f_n - \frac{1}{2} f_n^{(2)} \Delta x = D_+ f_n + \mathcal{O}[\Delta x] \tag{4.59}$$

$$f_n^{(1)} = \frac{1}{\Delta x} f_n - \frac{1}{\Delta x} f_{n-1} + \frac{1}{2} f_n^{(2)} \Delta x = D_- f_n + \mathcal{O}[\Delta x] \tag{4.60}$$

Both of these stencils have the same size error, proportional to Δx, but of opposite sign. If we take the average of the two methods, $D_0 = \frac{1}{2}(D_+ + D_-)$ we get the central difference method. This has an even smaller error:

$$f_n^{(1)} = \frac{1}{2\Delta x} f_{n+1} - \frac{1}{2\Delta x} f_{n-1} + \frac{1}{3} f_n^{(3)} \Delta x^2 = D_0 f_n + \mathcal{O}[\Delta x^2] \tag{4.61}$$

Note how calculating all three of these requires the same number of function calls (2), however by choosing the right stencil, we can decrease the error in our approximation

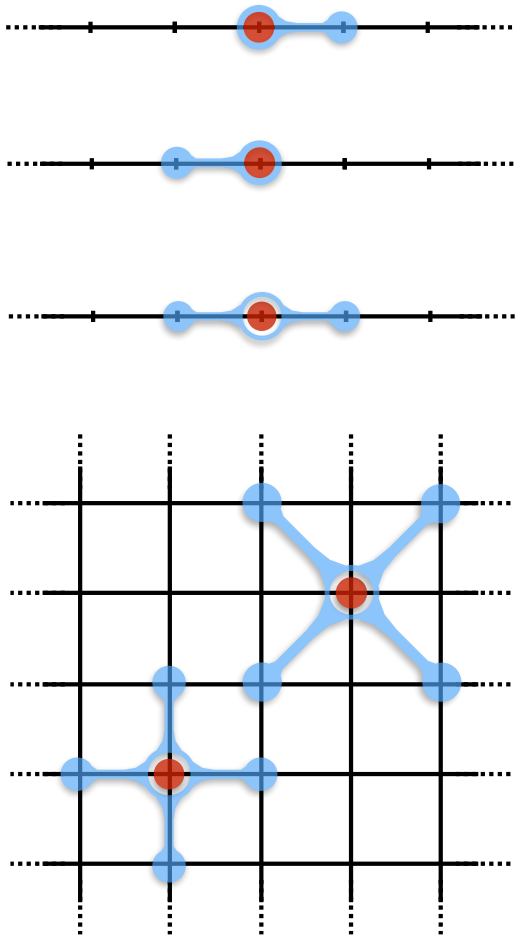

Figure 4.1: Top Panel: Three examples of the stencils discussed in this section. The top panel shows the Forward Difference stencil, which uses the values(blue) of the function at n and $n+1$ to approximate the derivative(red) at n. Similarly, the middle panel shows the backward difference method, and the bottom panel the Central Difference method.Bottom Panel: Two examples of stencils on a two dimensional lattice. Both break isotropy by picking out special directions. The maximally rotationally symmetric stencil at this order is a linear combination of these two.

method. We could, of course, have decreased the error by simply using larger stencils. This, however, would have lead to more function calls, sucking up computational resources. There is a trade off between accuracy and speed, so finding a stencil that affects one of those without touching the other is a thing to strive for when designing an approximation to derivatives.

Though central differences are appropriate for spatial derivatives, they cannot help us for solving equations in time. This is because they require future information at $t+1$ to approximate the time derivative at t. Backward difference methods are the choice we make for temporal derivatives.

What about higher order derivatives? Consider now the second derivative with a central stencil going out to nearest neighbors. We find:

$$\begin{aligned} f_n^{(2)} &= \frac{f_{n+1} - 2f_n + f_{n-1}}{\Delta x^2} + \mathcal{O}[\Delta x^2] \\ &= \frac{2}{\Delta x^2}\left(\frac{f_{n+1} + f_{n-1}}{2} - f_n\right) + \mathcal{O}[\Delta x^2] \end{aligned} \tag{4.62}$$

We have rewritten the central difference in an illuminating form. The second derivative is the difference between the value of the function at n and the average of the function around n. This is an interesting way to think about the Laplacian, though it does not hold for larger stencils.

4.3.4 Spatial Operators as Convolutions

We have written derivatives in terms of values of the function at neighboring points, weighed by coefficients. This is not unlike performing a convolution in the continuum:

$$\sum_m a_m^k f_n \simeq \frac{d^k}{dx^k} f(x) = \int \mathrm{d}x' \; K_k(x - x') f(x') \tag{4.63}$$

Here K_k is the *convolution kernel* for the k^{th} derivative, and we will refer to the coefficients a_m^k as the *convolution kernel* when we return to discritization. We can use this correspondence to guess good derivative approximations. To find the kernel we write f and K in terms of their Fourier transforms:

$$f(x) = \int_{-\infty}^{\infty} \mathrm{d}k \; \tilde{f}(k) e^{2\pi i k x} \qquad K(x) = \int_{-\infty}^{\infty} \mathrm{d}k \; G(k) e^{2\pi i k x} \tag{4.64}$$

Upon inserting 4.64 into 4.62 we find for the LHS:

$$\frac{d^m}{dx^m} f(x) = \int_{-\infty}^{\infty} \mathrm{d}k \; (2\pi i k)^m \tilde{f}(k) e^{2\pi i k x} \tag{4.65}$$

and for the RHS

$$\begin{aligned}
\int_{-\infty}^{\infty} \mathrm{d}x' K_k(x - x') f(x') &= \int_{-\infty}^{\infty} \mathrm{d}x' \int_{-\infty}^{\infty} \mathrm{d}k \; \tilde{G}(k) e^{2\pi i k (x - x')} \int_{-\infty}^{\infty} \mathrm{d}k' \; \tilde{f}(k') e^{2\pi i k' x'} \\
&= \int_{-\infty}^{\infty} \mathrm{d}k \; G(k) e^{2\pi i k x} \int_{-\infty}^{\infty} \mathrm{d}k' \tilde{f}(k') \int_{-\infty}^{\infty} \mathrm{d}x' e^{2\pi i (k' - k) x'} \\
&= \int_{-\infty}^{\infty} \mathrm{d}k \; G(k) e^{2\pi i k x} \int_{-\infty}^{\infty} \mathrm{d}k' \tilde{f}(k') \delta(k' - k) \\
&= \int_{-\infty}^{\infty} \mathrm{d}k \; G(k) \tilde{f}(k) e^{2\pi i k x}
\end{aligned} \tag{4.66}$$

Equating the two we must have that

$$G(k) = (2\pi i)^m k^m \tag{4.67}$$

So the convolution kernel is

$$\begin{aligned} K_m(x) &= \int_\infty^\infty \mathrm{d}k\ (2\pi i)^m k^m e^{-2\pi i k x} \\ &= (-1)^m \frac{d^m}{dx^m}\delta(x) \\ &= \lim_{\alpha\to\infty} (-1)^m \frac{d^m}{dx^m}\frac{\sin \pi\alpha x}{\pi x} \\ &= \lim_{\alpha\to\infty} \alpha^{m+1} b_m(\alpha x) \end{aligned} \tag{4.68}$$

The second line is clear from the definition, while in the third line we have used a particular limiting form for the Dirac Delta. The last line has rewritten the derivative in terms of an evaluated function. The trick now is to write things back in discretized form, and use $x = n\Delta x$, and $\alpha = 1/\Delta x$

$$\begin{aligned} f_n^{(k)} &\simeq \Delta x \sum_m K_k(m\Delta x) f_{n+m} \\ &= \sum_m \frac{b_k(m)}{\Delta x^k} f_{n+m} \end{aligned} \tag{4.69}$$

Voila! We have a correspondence between the convolution kernel and the derivative. For the second derivative we have

$$f_n^{(2)} = \frac{2}{\Delta x^2}\left(\cdots - \frac{1}{4}f_{n+2} + f_{n+1} - \frac{\pi^2}{6}f_n + f_{n-1} - \frac{1}{4}f_{n-2} + \cdots\right) \tag{4.70}$$

Unfortunetely, truncating this anywhere introduces errors at the zeroth order. The problem comes from the original definition of the kernel being nonvanishing over all of space. One can attempt to redo the calculation by setting a finite domain around each point, though this becomes much harder to attack analytically. The point here is that the convolution kernel of a finite difference method is doing for a grid what a Green's function does for the continuum.

4.3.5 The Laplacian

Moving to higher dimensions, we consider the Laplacian of a function. On the lattice we approximate the Laplacian with a multidimensional convolution kernel:

$$\nabla^2 f_{\vec{n}} = \sum_{i=1}^{d} \frac{\partial^2 f_{\vec{n}}}{\partial x_i^2} \simeq \sum_{\vec{i}} L^{\vec{i}} f_{\vec{n}+\vec{i}} \tag{4.71}$$

where the sum is over all the spatial directions.

The problem that arises with more than one dimension is the kernels inability to preserve rotational invariance, which stems from the fact that lattices are not rotationally invariant. The Laplacian is rotationally invariant, but the kernel is not, picking out directions between neighboring lattice points as being special. This breaking of isotropy causes simulated phenomena to align in these directions, introducing numerical artifacts that can contaminate simulations.

For example, in two dimensions one can consider both stencils in fig. 4.1. The

lower left and upper right have, respectively, stencils of the form:

$$L_+ = \frac{1}{\Delta x^2}\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad L_\times = \frac{1}{2\Delta x^2}\begin{bmatrix} 1 & 0 & 1 \\ 0 & -4 & 0 \\ 1 & 0 & 1 \end{bmatrix} \tag{4.72}$$

A general Laplacian can then be constructed as a convex linear combination of these two

$$\nabla^2 = \frac{1}{2}((1-\gamma)L_+ + \gamma L_\times) \tag{4.73}$$

The value for γ that maintains the most rotational symmetry is $1/3$ [68], so that the minimally nonisotropic finite difference Laplacian convolution kernel in two dimensions can be written as:

$$\frac{1}{12\Delta x^2}\begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix} \tag{4.74}$$

and has error of order $\mathcal{O}[\Delta x^4]$.

For our system we decided on using a lattice of size $L = 2^8$, with a grid spacing of $\Delta x = .25$. The resulting lattice has $N = 2^{20}$ lattice points.

4.3.6 The Leapfrog Method

With the Laplacian 4.74, we can now fully discritize the equations of motion 4.55. We use the leapfrog method, updating the field and its velocity at half time steps

apart. The updating equation for the velocity is

$$\begin{aligned}
\psi_{ij}^{t+1/2} &= \psi_{ij}^{t-1/2} + \dot{\psi}_{ij}^{t}\Delta t \\
&= \psi_{ij}^{t-1/2} + \Delta t\left(-\frac{\gamma}{2}(\psi_{ij}^{t+1/2}+\psi_{ij}^{t-1/2}) + \sum_{nm}[\nabla^2]^{nm}\phi_{i+n,j+m}^{t} - \phi_{ij}^{t} + (\phi_{ij}^{t})^3 + \sqrt{\frac{2\gamma T}{\Delta t \Delta x^2}}\right) \\
&\Downarrow \\
\psi_{ij}^{t+1/2} &= \frac{1-\gamma\frac{\Delta t}{2}}{1+\gamma\frac{\Delta t}{2}}\psi_{ij}^{t-1/2} + \frac{\sqrt{2\gamma T}}{\Delta x}\eta_{ij}\frac{\Delta t^{1/2}}{1+\gamma\frac{\Delta t}{2}} \\
&\qquad + \left(\sum_{nm}[\nabla^2]^{nm}\phi_{i+n,j+m}^{t} - \phi_{ij}^{t} + (\phi_{ij}^{t})^3\right)\frac{\Delta t}{1+\gamma\frac{\Delta t}{2}} \qquad (4.75)
\end{aligned}$$

Note that in the equation of motion for the velocity, there is a velocity present. Since all the fields are at time t in this equation, we need to write the velocity on the same time slice. This is accomplished by averaging the value of the field on the velocity fields half a time step behind and half a time step ahead of the present time step. This may seem strange, since we are using future information to define a quantity in the present, but the method can be explicitly solved for the future time. Meanwhile, for the field itself, we have

$$\begin{aligned}
\phi^{t+1} &= \phi^{t} + \dot{\phi}^{t+1/2}\Delta t \\
&= \phi^{t} + \psi^{t+1/2}\Delta t \qquad (4.76)
\end{aligned}$$

The scheme is displayed diagrammatically in figure 4.2

Choosing a good time step is crucial for the stability of the method. The size of the time step and the lattice spacing must satisfy the Courant Condition in order for numerical solutions to not diverge from analytical solutions. A neat way to think

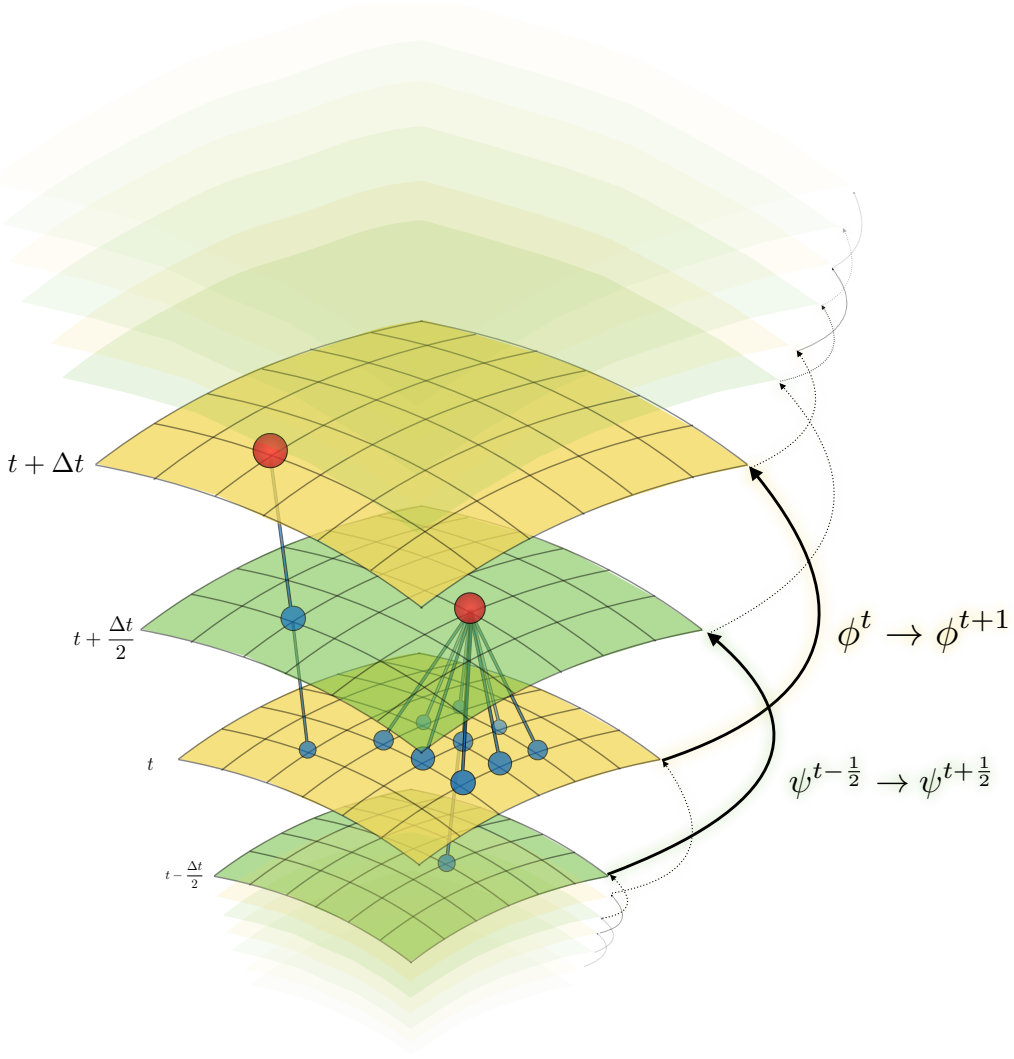

Figure 4.2: The leapfrog method works by evolving the field and its velocity at half time steps apart. Shown on the image is the general idea, as well as the stencils used to update both quantities. The velocity update requires information from the Laplacian of the field, and therefore needs more resources to run. The field then updates based on the updated velocity.

about the Courant condition is in terms of causality and light cones. Stencils represent a causal structure for the evolution of the field: they tell a given lattice points what lattice points in the past are directly causing its value in the present. If information travels at a finite speed, c, in the continuum, then this means that only points within the past light cone of an event can affect its current value. On the grid information travels at a speed Size[Stencil]$/\Delta t \sim \Delta x/\Delta t$, which defines the grid information light cone. If the grid information light cone is contained within the continuum light cone, then that means that the stencil is insufficient at relaying information fast enough on the grid to approximate the equations of motion. Hence, at the very minimum we must have that $\Delta x/\Delta t \geq c$ for points to update properly. This means that the size of the time spacing is bounded by the lattice spacing: $\Delta t \leq \Delta x/c$.

For analysis of our system to determine the wave equation is intractable analytically, even without the stochastic terms. For a standard diffusion equation we know that we must choose no larger time spacing than $\Delta t \sim \Delta x^2$. However, the stochastic terms actually end up helping us out in our case at finite temperatures, since they are much larger than rounding errors. We ended up choosing $\Delta t = \Delta x/10$.

4.4 Analysis

4.4.1 Raw Data

Our goal will be to examine the configurational entropy at equilibrium. Getting the system into an equilibrium state is tricky because of the structure of the vacuum manifold. We have seen that the free energy of the model has a critical temperature below which the vacuum consists of two field values. If we want to evolve the system

to equilibrium at a temperature below the critical, we could do so by starting it off in some arbitrary state and evolving it under the equations of motion until it naturally minimizes its free energy, a process known as a quench. Unfortunately as the system loses energy, different regions of space may end up falling into different vacua. This will lead to the formation of domain walls via the famous Kibble-Zurek mechanism [60],[59].

The presence of domain walls would be ok if they eventually decayed away. Periodic boundary conditions, however, lead to stripe states which at any finite temperature are VERY long lived [9]. These structures contribution to the two point correlation function dominate over the background fluctuations, and would therefore contaminate measurements of the background. To get around this from happening we need to either begin the field very close to its equilibrium state, which is very hard, or start the system at a very low temperature at the minimum of the Hamiltonian, and slowly raise the temperature quasi adiabatically, which is the option we choose.

To get to our target temperature T we initialize a random field at $T_0 = 0$, $\phi^0_{ij} = 1+\eta_{ij}, \psi^0_{ij} = 0$, where the η's are drawn from a normal distribution. The temperature is changed along a logistic trajectory for τ time steps:

$$T_t = \frac{e(e^2 - e + 1)}{(e-1)(e^2+1)} \frac{e^{\frac{4t}{\tau}} - 1}{e^{\frac{4t}{\tau}} + e} T \tag{4.77}$$

This trajectory uses half the time steps to change the temperature from T_0 to approximately T, and the remaining timesteps to ensure that equilibration of energy occurs. We chose $\tau = 1500$. To ensure good statistics, an ensemble of $\mathcal{N} = 128$ systems was used for a given temperature T.

The estimators for the mean and variance of the run were calculated:

$$\bar{\phi} = \frac{1}{N^2}\sum_{i,j}\phi_{ij} \quad \text{and} \quad \bar{\phi^2} = \frac{1}{N^2}\sum_{ij}(\phi_{ij} - \langle\phi\rangle)^2 \tag{4.78}$$

The ensemble means and variances of both the mean and variance were then calculated. The latter was divided by the temperature to get the susceptibility

$$\begin{aligned} \langle\phi\rangle = \frac{1}{\mathcal{N}}\sum\bar{\phi} \quad \text{and} \quad \sigma_\phi^2 = \frac{1}{\mathcal{N}}\sum(\sqrt{\bar{\phi^2}} - \langle\phi\rangle)^2 \\ \chi(T) = \frac{1}{T\mathcal{N}}\sum\bar{\phi^2} \quad \text{and} \quad \sigma_\chi^2 = \frac{1}{\mathcal{N}}\sum(\bar{\phi^2} - T\chi(T))^2 \end{aligned} \tag{4.79}$$

The susceptibility is useful at pinpointing the critical temperature because it diverges there in the thermodynamic limit.

To locate the critical temperature we then subdivided the temperature range $[.01, .94]$ into 64 into equally spaced intervals, and calculated the above quantities at each temperature. Finding the maximum in the susceptibility, we then further subdivided the intervals around it to get 6 more temperature values. The results are shown in figure 4.3.

The critical strip is defined as the region between the inflection point of of the mean field curve and the maximum of the susceptibility. A proper scaling analysis was done for smaller lattices, shown in figure 4.4. We are thus confident that the critical temperature lies in the temperature range $[.45, .483]$.

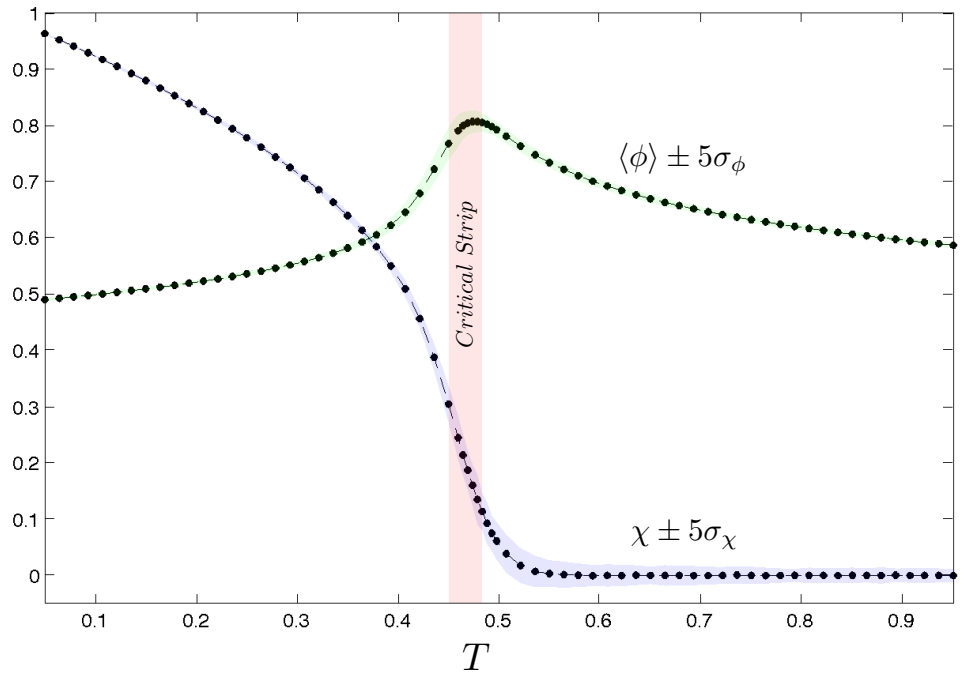

Figure 4.3: The ensemble mean(blue) and susceptibility(red) plotted with variances coming from an ensemble size of 128 runs. The standard deviations are displayed with decreasing color intensity out to 5σ's. The smallness of these errors is indicated by the tightness of the color bars around the curves. The maximum in the susceptibility and the inflection point in the mean field give us a good estimate for the location of the critical temperature, and we denote this region the critical strip.

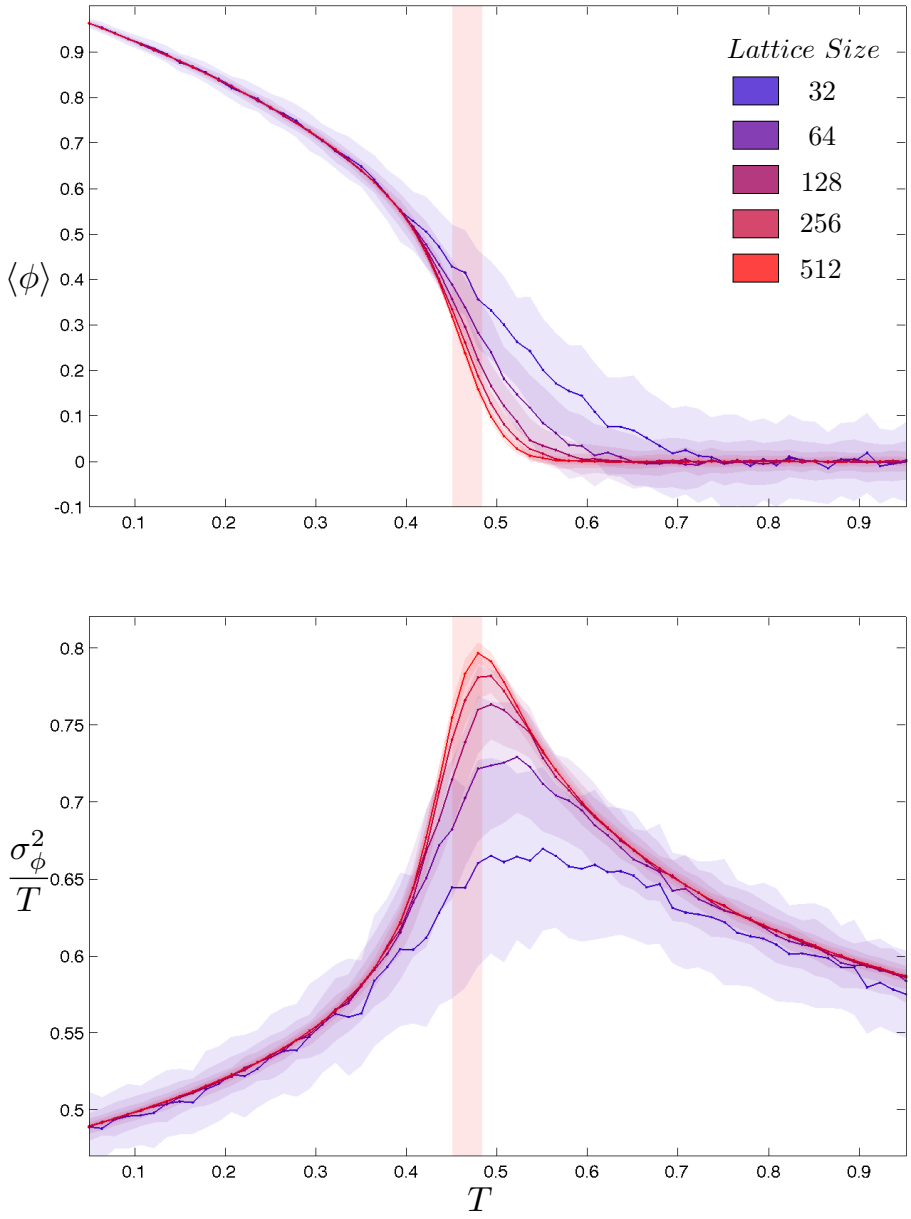

Figure 4.4: A scaling analysis performed on four smaller lattices indicates that the critical temperature is being approached in the thermodynamic limit. The top panel is the mean field and the bottom panel is the susceptibility.

4.4.2 Analysis of Fluctuations

To ensure equilibrium we calculate the average kinetic energy per lattice point, and find only small deviations from equilibrium $\left\langle \frac{1}{2}\psi^2 \Delta x^2 \right\rangle = \frac{1}{2}T$. We plot the fractional deviations in figure 4.5. Deviations from equipartition are no greater than 6% in all of our run.

To understand how the fluctuations are distributed at each temperature we cut up the field domain into 500 equally spaced values for each temperature run, and generate a frequency count which we normalize to give us a probability distribution for field values laying in some interval $[f, f + df]$, at a given temperature, $p([\phi, \phi + \Delta\phi], T) = \rho(\phi, T)\Delta\phi$ and $p([\psi, \psi + \Delta\psi], T) = \rho(\psi, T)\Delta\psi$.

The distributions for ϕ and ψ look very Gaussian, so we decided to look at how close they are by using the Kullback-Liebler divergence, 3.11. The velocity field has a divergence of ~ 4 parts in 10^7, so is pretty nearly gaussian, as was expected since the field is in equilibrium at each T. The field fluctuations, however, deviate from Gaussians. Their KL-divergence is plotted in Figure 4.6. The deviations away from Gaussianity for $p(\phi, T)$ are no greater than 4 parts in 10^3 for ϕ. These deviations come from the quartic contribution to the potential. For temperatures below the critical temperature, the asymmetry around the minimum of the effective potential introduces some kurtosis to the distribution. Fluctuations are more likely to point towards the origin than they are away. For higher temperatures, the symmetry of the effective potential is restored, but the deviation from a harmonic potential introduces a small flattening of the field distribution.

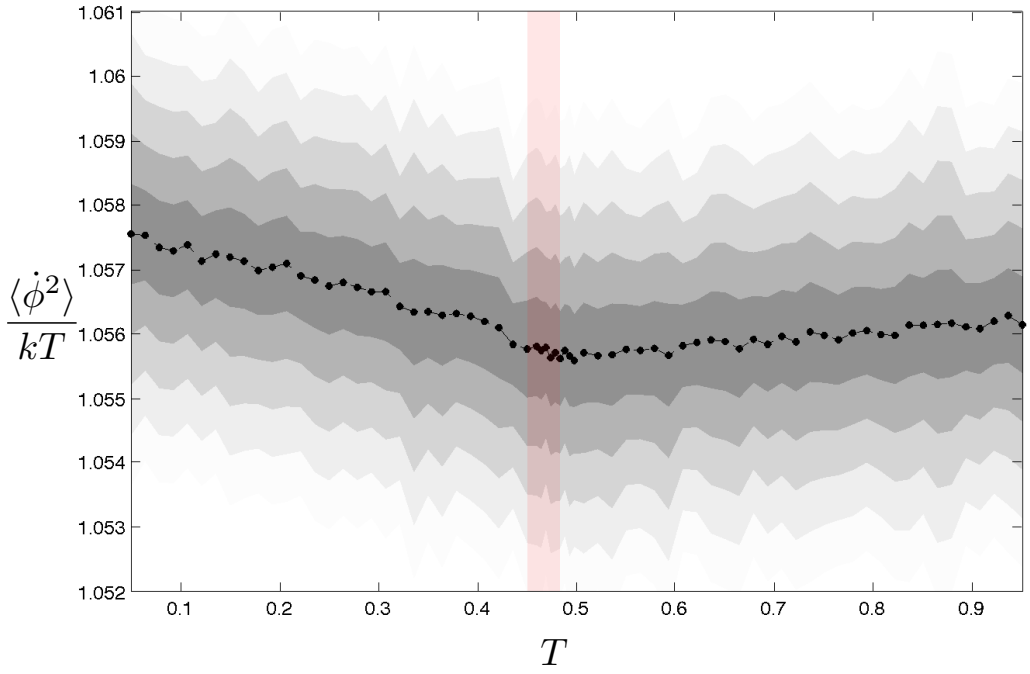

Figure 4.5: The deviation from equipartition of the field. Note the scale of the vertical axis. Deviations from equipartition are no greater than 6% for all of the runs we did.

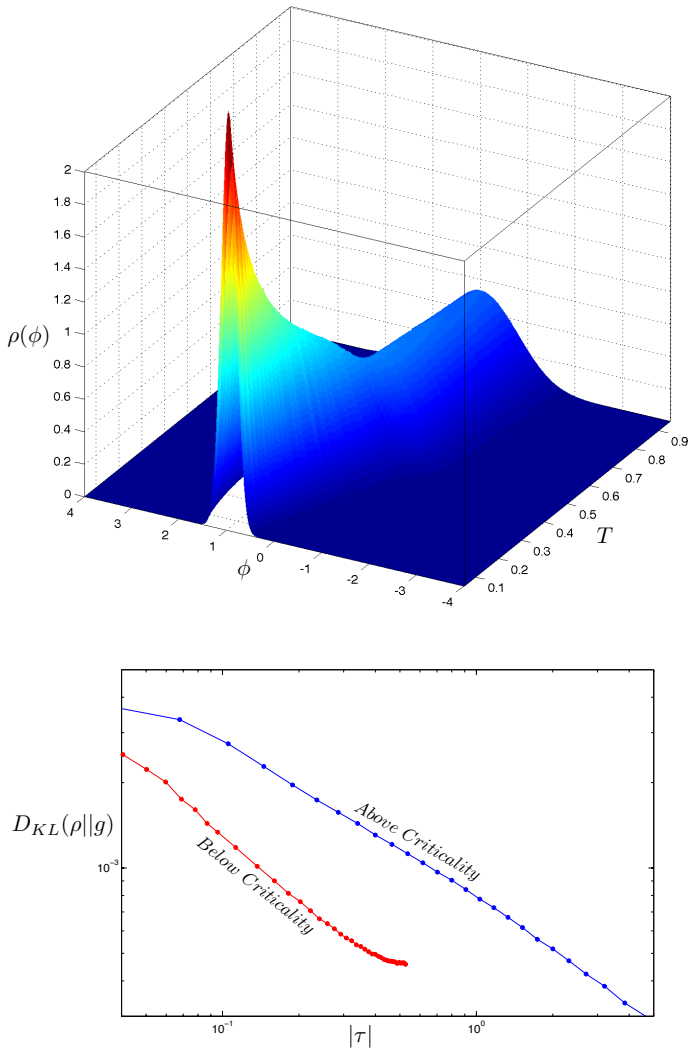

Figure 4.6: The field value probability distribution is on the left. As the temperature is increased the mean of the distribution moves towards 0, and spreads out. Though the distributions at each temperature may look Gaussian they are not. Their KL divergence around criticality is plotted against τ in the upper right panel. There is a clear distinction between the field distributions above and below criticality. We plot the difference between the actual distribution and a gaussian with the same moments for two temperatures in the bottom right panel. You can see that the differences for distributions below criticality is asymmetric, whereas it is symmetric for those above. For cold distributions this is due to the asymmetry of the potential away from 0, so fluctuations are more probable towards the origin than they are away. For hot distributions the quartic contribution to the potential flattens the distribution ever so slightly, causing the deviation from a Gaussian.

4.4.3 Power Spectra and Correlation Functions

Confident that our runs are producing data in line with theoretical expectations, we next moved to calculating correlators. Since a direct calculation of a 2-point correlation function is very costly, we use the fact that $C_2 = \mathcal{F}^{-1}|\mathcal{F}\phi|^2$. This allows us to extract the power spectrum as an intermediate step. Once we have both the 2-point correlation function and the power spectrum, we perform a radial average to get rid of any directional contributions.

By assuming the Ornstein-Zewike relation 3.2, the two point correlation function for $x \gg \Delta x$ can be modeled as:

$$C_2(r) \sim \frac{e^{-r/\xi}}{(r/\xi)^{\eta+d-2}} \tag{4.80}$$

were ξ is the correlation length. This form applies only close to the critical temperature. Very close to criticality the correlation length diverges, and we get an a power law. Farther from the critical point the correlation length gets smaller, and the suppressing exponential kicks in.

Correlation functions are calculated using DFTs, and imposing a cutoff in momentum space at 16 times the Nyquist frequency, just to be sure we weren't capturing any numerical artifacts. This means that structures smaller than a size of $8\Delta x$'s do not get resolved. We fit our data to the functional form 4.80 close to criticality, where we expect large correlation lengths. Fits for the correlation length do indicate a sharp rise up to about $30\Delta x$'s. The correlation length is then tested for its dependence on the reduced temperature, $\tau = T/T_c - 1$, namely $\xi \sim |\tau|^\nu$. These fits are not as easy to perform because they are very sensitive to how many points are included. The more

the better, but the farther we go from $\tau = 0$, the less we trust the power law fit that got us the correlation length. For the critical exponent above and below the critical temperature we found $\nu_- = -.52 \pm .09$ and $\nu_+ = -.81 \pm .06$, the above exponent consistent with the mean field theory value of $\nu = .5$. Results are displayed in figure 4.7

Fits for the values of η are displayed in figure 4.8. At and above criticality the value is consistent with mean field theory, $\eta = \frac{1}{4}$, though below criticality we do see a deviation. This is the same region where our fit for ν was off from mean field theory. We attribute this to parameter degeneracy between the correlation length and the critical exponent in the region where the two vacua are close to one another.

The power spectrum from equation 4.80 is

$$\mathcal{P} \sim \frac{\xi^{d-1}}{k} \left((1 - 2\pi i k\xi)^{(\eta-1)} - (1 + 2\pi i k\xi)^{(\eta-1)} \right) \tag{4.81}$$

It's good to look at the limits of this function for different size modes. Consider modes that are much larger than the correlation length, so that $k\xi \ll 1$. The spectrum can be expanded to as:

$$\mathcal{P} \sim \xi^d (1 - \frac{2\pi^2}{3} \left(\eta^2 + 5\eta + 6 \right) (k\xi)^2 + \cdots) \tag{4.82}$$

The power in these modes is nearly constant, or scale invariant. Structure has not had a chance to form because length scales in this regime are much larger than the correlation length. Far distant points haven't had a chance to communicate with one another.

On the other end of the extreme we have wavemodes that are much smaller than

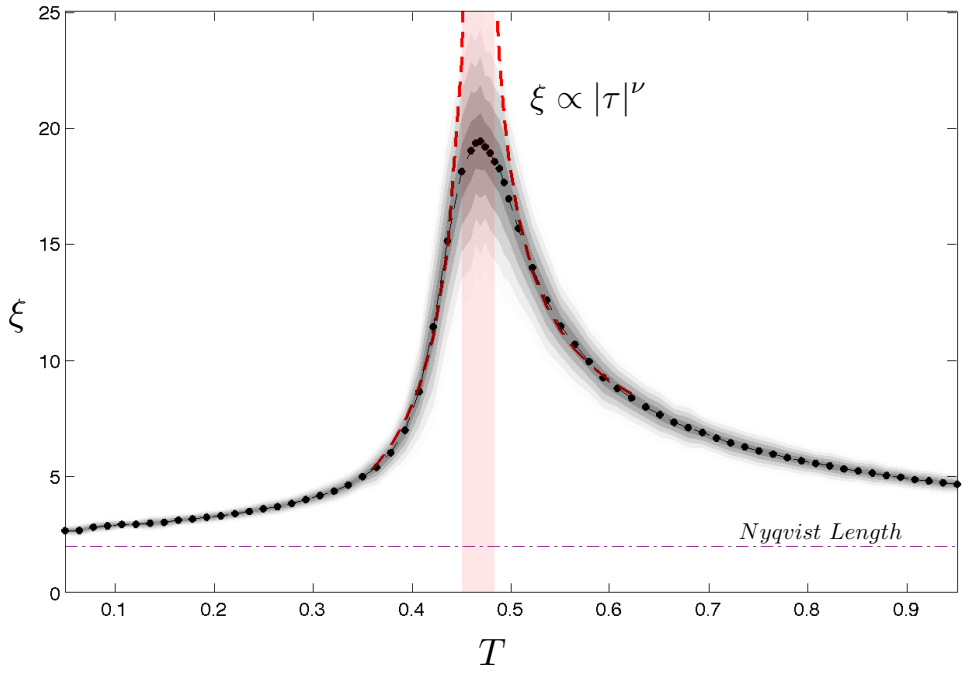

Figure 4.7: The correlation length is plotted for different temperatures. Red lines are fits for the correlation length dependence on τ. The dotted black line is the Nyquist length which is used as a cutoff in frequency space.

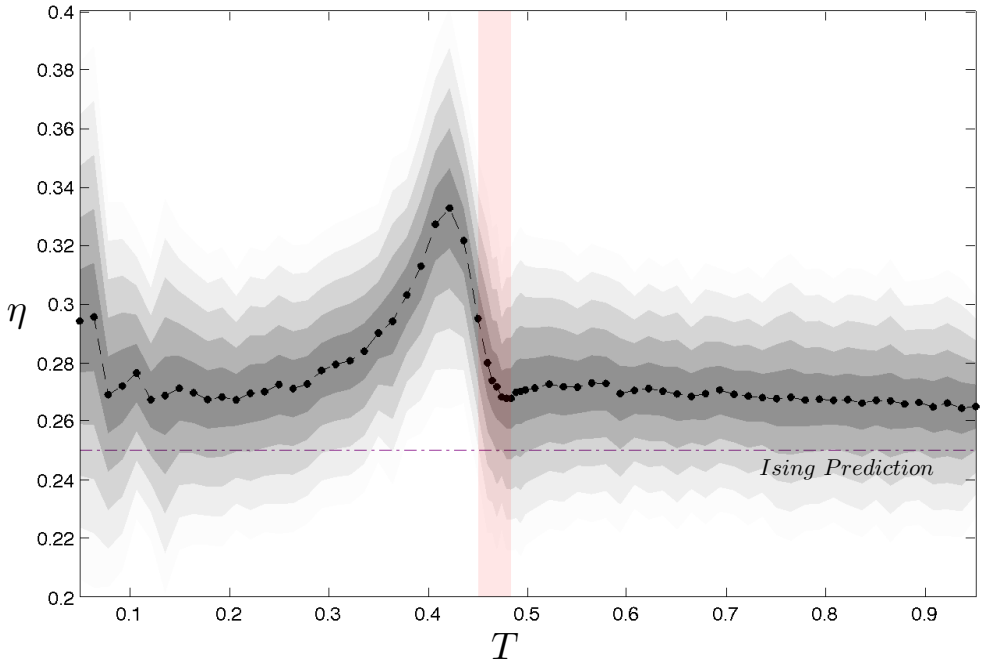

Figure 4.8: The values for the critical exponent at different temperatures. The dashed line represents the mean field theory prediction. Most of the range of values is consistent with mean field theory except in a region just below criticality. We attribute this deviation to parameter degeneracy in the functional form of the correlation function.

the correlation length scale, $k\xi \gg 2\pi$. Here it can be shown that the power spectrum obeys a power law of the form $k^{\eta-d}$, hence there is increasing power in decreasing length scales, indicative of the fact that the shorter the distances, the greater the correlations become.

4.5 Configurational Entropy

We use the power spectrum to construct the modal fraction, 3.8 at each temperature for each run in the ensemble, making sure to impose an ultraviolet cutoff due to the Nyquist frequency. The results are displayed in figure 4.9. A short detour into the white noise limit should be taken to understand the maximum CE on our lattice.

White noise has a two point correlation function which is a delta function, $C_2(r) = \sigma^2\delta(r)$. Points that are separated by non-zero distances are completely uncorrelated. [35]. This means that the Power spectrum is constant: all modes have equal power in them. The modal fraction of white noise is therefore just a uniform probability distribution, so the Configurational Entropy, using the discrete lattice definition, is:

$$S_C^{wn} = -\sum_{\Delta^d k} \frac{\Delta^d k}{V} \log \frac{\Delta^d k}{V} = \log N_{tot} = \log \pi (\frac{k_{max}}{k_{NQ}} N)^2 \tag{4.83}$$

For our lattice this is approximately 10.85. Numerical computations of the white noise CE limit seemed to indicate that our code was returning a value that was about .4 less than this. To test the accuracy of our simulations, we ran a noise analysis for small ensembles of several different sizes. Results are displayed in figure 4.10. The fractional error in the analytical and numerical results was seen to decrease with larger lattices, so we attribute the difference between the two as being a numerical

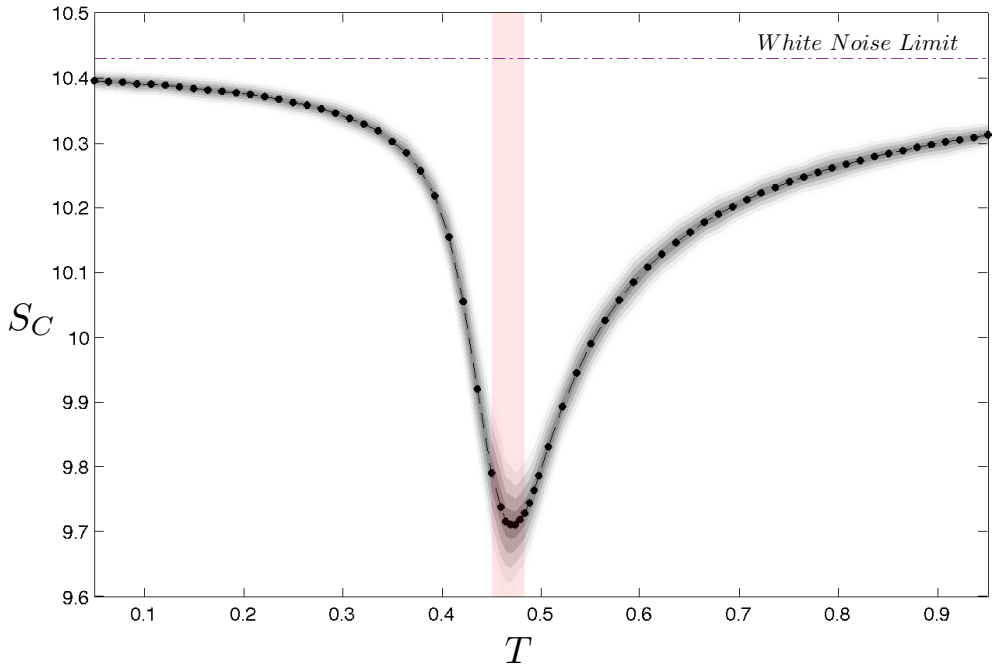

Figure 4.9: The configurational entropy for all the Landau model plotted against temperature. Close to criticality we have a a dip in the amount of CE relative to the maximal white noise limit. This indicates that the amount of hidden information concerning scale for an EA is decreasing sharply at criticality. The error bars go out to the 5σ level with decreasing color intensity.

artifact coming from the DFT.

The interpretation of this result needs a little bit of thought. Far from criticality, the correlation length is small, so the field is dominated by stochasticity in the Langevin equation, and looks very noisy. To an EA the noisy state is one which is maximally uninformative. Every mode is equally represented to the EA, so no particular scale jumps out at them.

Close to criticality, the correlation length grows as $|\tau|^{\nu}$. Starting at the smallest of scales, regions begin to become correlated, generating structure. Structures that grow beyond the correlation length scale may persist for a short period of time, but the stochastic kicks from the heat bath will serve to break apart any structure that gets much larger than the correlation length. The EA will recognize the appearance of coherent objects at these scales and her belief in *something interesting happening at a scale* ξ will start to become saturated relative to her belief in what is happening at larger scales. The strengthening of belief is accompanied by a certainty in events, and a corresponding decrease in the hidden information in the world. Configurational Entropy minimization at criticality is indicative of a transfer of information, once hidden in the world, into the EA.

To understand this further, let us look at the distribution of configurational hidden information present in all of the modes of the model.

4.5.1 Mode Structure

For each temperature T we calculated the CE density. Results for the CE are plotted in Fig. 4.9. The CE drops sharply in the critical strip where the phase transition occurs. There is increased spatial coherence at larger scales as criticality is approached,

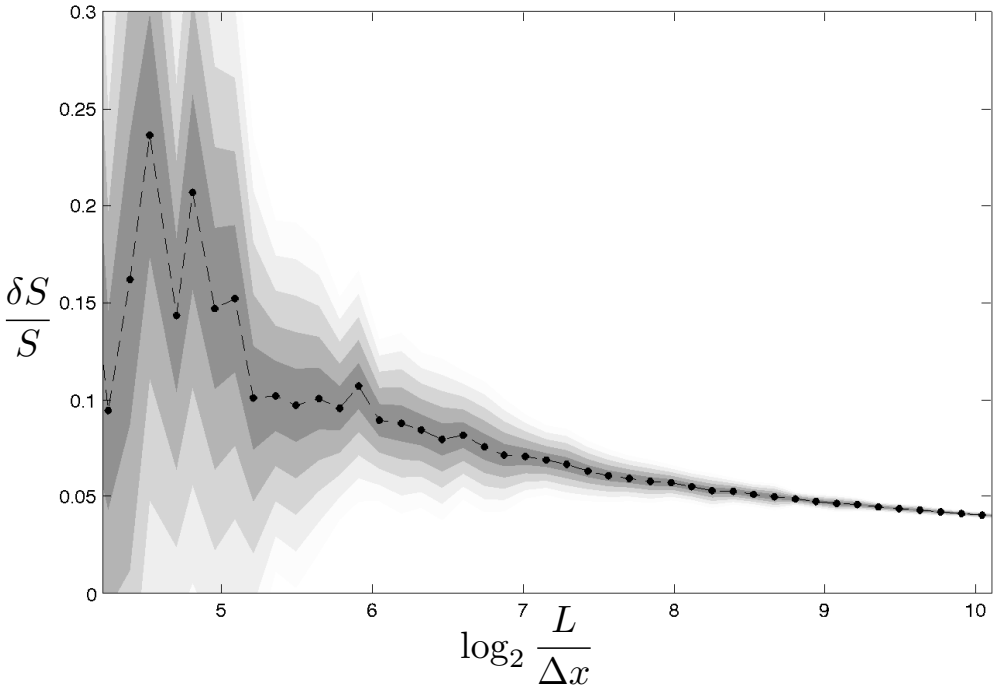

Figure 4.10: A scaling analysis of the CE of white noise comparing the difference between analytical and numerical CE, δS, to the analytical value, S.. We ran ensembles of size 32 on several lattices and calculated their CE. The figure shows the fractional CE for the ensemble with up to 5σ deviations. For larger lattices, the fractional CE is approaching zero, implying convergence to the analytical limit.

and thus a larger departure from the maximum entropic state ("white noise limit" line) characterized by a scale-free regime, as we show below. In other words, the critical point maximizes information about the spatial organization of the system. Compressibility is at a maximum, given that a few long wavelengths dominate the probability distribution: increased spatial coherence is equivalent to increased compressibility. *Minimizing CE implies minimizing shape complexity.*

This stored information reduces the uncertainty in the available phase space for the evolution of the system, preserving coherence and hence leading to the longevity of large domain structures. This is the information-theoretic narrative of the critical slowing down near T_C [42]. The CE is thus a measure of *information storage* within the context of phase transitions.

To explore information transfer between modes as the system cycles through different equilibrium states, we investigate the behavior of the CE density at different temperatures. As can be seen from Fig. 4.11 there are 3 regimes where different scaling laws, of the form $k^{-\sigma}$, hold:

(a) **Scale Free** ($\sigma = 0$): away from criticality and for low-k modes;

(b) **Turbulent** ($\sigma = 5/3$): at length scales close to the correlation length (denoted by a dashed vertical line);

(c) **Critical** ($\sigma = 7/4$): at or very near criticality and for scales smaller than the correlation length.

These regions shed light on the scales at which information is being stored and transferred, as we discuss next.

4.5.2 Scale-Free Regime

The scale-free region only applies for low wave numbers, which correspond to large spatial scales. As we have seen, in information theory the uniform probability distribution maximizes entropy. It carries no information, in that it predicts all outcomes are equally likely: the default position when nothing is known about a system. Similarly, the scale-free region implies that the uniform modal fraction at these scales is storing no information about the fluctuations. At different temperatures, fluctuations at the largest spatial length-scales, where the scale-free regime holds, are uncorrelated: information must therefore be stored elsewhere in the spectrum.

At low temperatures the CE clearly approaches the white noise limit, indicating that fluctuations get more and more uncorrelated at all scales. This seems reasonable since at low temperatures the correlation length is small and fluctuations are uncorrelated at most spatial scales. There is a departure from pure white noise, though, given that at any finite temperature there is a probability that some correlations at low wave numbers will be excited, a trend that grows as criticality is approached from below and from above.

Going back to the alphabet analogy, a system far from criticality is nearly incompressible. Equipower modes take the same number of bits to encode, so there isn't a more efficient way of representing them. The informational narrative far from criticality is then one of ignorance: the state of the system is nearly that of white noise.

4.5.3 Turbulent Regime

Searching for scaling laws as the temperature was increased, we discovered a surprising regime which obeys a power law identical to Kolmogorov turbulence in fluids. Unlike normal turbulence, which describes how an energy input at large spatial scales gets transported to smaller and smaller scales until it is dissipated away at the viscous length scale, the *informational turbulence* encountered here is a bottom-up phenomenon. Due to the nonlinear interactions of the order parameter and the free-energy cost of field gradients, small correlated domains tend to join together to become larger and larger. However, this trend cannot run away to organize the entire system because the scale-free fluctuations flowing into the system from the heat bath act to break them apart.

Dimensional analysis, generalized from Kolmogorov's original argument, can be used to find this scaling law. Within our standard formalism, CE is an information-theoretic measure with no units. We restore physical units by multiplying the CE by the Boltzmann constant, and dividing it by the mass of the system: $s_C = \frac{k_B}{M} S_C$. Note the units of this CE entropy-density: $\left[\frac{\partial s_C}{\partial k}\right] = [\text{Length}]^3\,[\text{Time}]^{-2}\,[\text{Temperature}]^{-1}$. Following Kolmogorov we assume that the flow of energy between scales per unit mass , $\dot{\epsilon}$, is scale-independent. This energy is supplied by the heat bath and flows to larger scales. Since in the thermodynamic limit the only other dimensionful constant is the critical temperature, we posit an *ansatz* of the form:

$$\frac{\partial s_C}{\partial k} = A T_C^{\alpha} \dot{\epsilon}^{\beta} k^{\gamma}, \tag{4.84}$$

where A is a dimensionless proportionality constant. Consistency demands that $\alpha =$

$-1,\ \beta = 2/3,\ \gamma = -5/3$, and we obtain the scaling law in the turbulent regime.

As criticality is approached, a resonant cooperation between deterministic and stochastic forces in the system drives information-carrying modes to larger and larger spatial scales. The turbulent regime determines the boundary of this resonant behavior, thus defining an information-flow boundary between the scale-free (minimal information) and the critical regime (maximal information) spliced by correlation-length sized fluctuations (vertical dashed line in Fig. 4.11).

4.5.4 Critical Regime

At criticality the effective temperature-dependent potential becomes quartic, with no curvature at the minimum. The corresponding divergence in the correlation length implies that fluctuations separated by large distances can exchange information, leading to larger correlated domains. Interpreting the CE as a measure of information storage, information is thus being maximally stored in these low-k ordered structures. In the infinite-volume limit, this scaling law would cover all modes ($k \to 0$).

To understand the scaling law recall the expressions for the two-point correlation function and power spectrum close to criticality,

$$\langle \phi\phi' \rangle \propto \frac{e^{-r/r_c}}{(r/r_c)^{\eta}} \xrightarrow{r_c \to \infty} \left(\frac{r}{r_c}\right)^{-\eta}; \tag{4.85}$$

$$\mathcal{P}(k) \propto \frac{r_c^{d-1} \sin\left[(d-1-\eta)\arctan kr_c\right]}{k(1+k^2 r_c^2)^{\frac{d-1-\eta}{2}}} \xrightarrow{r_c \to \infty} \frac{(kr_c)^{\eta}}{k^d}, \tag{4.86}$$

where η is the critical exponent of the Ising universality class to which this Landau model belongs. r_c is the correlation length, which diverges at criticality. The CE

density at T_C can then be expanded, with $\eta = \frac{1}{4}$ and $(d = 2)$, as

$$\frac{\partial S_C}{\partial k} \propto k^{-7/4} + \mathcal{O}(k^{-7/4} \log k). \tag{4.87}$$

The ratio between the first term to the second goes as $\log r_c$ close to criticality. We can thus neglect the second term and obtain the scaling law in the critical regime [85]. This scaling persists away from (but near) T_C, though suppressed to smaller and smaller spatial scales.

Since this regime obtains only close to criticality and CE is minimized in the critical strip, we conclude that the modes obeying this scaling law are storing the majority of the information encoded in the system. Our numerical calculations show that modes obeying the critical scaling are comparable to or smaller than the correlation length, while turbulent modes are larger. The presence of these modes causes a drop in the CE of the system, which we interpret as an increase in the compressibility of its states.

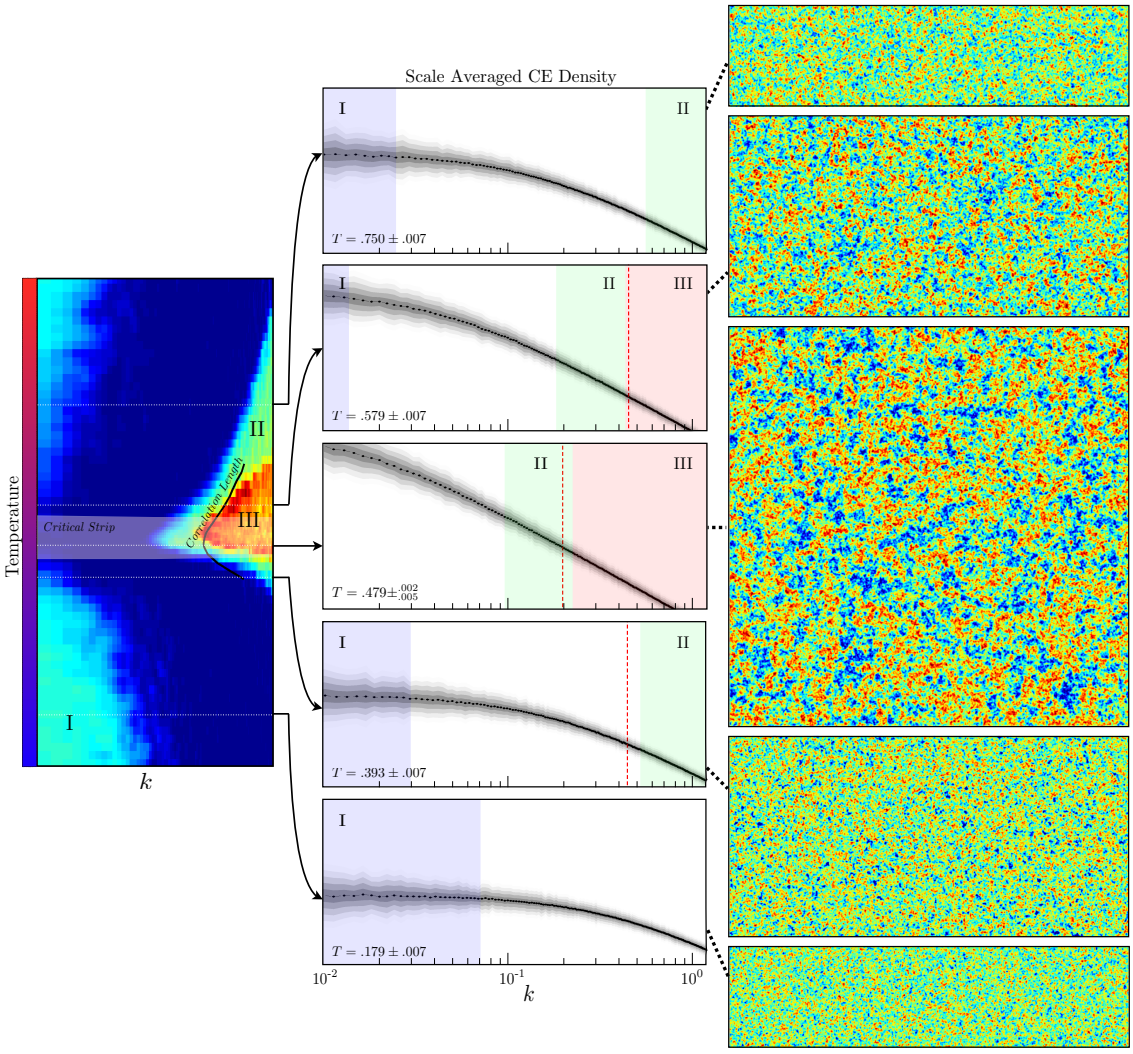

Figure 4.11: For different temperatures, there are three regions with different scaling behaviors: scale free (I), turbulent (II), and critical (III). At large scales the scale-free spectrum dominates. As the temperature approaches the critical temperature, the critical region moves from small to large scales. Between these two regions is a turbulent region. In the top panel one can examine the distribution of configurational entropy density at many scales. Close to criticality, the correlation length diverges, leading to large correlated domains. Small scales become critical.

Chapter 5

Walk Through the Fire

Our protagonists find themselves walking through the forest on a winter's night. The air is cold and crisp, and from above them the light of an infinitude of tiny pinpricks shines down.

Epistemicus: The winter here is almost unbearable. I'm wearing three layers, and I'm still not quite warm enough.

Oontolicus: It's the price we pay for this.

Both of them look up to gaze at the stars.

Oontolicus: It's amazing how clear the nights are here.

Epistemicus: The air is so dry, just look at Betelgeuse there in Orion. Not a flicker due to the atmosphere.

Oontolicus: I could just sit here for eons and stare at the sky. That's going to be spectacular show if we wait long enough.

The air is cold, and their breath condenses into a fog as it escapes their lungs. The sky stares back at them, ambivalent to the cold.

Epistemicus: Betelgeuse is quite large; over ten times the mass of our Sol, and a

radius nearly 1000 times as big.

Oontolicus: It's sure to light up the entire constellation when it runs out of fuel to support all that substance. All we have to do is wait a few eons...

Epistemicus: I'm cold and don't want to wait that long.

Oontolicus: Hahaha... ok, let's go make some tea at home.

The stability of localized field configurations will be the main focus of this chapter. To this end, the deviation from expectation form of the configurational entropy will be examined. Unstable complex field configurations known as Q-balls will be developed and analyzed first, followed by an analysis of gravitating polytropic fluids; this analysis was first done in my 2013 work, *Information-Entropic Stability Bound for Compact Objects: Application to Q-Balls and the Chandrasekhar Limit of Polytropes*, with Marcelo Gleiser [84]. Polytropes have direct application in astrophysics, describing degenerate stars known as white dwarfs.

5.1 Q-balls

Q-balls are non topological solitons first studied by Sidney Coleman, [16], and have since been studied extensively in the context of gauge field theories, supersymmetric extensions to the Standard Model, and as candidates for Dark Matter [80] [66] [4] [67] [62] [5] [8] [43] [32] [33] [63]. Their properties have even caught the attention of moviegoers: in 2007 the movie *Sunshine* had one at the center of its apocalyptic tale of a premature Solar demise [26].

5.1.1 The Potential

Consider a complex scalar field in a sextic potential. The Lagrangian density for this model reads:

$$\mathcal{L} = \partial_\mu \Phi^* \partial^\mu \Phi - V(|\Phi|) \tag{5.1}$$

$$V(|\Phi|) = m^2|\Phi|^2 - \frac{1}{2!}\mu^2|\Phi|^4 + \frac{1}{3!}\lambda^2|\Phi|^6 \tag{5.2}$$

The potential here is very interesting. It has a minimum at the origin, and one can interpret this model as two equal mass real scalar fields with interactions. As μ^2 changes from 0 up to $\sqrt{2}m\lambda$, this continues to be the unique vacuum state with a $U(1)$ symmetry. At $\mu^2 = \sqrt{2}m\lambda$ a new critical point forms at $|\Phi| = 2^{1/4}m$. This is a plateau, and is an unstable equilibrium for the field. As μ^2 continues to grow, a metastable vacuum appears at $|\Phi| = \frac{\mu}{\lambda}\sqrt{1+\sqrt{1-\frac{2m^2\lambda^2}{\mu^4}}}$ along with an unstable false vacuum at $\frac{\mu}{\lambda}\sqrt{1-\sqrt{1-\frac{2m^2\lambda^2}{\mu^4}}}$. When μ^2 reaches the value of $\sqrt{\frac{8}{3}}m\lambda$, the metastable vacuum becomes stable, changing roles with the vacuum at the origin as being the true vacuum.

We will only consider $\mu^2 \in [\sqrt{2}m\lambda, 2\sqrt{\frac{2}{3}}m\lambda]$, where a metastable vacuum exists. Q-balls will be field solutions that interpolate between the true and metastable vacuum, and can thus be thought of as non-perturbative excitations of the true vacuum at $|\Phi| = 0$. In the true vacuum we can rewrite the model as two interacting real scalar fields using $\Phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$. Perturbing both of these at the minimum of the potential reveals the masses of both field quanta are just m. In the metastable vacuum it is more convenient to use a polar form, $\Phi = \frac{1}{\sqrt{2}}\rho e^{i\phi}$. In the metastable vacuum the tangential field is massless due to the symmetry of the potential. The

other degree of freedom acquires a mass:

$$m_{MS}^2 = \frac{2\mu^4}{\lambda^2}(1 - \frac{2m^2\lambda^2}{\mu^4} + \sqrt{1 - \frac{2m^2\lambda^2}{\mu^4}}) \tag{5.3}$$

which ranges from 0, when the metastable vacuum first forms, all the way up to $4m^2$ by the time it becomes stable. A diagram of the potential and what it looks like at critical values of μ^2 is displayed in figure 5.1.

This theory has a global $U(1)$ symmetry; it is unchanged under the phase shift $\Phi \to e^{i\theta}\Phi$ for any constant θ. By Noether's theorem, this implies a conserved current in spacetime of the form

$$\mathcal{J}^\mu = i\left(\Phi\partial^\mu\Phi^* - \Phi^*\partial^\mu\Phi\right) \tag{5.4}$$

$$\partial_\mu\mathcal{J}^\mu = 0 \tag{5.5}$$

Integrating the continuity equation 5.5 over a spacetime volume between two parallel spacelike hypersurfaces gives the conserved charge:

$$Q = \int \mathrm{d}^d\mathbf{r}\ \mathcal{J}^0 = i\int \mathrm{d}^d\mathbf{r}\ \left(\Phi\dot{\Phi}^* - \Phi^*\dot{\Phi}\right) \tag{5.6}$$

This conserved charge is where Q-balls derive their name. It's conservation will ensure that certain spherically symmetric excitations of the vacuum are prevented from decaying away. Because this barrier is not due to a topological obstruction, Q-balls are non-topological solitons.

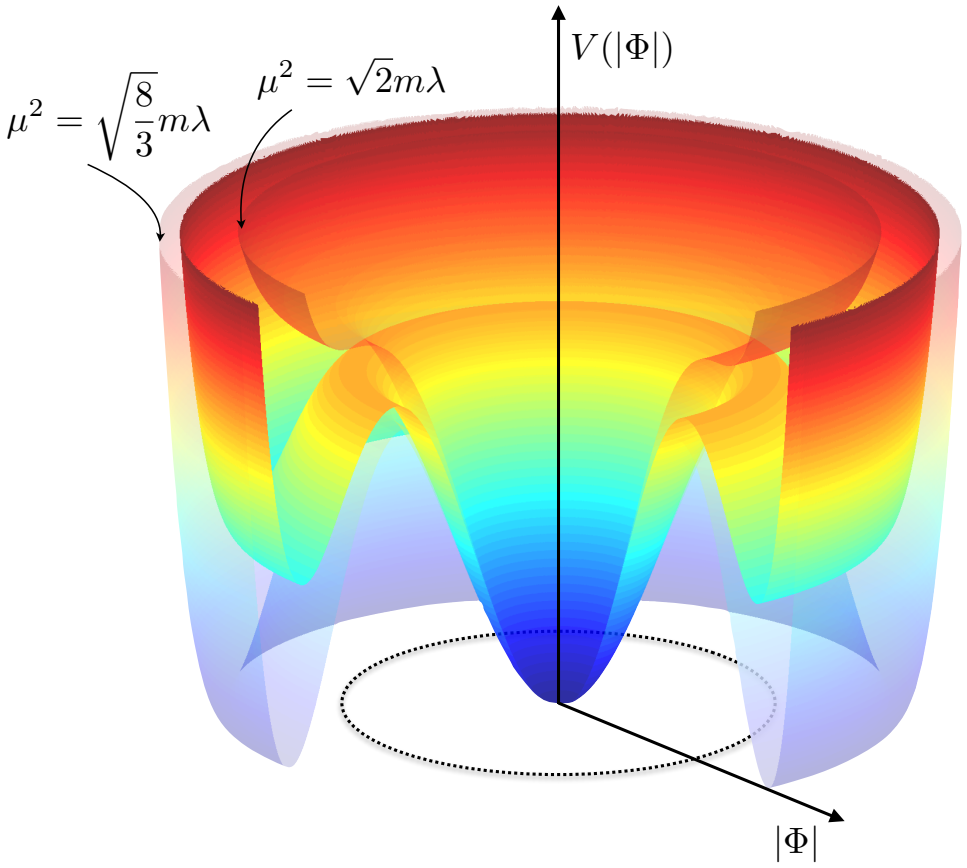

Figure 5.1: The potential that the complex field lives in. As μ^2 is changed, the shape of the potential changes. The minimum at the origin represents the true vacuum, which has a $U(1)$ symmetry. The ring at finite field value represents a metastable vacuum, with a broken $U(1)$ symmetry. Q-balls are spherically symmetric field configurations that interpolate between these two vacua.

5.1.2 Finding a Q-Ball

The Hamiltonian of the Lagrangian 5.1 reads

$$H[\Phi] = \int \mathrm{d}^d\mathbf{r} \ \left(\dot{\Phi}^*\dot{\Phi} + \nabla\Phi^* \cdot \nabla\Phi + V(|\Phi|)\right) \tag{5.7}$$

We will be interested in solutions that minimize this energy functional. We can incorporate the conservation of charge via a Lagrangian multiplier, ω:

$$\begin{aligned} H[\Phi,\chi] &= H[\Phi] + \omega\left(Q - i\int \mathrm{d}^d\mathbf{r}\ (\Phi\dot{\Phi}^* - \Phi^*\dot{\Phi})\right) \qquad (5.8)\\ &= \int \mathrm{d}^d\mathbf{r} \ \left(\dot{\Phi}^*\dot{\Phi} - i\omega\Phi\dot{\Phi}^* + i\omega\Phi^*\dot{\Phi} + \nabla\Phi^* \cdot \nabla\Phi + V(|\Phi|)\right) + \omega Q \\ &= \int \mathrm{d}^d\mathbf{r} \ \left(\dot{\Phi}^*\dot{\Phi} - i\omega\Phi\dot{\Phi}^* + i\omega\Phi^*\dot{\Phi} + \nabla\Phi^* \cdot \nabla\Phi + V(|\Phi|)\right) + \omega Q \\ &= \int \mathrm{d}^d\mathbf{r} \ \left(|\dot{\Phi} - i\omega\Phi|^2 + \nabla\Phi^* \cdot \nabla\Phi + V(\Phi) - \omega^2|\Phi|^2\right) + \omega Q \end{aligned}$$

(5.9)

Minimizing this is not too difficult. First notice that the kinetic energy term is positive definite. To minimize this we simply set it equal to zero to get our ansatz for the field distribution:

$$\dot{\Phi} - i\omega\Phi = 0 \quad \Rightarrow \quad \Phi(\mathbf{r},t) = \frac{1}{\sqrt{2}}\rho(\mathbf{r})e^{i\omega t} \tag{5.10}$$

So we see that the radial field is static, and the tangential field has a constant velocity. We don't mind since the tangential field is massless, and can move along the metastable vacuum at no cost in energy. The tangential field has vanished from the

Hamiltonian and what remains is:

$$H = \int \mathrm{d}^d\mathbf{r} \ \left(\frac{1}{2}\nabla\rho\cdot\nabla\rho + V(\frac{\rho}{\sqrt{2}}) - \frac{1}{2}\omega^2\rho^2\right) + \omega Q \tag{5.11}$$

Varying with respect to the radial field, we arrive at the configuration equation:

$$\nabla^2\rho = -\frac{dU}{d\rho} \tag{5.12}$$

Where $U = -V + \frac{1}{2}\omega^2\rho^2$ is an effective potential incorporating the Lagrange Multiplier term.

As a final requirement we impose spherical symmetry on the solution. We want to describe a ball after all. Plugging in the Laplacian we get:

$$\frac{d^2\rho}{dr^2} = -\frac{d}{r}\frac{d\rho}{dr} + \frac{dU}{d\rho} \tag{5.13}$$

There is a striking similarity here between a kinetic equation if we think about r as a temporal coordinate. The LHS is then the acceleration of the particle, and the RHS is a dampening term alongside force coming from the inverted potential U. The dampening term has a viscosity that is inversely proportional to the radial coordinate. We can now understand what the Lagrange multiplier is doing to the field by looking at the inverted potential:

$$U(\rho) = \frac{1}{2}(\omega^2 - m^2)\rho^2 + \frac{1}{8}\mu^2\rho^4 - \frac{1}{48}\lambda^2\rho^6 \tag{5.14}$$

Without the ω^2 term this potential would have a maximum at the origin. The origin is where the true vacuum lives, so we require that the solution of our Q-ball be such that far from its center we are at the true vacuum, $\lim_{r\to\infty}\rho(r) = 0$. But if the true vacuum is the global maximum of the potential, then there is no way that the solution could start anywhere else and eventually roll *down* to the origin. There is no *up* from the true vacuum. The Lagrange multiplier acts to lift the local maximum, raising it above the true vacuum, and allowing for the possibility of a solution. The process is illustrated in figure 5.2. Now the question remains, how do we find solutions?

5.1.3 On the Stability of Q-Balls

Before we attack the problem of a numerical technique for solving equation 5.13, let's dimensionalize our model so that we can work with dimensionless quantities and determine the parameter space of the model. Start with Hamiltonian 5.11, and transform the coordinates, fields, energy and charge as:

$$x^\mu \to \frac{1}{m}x^\mu \tag{5.15}$$

$$\omega \to m\omega \tag{5.16}$$

$$\rho \to \sqrt{\frac{2\sqrt{2}m}{\lambda}}\rho \tag{5.17}$$

$$H \to \frac{\lambda m^{3-d}}{2\sqrt{2}}E \tag{5.18}$$

$$Q \to \frac{\lambda m^{2-d}}{2\sqrt{2}}Q \tag{5.19}$$

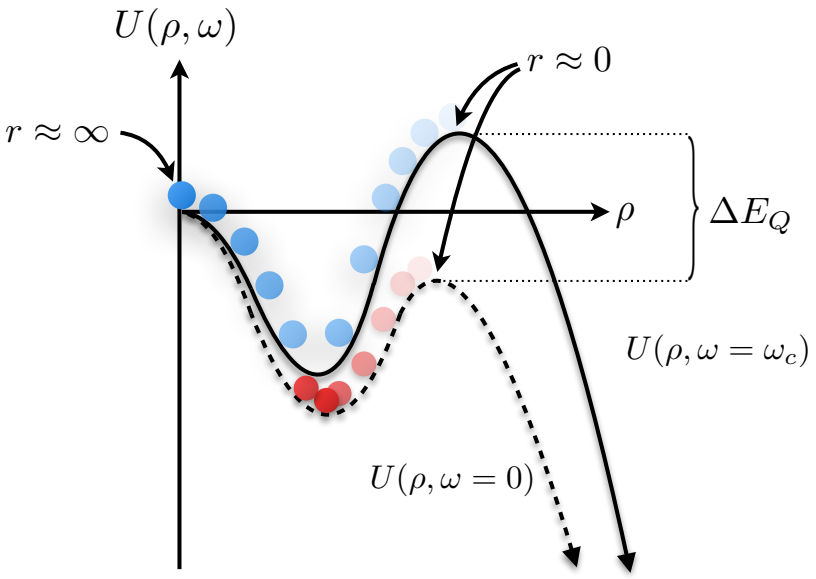

Figure 5.2: The effect of the Lagrange Multiplier, which is related to the charge of the Q-ball, acts to lift up the effective potential. There is a minimum height that the potential needs to be raised in order for solutions to exist, ΔE_Q. This energy barrier corresponds to some critical value of ω. When $\omega \geq \omega_c$, solutions will exist. The red ball's trajectory corresponds to a potential that is insufficient to get the solution to the true vacuum, which is located at the origin. The blue ball's trajectory starts at a high enough effective potential for a solution to exist and end in the true vacuum at spatial infinity.

and then define $b = \sqrt{2}\frac{\mu^2}{m\lambda}$. The model takes on a dimensionless form with two parameters, (b, ω), where we have minimized for the Lagrange multiplier:

$$E = \int \mathrm{d}^d\mathbf{r} \ \left(\frac{1}{2}\nabla\rho\cdot\nabla\rho + \frac{1}{2}(1-\omega^2)\rho^2 - \frac{1}{4}b\rho^4 + \frac{1}{6}\rho^6\right) \tag{5.20}$$

$$Q = \omega\int \mathrm{d}^d\mathbf{r} \ \rho^2 \tag{5.21}$$

It is necessary, though not sufficient, for the maximum of the effective potential to be greater than 0. This puts a lower bound on ω, namely

$$\omega > \sqrt{1 - \frac{3b^2}{16}} \tag{5.22}$$

Because of friction effects, ω's larger than this will not guarantee a Q-ball, but those smaller than this value definitely can't form. What about an upper bound?

The stability of a Q-ball rests on it being a lower energy state. A lower energy state of what though? Quantizing the field leads to the interpretation that the charge is related to the difference between particles and antiparticles of field quanta. Particles and antiparticles carry opposite charge, so the Q-ball is characterized by the surplus of one over the other. The question now becomes when is it more favorable to have free field quanta as opposed to a bound Q-ball? In dimensionless units the energy of the Q-ball is E. The energy of free field quanta is just their number (Q) times the mass of an individual particle (m, which in dimensionalized units is 1). We therefore are led to examining the binding fraction of these configurations:

$$\epsilon = \left|\frac{E}{Q}\right| \tag{5.23}$$

When this quantity is less than 1 then it is energetically favorable for a classical Q-ball to form. When it reaches one the Q-balls become unstable under fission. Rather than just exploding into a flurry of particles (or antiparticles) Q-balls just above the binding energy limit from metastable, non localized objects that have come to be known as Q-clouds [3]. We shall examine the Configurational entropy for both Q-balls and Q-clouds, but first let's discuss a numerical procedure for finding them.

Numerics

We use a 4th order Runge Kutta method to integrate the two first order equations:

$$\frac{d\psi}{dr} = -\frac{2}{r}\psi + (1-\omega^2)\rho - b\rho^3 + \rho^5 \tag{5.24}$$

$$\frac{d\rho}{dr} = \psi \tag{5.25}$$

$$\psi(0) = 0 \quad \rho(0) = \rho_c \tag{5.26}$$

We discretize the parameters (ω, b) into steps 10^{-3} in size over the domain established int he prior section. To find the core value, ρ_c, of the Q-ball we use a shooting method. We initialize the value of the core, ρ_0 midway between the two values $[\rho_-, \rho_+]$ that define where the potential is zero,$V(\rho_-) = 0$, and the metastable minimum,$V'(\rho_+) = 0$. We then run the RK4 method on 5.24, and look at the far r behavior. If the solution begins to move away from 0 on the negative side, then we must have initialized the core at too high a value. Hence we move $\rho_+ \to \rho_0$, keeping the lower bound the same, and reinitialize the core,ρ_1, at the midpoint of the refined region. If the solution begins to move away from 0 on the positive side, then we must have initialized the core at too low a value. In this case we move the lower bound up, $\rho_- \to \rho_0$, and

reinitialize the core midway between the new finer range. We repeat this process until the core value is pinpointed to 64-bit floating point accuracy. This numerical procedure is typically referred to as a shooting method. [79]

5.1.4 Configurational Entropy

In Figure 5.3 we plot the solution landscape of Q-balls as a function of b and ω. The dashed lines denote contours of constant ratio E/Q. Q-balls exist within the central region. The bold black line denotes the classical stability limit $E/Q = 1$. We also indicate the Q-cloud region, within the classically unstable area above the $E/Q = 1$ line.

In Figure 5.4 we plot the ratio E/Q of Q-balls for several allowed values of b as a function of their configurational entropy. The CE is computed from the energy density $\rho(r)$ for each Q-ball solution. Each point corresponds to a solution of Eq. **??**. The curves start at the lower left and run upwards with increasing ω. Q-balls are classically stable when $E/Q < 1$. There is a clear correlation between the binding energy and the configurational entropy: Figure 5.3 shows the maximal CE tracing the line of instability; Figure 5.4 shows that the maximum CE overshoots the line of instability by no more than 3%.

5.2 Polytropic Stars

5.2.1 The Model

Consider a space filled with a three unknown scalar fields: a classical matter density distribution, $\rho = \rho(\mathbf{r})$, a pressure distribution $P = P(\mathbf{r})$, and a gravitational potential

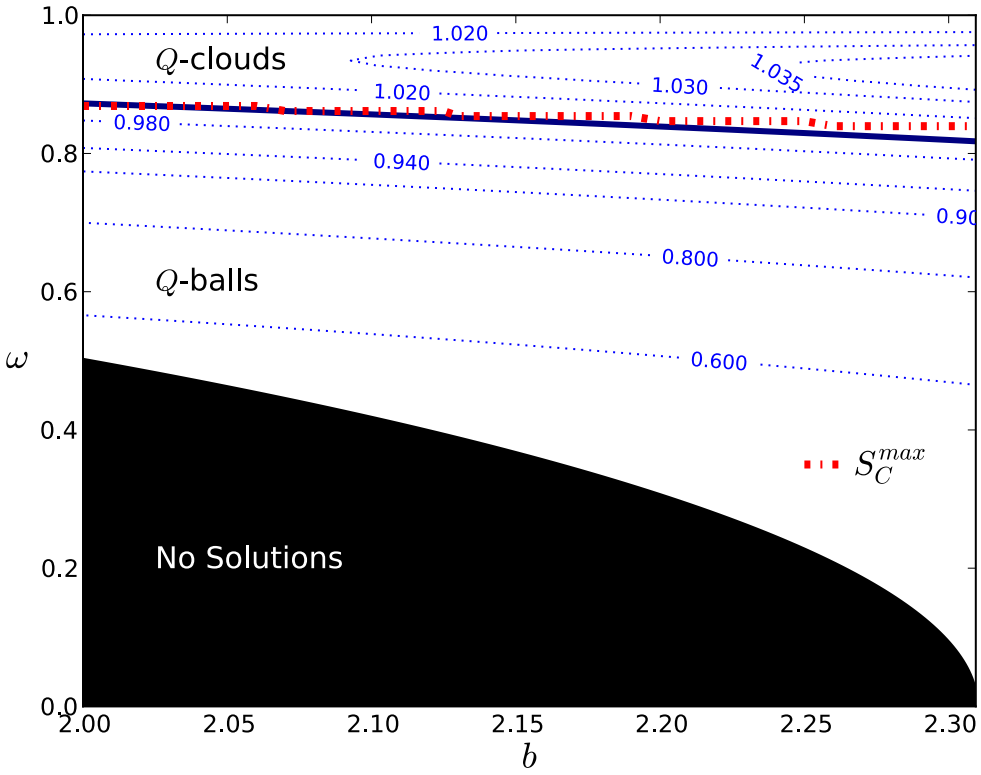

Figure 5.3: The solution landscape of Q-balls. The dotted (blue) lines represent contours of E/Q. The shaded region is forbidden by the inequality $\omega_c \geq \sqrt{1-3b^2/16}$, while the bold continuous line represents $E/Q = 1$. The thick dotted (red) line is the maximum of the configurational entropy (cf. Figure 5.4). Its near overlap with the $E/Q = 1$ line confirms that the CE provides an efficient measure of Q-ball stability.

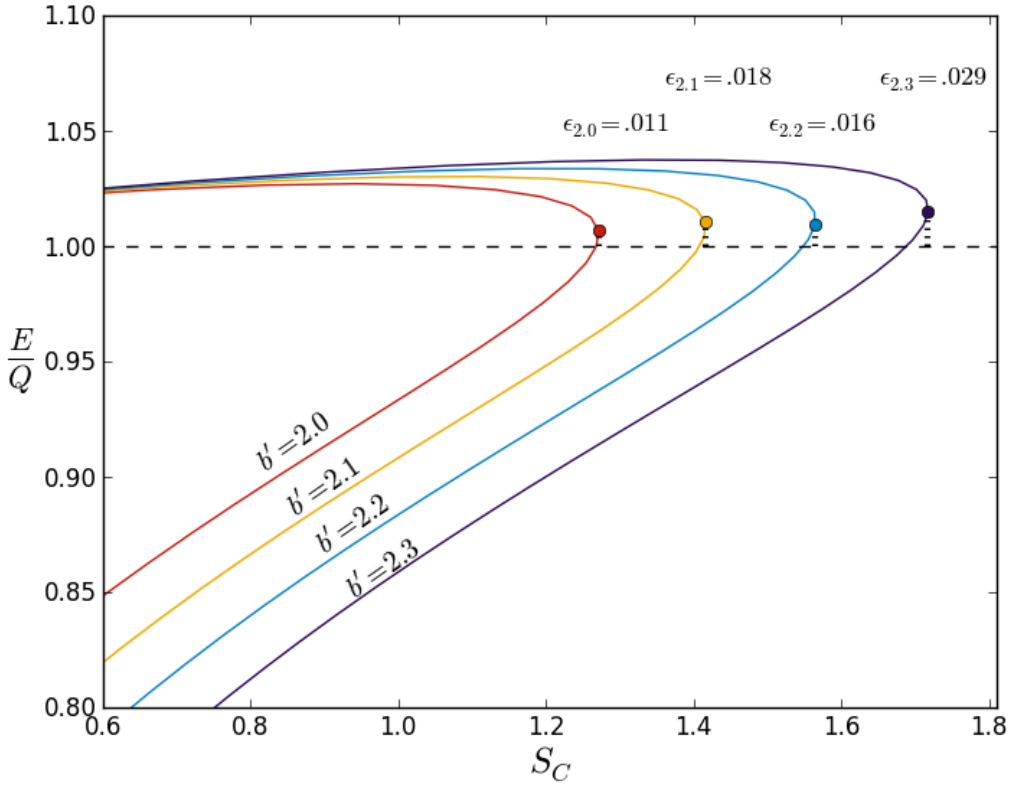

Figure 5.4: Q-ball instability ratio versus configurational entropy parametrized by the value of ω for several values of b. Percent errors indicate the deviation of the maximum CE from the instability line (dashed).

$\phi = \phi(\mathbf{r})$. How are these quantities related?

If the pressure gradients oppose the gravitational forces, then there will be no net force on individual matter elements, and the distribution is said to be in *hydrostatic equilibrium*. This condition is written:

$$\nabla P = -\rho \nabla \phi \tag{5.27}$$

Meanwhile the gravitational field is described by *Newton's Theory of Universal Gravitation*, so is sourced by the matter density via Poisson's equation:

$$\nabla^2 \phi = 4\pi G \rho \tag{5.28}$$

Lastly, we assume that there exists a closure relation between the three unknown fields known as an *Equation of State* of the form:

$$\mathcal{E}[\rho, P, \dots] = K \tag{5.29}$$

where K is a constant. If we can isolate the pressure in the equation of state then this allows us to write $P = P(\rho)$ when we are lazy, though this is an abuse of notation.

We will be interested in all density distributions that satisfy these constrains, in particular spherically symmetric ones. Before imposing this symmetry contraint, note that we can combine 5.27 with 5.28 to get rid of the gravitational potential:

$$\nabla \cdot \frac{\nabla P}{\rho} = -4\pi G \rho \tag{5.30}$$

Next we shall consider the equation of state. We will be interested in substances that

obey a polytropic equation of state such that:

$$\mathcal{E} = P\rho^{-\gamma} \quad \Rightarrow \quad P = K\rho^{\gamma} \quad \Rightarrow \quad \nabla P = \gamma K \rho^{\gamma-1} \nabla\rho \tag{5.31}$$

If the processes that occur in the substance are restricted to maintaining constant entropy, then the exponent γ is the adiabatic index. Other types of processes can also be represented in this form, including isothermal ($\gamma = 1$), isobaric ($\gamma = 0$), and isochoric ($\gamma = \infty$). We will assume only isentropic processes occur, so the substance can be considered at equilibrium. Many systems can be so described. Nonrelativistic degenerate electron gases in a matrix of nucleons can be modeled by $\gamma = \frac{5}{3}$ and $K \approx \frac{\hbar^2}{15\pi^2 m_e}\left(\frac{3\pi^2 Z}{m_N A}\right)^{5/3}$. Relativistic degenerate electron gases, on the other hand, can be modeled with $\gamma = \frac{4}{3}$ and $K = \frac{hc}{8}\left(\frac{3Z}{\pi m_N^4 A}\right)^{1/3}$ [93], [82]. This means that polytropic fluids are very convenient for describing white dwarfs as they approach the Chandrasaekhar limit.

Plugging 5.31 into 5.30 and performing a little algebra we can get:

$$\nabla^2 \rho^{\gamma-1} = \frac{4\pi(\gamma-1)G}{\gamma K}\rho \tag{5.32}$$

Next we perform, as usual, a dimensionalization procedure. We introduce a dimensionless field θ, such that $\theta(0) = 1$, and transform:

$$\mathbf{r} = \alpha\xi = \sqrt{\frac{\gamma K \rho_c^{\gamma-2}}{(\gamma-1)4\pi G}}\xi \tag{5.33}$$

$$\rho(\mathbf{r}) = \rho_c \theta(\xi)^{\frac{1}{\gamma-1}} \tag{5.34}$$

The transformation coefficient of the coordinates is known as the scale length, so that

ξ is counting the number of scale lengths. With these transformations, our equation becomes:

$$\nabla^2\theta = -\theta^{\frac{1}{\gamma-1}} \tag{5.35}$$

Finally we want to describe localized, spherically symmetric matter distributions in 3-d, so the resulting equation we will study with boundary conditions is:

$$\frac{1}{\xi^2}\frac{d}{d\xi}\xi^2\frac{d}{d\xi}\theta + \theta^{\frac{1}{\gamma-1}} = 0 \tag{5.36}$$

$$\theta(0) = 1 \quad \theta'(0) = 0 \quad \theta(R) = 0 \tag{5.37}$$

This equation is known as the Lane-Emden equation [64]. Solutions are obtained numerically.

5.2.2 Solutions to Lane-Emden

There are three known analytical solutions for the Lane-Emden equation 5.36.

For the isochoric case, $\gamma = \infty$, and we have a constant density object. Lane-Emden gives us:

$$\theta = 1 - \frac{1}{6}\xi^2 \tag{5.38}$$

Thus the radius of this polytrope is $\sqrt{6}$ scale lengths. This object is rather unrealistic, but a good starting point. For the case $\gamma = 2$, which is just above the nonrelativistic

degenerate electron gas, the exact solution is:

$$\theta = \frac{\sin \xi}{\xi} \tag{5.39}$$

And has a radius of π scale lengths. Finally for $\gamma = 4/5$, which is just below the relativistic degenerate electron gas, the solution is:

$$\theta = \frac{1}{\sqrt{1 + \frac{1}{3}\xi^2}} \tag{5.40}$$

which has an infinite radius [15]. This value of γ is the boundary between solutions with a finite and an infinite radius, though it does have a finite mass. Polytropes with indices lower than this value have been found to fit well for globular clusters, though the interpretation is strange, since these have formally an infinite mass [48]. There is a nice relationship between the radius of a polytrope and its total mass:

$$M^{2-\gamma} R^{3\gamma-4} = \frac{K}{G N_\gamma} \tag{5.41}$$

Where the constant depends only on the adiabatic index. For nonrelativistic electron gases we have $\frac{M}{M_\odot}\left(\frac{R}{R_\odot}\right)^3 \approx \left(\frac{Z}{A}\right)^4 6.65 \times 10^{-5}$, where we have scaled to solar units. We defer the ultrarelativistic calculation till the next section, where we examine something very interesting happening for $\gamma = \frac{4}{3}$.

5.2.3 The Chandrasekhar Limit

The kinetic energy for a virialized gravitational system reads:

$$K = \frac{1}{3}\frac{1}{\gamma - 1}U \tag{5.42}$$

One can quickly see that $\gamma \to \frac{4}{3}$ reproduces the non relativistic result $2K + U = 0$. Meanwhile for relativistic systems $\gamma \to \frac{5}{3}$, so $K+U = 0$. As the system becomes more and more relativistic, the total energy approaches 0 and becomes unstable. A degenerate electron gas will no longer be capable of maintaining hydrostatic equilibrium when it becomes relativistic.

Plugging in the values for an ultrarelativistic electron gas into the Mass-Radius relationship we find a unique value:

$$M = \left(\frac{K}{GN_{\frac{4}{3}}}\right)^{\frac{3}{2}} = \left(\frac{\hbar^2}{12\pi^2 G}\right)^{\frac{3}{2}} \left(\frac{3\pi^2}{m_N}\right)^2 \frac{1}{N^{\frac{3}{2}}_{\frac{4}{3}}} \left(\frac{Z}{A}\right)^2 \approx 5.82 \left(\frac{Z}{A}\right)^2 M_\odot \tag{5.43}$$

For a white dwarf that no longer has any hydrogen in it, and is composed primarily of Helium we have $M \approx 1.45 M_\odot$. This is known as the Chandrasekhar mass. If a non relativistic white dwarf begins accreting mass, then its electrons will slowly become more and more relativistic. When the mass reaches this value the electrons will be relativistic and the polytropic instability will set in, resulting in a supernova.

5.2.4 Configurational Entropy

We wanted to see what configurational entropy does in the system described above and to see if it is sensitive to the instability. We calculated the density profiles for

polytropes in the range $\gamma \in (\frac{4}{5}, 2)$ numerically, using a fourth order Runge-Kutta scheme on the set of first order equations:

$$\frac{d\psi}{d\xi} = -\xi^2 \theta^{\frac{1}{\gamma-1}} \tag{5.44}$$

$$\frac{d\theta}{d\xi} = \xi^{-2}\psi \tag{5.45}$$

$$\theta(0) = 1 \qquad \psi(0) = 0 \tag{5.46}$$

We use a step size of 10^{-3} in both ξ and γ. Solutions are displayed in figure 5.5. We then took the Fourier transform of the profiles to calculate the modal fractions and the configurational entropies, with several caveats.

Since these are spherically symmetric profiles, localized objects in space, the natural measure to use to analyze them is the deviation from expectation form of the CE. This means that a cutoff needs to be imposed. Since we are focusing on the polytrope, the natural scale is it's radius.

There is a natural interpretation here if we think back to an epistemic agent examining the polytrope. She could look at a very large volume, but then the size of the polytrope will become the scale that she is drawn to. After realizing that the polytrope has structure, she will focus on the polytrope to find any new scales of interest. This focusing on a localized object is the equivalent of imposing a cutoff on CE. Modes that are larger than the space being focused on are no longer relevant to the EA. The cutoff at a particular scale is not the same thing as imposing periodic boundary conditions as in the Landau model: the modes here are still a continuum. Secondly, using the continuum form of the CE demands that we regularize the integral.

The results are shown in figure 5.6. The Configurational entropy is at a minimum

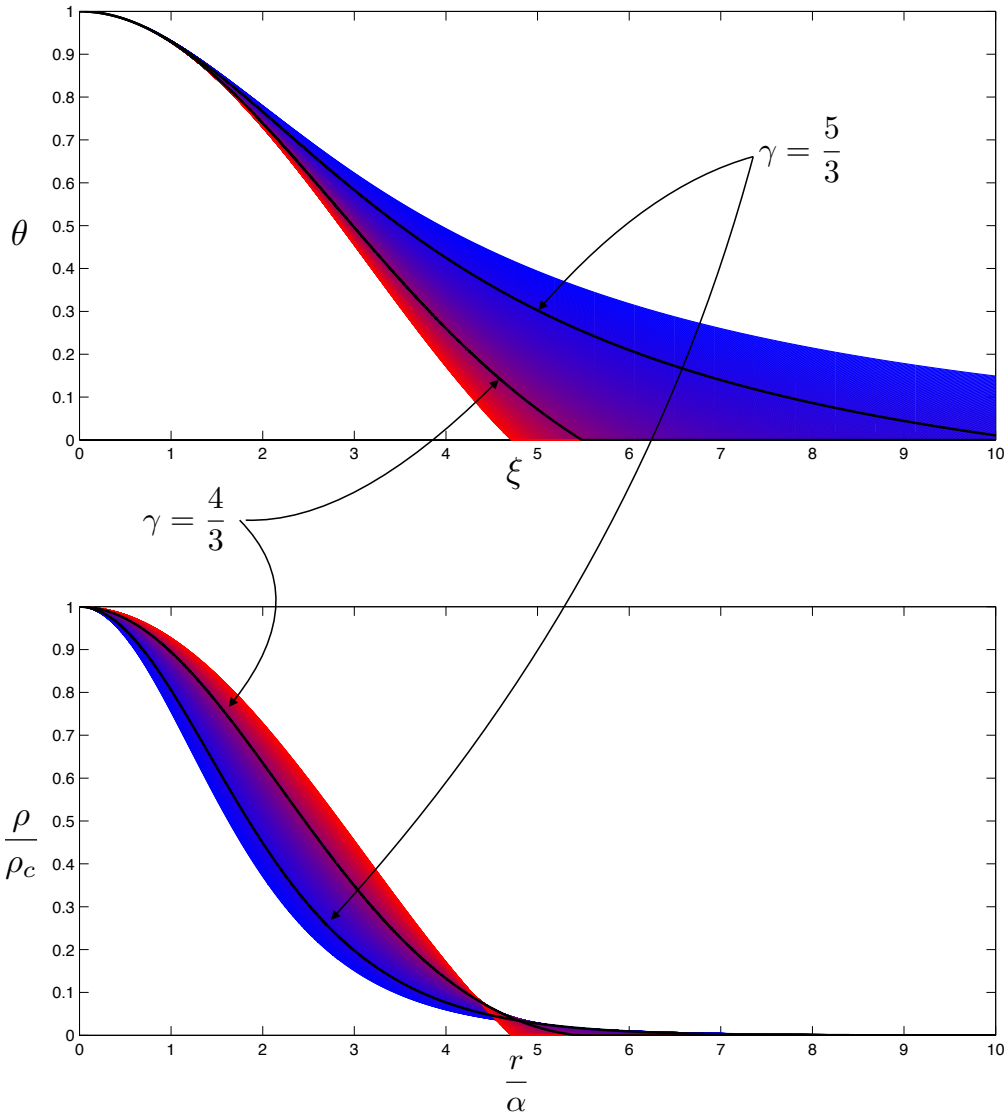

Figure 5.5: Solutions of the Lane-Emden equation for Polytropes in the range $[1.25, 2]$ increasing from red to blue. The two black profiles are the nonrelativistic white dwarf ($\gamma = \frac{5}{3}$) and the relativistic white dwarf ($\gamma = \frac{4}{3}$). The top panel shows the Lane-Emden function, θ, wheras the bottom panel displays the density fraction in scale lengths.

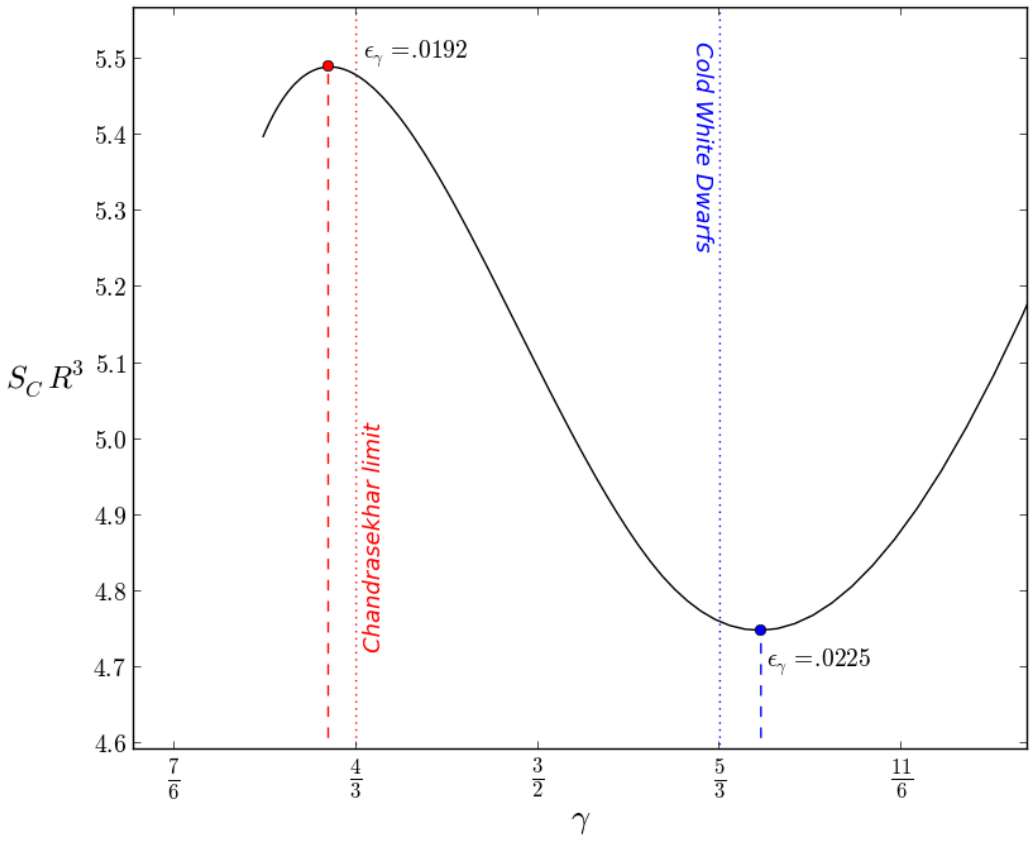

Figure 5.6: The CE of polytropes at different polytropic indices. The maximum CE comes very close to the value of $\gamma = \frac{4}{3}$, with a relative error of only .0192. The minimum falls to within a relative error of .0225 of $\gamma = \frac{5}{3}$.

very close to the nonrelativistic polytrope, and at a maximum very close to the relativistic, and unstable, polytrope. Since we are using the deviation from expectation form of CE, these results are striking. Stable polytropes have a minimum deviation. This would be expected since cold white dwarves are incredibly stable objects. The relativistic polytrope, on the other hand is highly unstable, which means its behavior may differ significantly from expectations.

Chapter 6

Where Do We Go From Here?

Where our protagonists find themselves to be unwitting participants in a thesis.

Epistemicus: This whole time, Damian was using us to give his readers a little literary enjoyment while trying to build a foundation for the work he's been doing in grad school.

Oontolicus: I know! Without even asking us, the nerve...

Epistemicus: Well, if we're to be participants in this endeavor we should at least point out the holes in his reasoning.

Oontolicus: It seems only fair.

The two pour themselves some tea and settle down next to the fireplace.

Epistemicus: I did enjoy his desire to examine the connection between logic, belief, and information. However, he did seem to focus solely on sentential logic. What about predicate logic?

Oontolicus: Ah, yes. All humans are mortal. Socrates is a human, so Socrates is mortal. The prototypical syllogism of predicate logic.

Epistemicus: Precisely! Can any of this be extended to languages with quantifiers?

Oontolicus: Sentential logic speaks about the properties of the World, encapsulating those properties in sentences. In a sense, a statement is about that one object, namely the World. But what about statements that discuss the properties of multiple objects?

Epistemicus: That's what the quantifiers *there exists*, $\exists$, and *for all*, $\forall$, allow us to do: speak about the properties of multiple objects. It is unclear as to how to extend the Cox axioms of consistency and common sense to the quantifiers. O, for that matter, logics higher than either the zeroth or first order types. Clearly more work needs to be done

Oontolicus: Even if we stick to sentential logic, how do we store all the sentences of a language in a physical substrate? There're more sentences than there are atoms in the Universe, so is it even sensible to say that a real thinking thing is anything like an EA?

Epistemicus: Maybe they only store some fraction of the language... somewhere.

Oontolicus: That might work... but if sentences are being represented by patterns in some substrate, the updating procedure should obey the known laws of physics. In that case, beliefs might not update simultaneously. *Inference propagation fronts* might emerge in the physical substrate, leading to contradictions that need to be resolved.

Epistemicus: Clearly, there are many ways in which we could try to generalize these ideas in future work. I'd be interested to see if the use of a generalized mean can be incorporated into the definition of total hidden information so that the resulting entropy has the Renyi for instead of just Shannon.

Oontolicus: Right, or what if the total hidden information for independent systems

deviates from the additive property. Then the proper measure to use would be the Tsallis entropy. Clearly there are more generalizations lurking in the shadows if we think long enough.

The two sip their tea with small hints of smiles hidden on their faces.

Oontolicus: Then there's his biological connection to CE in Chapter 3.

Epistemicus: I was just thinking about that. Should we start the fire?

Oontolicus: Oh, yes, that would be nice...

Epistemicus gets up to grab some pieces of firewood and throw the in the fireplace. After setting up some kindling, he starts a small flame underneath the wood and sits back down.

Epistemicus: It seems to me that total experience should be predicated by the sensitivity of perceptual organs...

Oontolicus: Haha, now you're thinking like a biologist; I agree wholeheartedly.

Epistemicus: We know that our senses don't span all possible length scales, and the Weber-Fechner law breaks down at the edge of perception. Mathematically speaking, one could model that biological limitation with a window function, and then perhaps explore the relationship between biology and hidden information.

Oontolicus: How does limited perception limit what we CAN believe?

Epistemicus: Exactly.

The two stare at the flames for a long time, watching the pieces of wood disintegrate, first turning into bright embers only to slowly fade away into imperceptible ash. The fire crackles and warms our protagonists from without, not unlike the tea they sip waring them from within.

Epistemicus: There's a lot left to be done.

Oontolicus: Undoubtebly... I wouldn't mind being there to see what comes of these ideas.

Epistemicus: Agreed. I'm certain this isn't the last time Damian will use us in his writing. But, you know, next time I don't believe we'll be here in the wilderness.

Oontolicus shifts his gaze from the fireplace to the window. Outside the snow has become a blanket covering all the world. Thick layers of it rest on the branches of evergreens, while the birch trees jut out from the forest floor like thin white extensions of the snow itself.

Oontolicus: I will miss this Kingdom of New Vermontshire. It has grown me.

Epistemicus: As will I, my old friend. As will I.

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